Owning the Chokepoint: ASML, Alliance Dependence, and the Limits of Europe's Leverage over Washington

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Abstract

Since 2019 the Netherlands, home of ASML, the only maker of extreme ultraviolet lithography machines, has aligned its export controls with US demands while extracting little in return, as has Japan since 2023. Yet both resist some demands, such as limits on servicing their installed base in China. In our game-theoretic model, a security-dependent chokepoint holder bargains with its patron while the patron's rival bids for it. We prove each result and check it numerically. The holder's exit premium, the rival's credible offer net of the credible costs of exit, is a floor on what it obtains. Without rupture risk or agenda power, and without hold-up, its leverage is the positive part of that premium. The patron's demands are bounded by the holder's slack, which all its conditional threats share. Conditionality is the price of concessions, and a wasting slack gives the patron a preventive motive to demand control. The sign of Europe's premium turns on China's credible promises and, if Washington's threat to withhold frontier AI is credible, on that instrument, which on our orders of magnitude weighs more than how conditional the US guarantee is.

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1 Introduction

In *Europe 2031*, a scenario published in June 2026 by eight authors writing in a personal capacity, the story's climax comes in March 2031, when the US vice president offers the leaders of the Netherlands, Germany and France a joint Dutch-American holding company in which Washington holds a controlling vote over ASML's output, customers and technology transfer (Juijn et al., 2026). ASML, the Dutch firm that alone builds extreme ultraviolet (EUV) lithography machines, is what the scenario's summary page calls "the only card Europe still holds". Three years earlier in the story, the Dutch end ASML's remaining business in China under US pressure, and one character reflects that "we didn't even manage to negotiate a single thing in return". The scenario is fiction (Appendix D). Its premise describes the record.

Since 2019 The Hague has aligned its export controls in steps, from 2023 each following a US move, and in November 2025 the EU put the most advanced immersion lithography tools on its control list (European Commission, 2025a). What The Hague received was a status. In December 2024 the Bureau of Industry and Security (BIS) placed the Netherlands on Supplement No. 4, whose listed countries' exporters are spared new US licence requirements on chipmaking equipment because their own governments can control the items (Bureau of Industry and Security, 2024). BIS can withdraw the status by amending a rule, and we code it as a threat left unexercised, a reward for self-binding and not a transfer (Section 2 and Section 8.2).

Yet the holders of these chokepoints resist some demands. In April 2026 the MATCH Act, introduced in both chambers, proposed US controls on all immersion lithography and on servicing listed Chinese fabs, to be applied to allies whose own controls were not certified as equivalent within 150 days (Baumgartner, 2026b; Ricketts, 2026). On 12 May the Dutch trade minister told parliament that "the cabinet is against the extraterritorial effect of the American proposal" (Sjoerdsma, 2026, our translation). As of June 2026 both the Netherlands and Japan still let their firms service equipment in Chinese fabs (Zakaria, 2026). Tokyo had put 23 types of chipmaking equipment under licence in 2023 while stating that it was "not aligning itself with the United States' measures" (Allen, 2023; Scanlan-Dyas et al., 2023).

Holders that Washington needs thus comply for little, and selectively. Theories of asymmetric and weaponized interdependence give power to the less dependent party and to the state with jurisdiction over a hub (Hirschman, 1945; Keohane and Nye, 1977; Farrell and Newman, 2019). On that account Washington, which has no substitute for EUV lithography, should be paying The Hague. Accounts of alliance dependence and hierarchy treat compliance as the price of protection already received (Morrow, 1991; Lake, 2009). They fit the compliance and leave the resistance on servicing and extraterritorial reach unexplained. Models of rival bidders hold that a client courted by two patrons is paid what the losing bidder would offer (Groseclose and Snyder, 1996; Bueno de Mesquita and Smith, 2016), as Kyrgyzstan was paid for the Manas base in 2009 (Cooley, 2012). A bidding account needs a rival offer conditioned on exit. China is ASML's largest market (ASML Holding N.V., 2026a), but we have found no Chinese offer conditioned on Dutch exit, and The Hague has not been paid.

We argue that a chokepoint holder's exit premium is a floor on what it obtains from its patron, and that the patron's demands are bounded by the holder's slack. The exit premium is the rival's credible offer net of the credible costs of exit, including the value the holder places on alignment; slack is the amount by which the premium falls short of zero. Unless talks risk rupture while the holder has agenda power, or the holder can impose costs on the patron while the two bargain, the holder's leverage is exactly the positive part of its premium (Propositions 1 and 6). A holder with a negative premium then gets nothing for its chokepoint, however much its patron needs it. The floor is a known result: the outside-option principle (Shaked and Sutton, 1984; Sutton, 1986; Binmore et al., 1989) and the one-voter case of sequential vote buying (Groseclose and Snyder, 1996; Dekel et al., 2008). The bound on demands organizes the paper. Clayton et al. (2026b) write that "threats generate power to the extent that they generate slack in the participation constraint". We take their term and ask what slack buys a patron when each of its threats must be sequentially rational, when the holder's outside option is a rival's bid, and when a threat that is carried out is spent. Our answer is a slack principle: whatever instruments back them, the conditional parts of a patron's threats draw on one budget, and every demand the patron makes (on sales, servicing, hedging or control) competes for it. Under rupture risk, whatever agenda power wins Europe above the positive part of its premium, Washington's credible conditional threats can claw back, down to the premium itself (Remark 1). The patron must ration its demands. A demand meets resistance when its cost to the holder exceeds the bite that remains, and China can raise that cost: on our reading, its 2026 decree on supply-chain security attaches a counter-threat to any order to stop servicing Chinese fabs (State Council of the People's Republic of China, 2026).

After the background and the literature (Sections 2 and 3), Section 4 sets out the game. The United States and China bid for Europe's alignment under one commitment technology, and China's offer is Europe's outside option in bargaining with Washington that can rupture (Lemma 1 and Proposition 1). Washington's dependence on ASML enters through five channels: agenda power under rupture risk, the rival's valuation, self-harming sanctions, the silicon shield and hold-up (Corollary 1). The first and the last can raise Europe's transfer above the positive part of its premium; the other three work through the premium. Section 5 proves the slack principle. In every subgame perfect equilibrium of any sequence of ultimatums, Europe's total loss is bounded by the bite of the credible threats, the unconditional parts plus the smaller of the conditional parts and the slack (Proposition 2), and demands are rationed from a capacity that shrinks as the premium rises toward zero (Corollary 2). China executes a counter-threat only to crowd out another US demand, and if Washington has a divisible sunk demand beyond China's reach, the counter-threat taxes the patron and leaves Europe's loss unchanged (Proposition 3). Because conditionality has a cost in allied assurance, a patron makes its guarantee more conditional than deterrence requires only to create slack it can spend, so conditionality is the price of concessions, and a stronger Chinese bid can be met with a more conditional guarantee instead of a payment (Proposition 4).

Section 6 shows that a wasting slack gives Washington a preventive motive to demand control of the chokepoint while its threats still bite (Proposition 5). An autonomy investment that Europe

cannot commit to forgo strengthens the motive, and a credible Chinese threat against a transfer can protect European ownership without being carried out (Corollary 3). A cut is spent once executed, and extending a conditional cut into dates at which Europe's premium is positive adds nothing to its bite (Lemma 2), so demands come in windows (Corollary 4). If Washington is unsure what a transfer would cost the Dutch government, an ultimatum can be refused and the threat carried out (Proposition 7). Section 7 separates the Dutch licensing authority, the firm and the Council. A benefit aimed at The Hague alone buys alignment at a discount (Remark 3), Chinese penalties for complying with US measures can make ASML ship where The Hague licenses it (Remark 4), and Washington defeats a countermeasure most cheaply by buying a blocking minority before the Council's determination (Remark 5). Interrupting servicing during a dispute helps Europe only if the US buffer of tools and know-how is short, only for the servicing Washington cannot compel, and only under a rule fixed before offers are made (Proposition 6 and Corollary 5), which on our reading the Union's current instruments do not supply.

Section 8 is a plausibility probe. It derives predictions that separate the slack logic from rival accounts (Table 2) and reads the Dutch record, the Nexperia dispute of 2025 and other holders through the model (Sections 8.3 and 8.4). The record is consistent with the jurisdictional channel. Washington has coerced mainly with statuses it can grant and withdraw by rule, and its jurisdiction over ASML's inputs and staff can make the premium negative without any security term. We found no public record linking European chip controls to US force posture, so the security channel is largely untested. On our orders of magnitude, if Washington's threat to withhold frontier AI is credible, the AI instrument is likelier than the conditional part of the guarantee to decide the sign of Europe's premium. Section 9 draws implications and states the scope conditions.

We contribute to four literatures. In geoeconomics, where Clayton et al. (2026a) pool enforcement threats that are never spent and the bidding models cited above have carrots only, we offer a theory of how a patron rations cut-off threats across demands as an ally's outside option improves. In alliance politics, where abandonment threats are known to work when they are conditional and paired with assurance (Gerzhoy, 2015; Blankenship, 2023), we make conditionality a costly choice whose value to the patron is the slack it creates against the rival's credible offer. In work on shifting power, preventive war answers a rising state's inability to commit (Fearon, 1995; Powell, 2006; Krainin, 2017; Debs and Monteiro, 2014), and control rights are a remedy for hold-up (Grossman and Hart, 1986; Lake, 1996; Cooley and Spruyt, 2009). We show that the conditional part of a patron's coercive power to demand control is capped by the holder's current slack, so a coerced demand comes in windows while that slack lasts. When Washington values control little and Europe has no autonomy investment that a transfer would forestall, an accepted transfer is inefficient in the way preventive war is: Europe bears a sovereignty cost to spare Washington rents that are only transfers. Finally, to the two conditions for weaponizing a hub in Farrell and Newman (2019), jurisdiction and the institutions to use it, we add a third for a holder that bargains with a state that depends on the hub (Section 10).

2 The Chokepoint and Its Owners

2.1 A nested chokepoint

ASML is the only firm that makes extreme ultraviolet (EUV) lithography machines, on which the leading-edge logic processes used for the most advanced US AI accelerators depend (Miller, 2022; ASML Holding N.V., 2026a). China, which cannot buy them, buys immersion deep-ultraviolet (DUV) tools instead, and Fedasiuk and Torres (2026) estimate that Chinese entities took 70 percent of ASML's immersion DUV sales in 2024. Exposing the same layer several times (multiple patterning) lets these tools print features that would otherwise need EUV. SMIC makes 7nm logic this way, and Huawei's Ascend accelerators are built on that node; without domestic EUV, Allen (2025) expects China to stay at about 7nm. Customers in China accounted for 26.3 percent of ASML's net sales in 2023, 36.1 percent in 2024 and 29.1 percent in 2025 (ASML Holding N.V., 2026a). Servicing follows each sale: installed-base management made up 25.1 percent of ASML's 2025 sales (ASML Holding N.V., 2026a), and several hundred ASML immersion tools in China need specialized maintenance about every six months, often from ASML staff on site (Fedasiuk and Torres, 2026).

ASML's supply chain runs through Germany and the United States. Carl Zeiss SMT is its sole supplier of critical optics (ASML Holding N.V., 2026a). At the end of 2025 ASML had 8,242 employees in the United States, 18.7 percent of its workforce (ASML Holding N.V., 2026a), and US subsidiaries that include Cymer, the maker of its EUV light sources (ASML Holding N.V., 2026b). Under US export rules these entities and their US staff are "U.S. persons" (Bureau of Industry and Security, 2026), subject to US rules that restrict support for advanced-node fabs in China (ASML Holding N.V., 2026a).

Washington also reaches the machines. An October 2023 rule set the de minimis threshold of US-origin content at zero for certain immersion tools bound for China (Office of the Federal Register, 2026; U.S. Department of Commerce, Bureau of Industry and Security, 2023). A rule effective on 2 December 2024 added a foreign direct product (FDP) rule and a de minimis rule specific to semiconductor manufacturing equipment (SME). Exporters located in the 33 countries of its new Supplement No. 4, among them 25 EU members and Japan but not South Korea or Taiwan, are excluded from the new licence requirements for items caught only by these two rules, because the listed governments "have the authority to control key SME items of concern" (Bureau of Industry and Security, 2024). BIS can amend the list by rule (Section 8.2 codes the status).

2.2 The control sequence

Licences for ASML's shipments to China are a Dutch decision (Section 7.1). From 2019 The Hague left ASML's application to ship an EUV machine to China undecided, reportedly after US lobbying; the application was still pending in October 2020 (NL Times, 2020; Kaag, 2020). From 2023 each step followed a US move. National licensing of the most advanced immersion tools (NXT:2000i and later) took effect on 1 September 2023, after the US rules of October 2022 (Minister voor Buitenlandse Handel en Ontwikkelingssamenwerking, 2023); licensing of the older NXT:1970i and

NXT:1980i moved from Washington to The Hague on 7 September 2024, after the US rule of October 2023 (Minister voor Buitenlandse Handel en Ontwikkelingshulp, 2024; ASML Holding N.V., 2024); and controls on metrology and inspection systems took effect on 1 April 2025, after the December 2024 rule (Minister voor Buitenlandse Handel en Ontwikkelingshulp, 2025). On 15 November 2025 Commission Delegated Regulation (EU) 2025/2003 put the most advanced immersion tools on the EU list (European Commission, 2025a). ASML reported that this "did not mean a change", and the licence decision stayed in The Hague (ASML Holding N.V., 2026a).

The US-facing direction is collective. EUV and the most advanced immersion tools ship to US fabs under the Union general export authorisation EU001, which a member state may withdraw from an exporter only for doubts about its ability to comply (European Parliament and Council of the European Union, 2021). Only the tools left on the Dutch national list, the NXT:1970i and NXT:1980i class, still need a Dutch licence to leave the EU (Staatssecretaris van Buitenlandse Zaken, 2025). Turning the listed tools against Washington would take an EU decision (Section 7.3). Ownership is guarded by a foundation that can acquire preference shares halving ordinary shareholders' votes (ASML Holding N.V., 2026a) and by notification duties under the Vifo Act (Staten-Generaal, 2022; Kingdom of the Netherlands, 2023).

Servicing of the installed base in China remained permitted at least into mid-2026 (Zakaria, 2026). The MATCH Act, introduced in April 2026 and not enacted by late October, would ban immersion-tool sales anywhere in China and servicing at listed Chinese fabs, and would give allies 150 days to align before Washington acted alone (Baumgartner, 2026a,b). The Dutch government told parliament it opposed the bill's extraterritorial reach (Sjoerdsma, 2026), and the trade minister said it "could potentially put our national security in American hands" (NL Times, 2026b).

2.3 Europe's dependencies

Security. Europe relies on the United States for nuclear deterrence, and replacing US conventional support would cost about EUR 250 billion a year (Burilkov and Wolff, 2025), a replacement cost rather than the crisis-weighted stake S_k of Section 4.3. The guarantee is graduated. Its core is statutory: Section 1250A of the FY2024 National Defense Authorization Act bars withdrawal from the North Atlantic Treaty without two-thirds of the Senate or an Act of Congress (United States Code, 2023). Posture is cheap to move. On 1 May 2026, after Chancellor Merz criticized US strategy on Iran, Defense Secretary Hegseth ordered about 5,000 troops out of Germany (Martinez, 2026). Dependence also runs the other way. Spain was already refusing US use of jointly operated bases for the Iran war when, on 30 March 2026, it closed its airspace to the US aircraft involved (Roque, 2026).

Frontier AI. The AI Diffusion Rule of January 2025 put ten EU members in its favoured group and held the other seventeen to country allocations (Bureau of Industry and Security, 2025). Two days before compliance was due, BIS announced its rescission, saying the rule would have "undermined U.S. diplomatic relations with dozens of countries by downgrading them to second-tier status" (U.S.

Department of Commerce, Bureau of Industry and Security, 2025a); by late October 2026 neither the formal rescission nor a replacement had appeared in the Federal Register. In the August 2025 joint statement the EU said it intended to buy at least $40 billion of US AI chips and to align its technology-security requirements with US ones, and Washington said it would then facilitate those exports (The White House, 2025c). The National Security Strategy of December 2025 holds out "more favorable treatment on commercial matters, technology sharing, and defense procurement" to countries that take more responsibility for their regional security and "align their export controls with ours" (The White House, 2025d).

China. Beijing has built instruments against those who restrict it. Its licensing of seven medium and heavy rare earths, in place since 4 April 2025, is still in force (Ministry of Commerce of the People's Republic of China and General Administration of Customs of the People's Republic of China, 2025a; Benchmark Mineral Intelligence, 2026). Its broader, extraterritorial controls of October 2025 (Mayer Brown, 2025) were suspended after the Busan summit of 30 October 2025, a pause the two governments agreed in September 2026 to extend to 10 January 2027 (The White House, 2025a; Ministry of Commerce of the People's Republic of China and General Administration of Customs of the People's Republic of China, 2025b; Ministry of Commerce of the People's Republic of China, Department of American and Oceanian Affairs, 2026); Section 8.5 asks what a lapse would bring. Two State Council decrees of April 2026 authorize countermeasures against foreign firms that cut off normal trade with Chinese entities (Decree 834) and penalties for those who implement foreign extraterritorial measures (Decree 835) (State Council of the People's Republic of China, 2026; Li et al., 2026). Decree 835 was first used against Brussels: on 15 May 2026 Beijing barred anyone from implementing the Commission's investigative measures against Nuctech under the Foreign Subsidies Regulation (Cleary Gottlieb Steen & Hamilton LLP, 2026; Ministry of Commerce of the People's Republic of China, 2026c).

3 Related Literature

3.1 Interdependence and chokepoints

Hirschman (1945) located influence in the losses a partner would suffer if trade stopped, and Keohane and Nye (1977) separated the sensitivity and vulnerability that Section 7.4 uses. Farrell and Newman (2019) moved the argument to networks and attached two conditions: a state must have "physical or legal jurisdiction over hub nodes" (p. 56) and the "legal and regulatory institutions" to exploit them (p. 57), and where hubs span jurisdictions the states involved must coordinate (pp. 56, 58). Farrell and Newman (2023) trace how Washington built that capacity. Kalyanpur and Newman (2019) show that jurisdiction is itself a policy choice, since harmonized EU rules let European firms escape the reach of US securities law. In semiconductors the foreign direct product rule forced foreign chipmakers to choose between US tools and Huawei as a customer (Bown, 2020), and US firms routed sales through third countries until that rule closed the gap (Chen and Evers,

2023, 194–195). Treating a toolmaker as a hub extends Farrell and Newman's argument beyond finance and the internet. ASML meets the network condition, but Washington shares jurisdiction over it (Section 2), and the European institutions that could turn it against the United States are collective (Section 7.3).

These works disagree about whether a hub yields lasting power. For Farrell and Newman the power comes from network position and the institutions behind it, and targets' responses come later (Farrell and Newman, 2019, 76). Gholz and Hughes (2021) find that Japan absorbed China's 2010 rare-earth embargo through market adjustment and argue that cost measures that ignore market structure overstate a sender's power. Work on chokepoints as depleting assets sides with Gholz and Hughes: activation induces indigenization and rerouting (Tan et al., 2025; Rajan, 2026). We let the exit premium drift and ask what a patron does while it lasts (Section 6). Wagner (1988) supplies the discipline: asymmetric interdependence yields influence only when trading economic resources for political concessions beats bargaining over the economic gains alone, however asymmetric the relationship.

Whether Europe can act on its side of the hub is also disputed. Meunier and Nicolaidis (2019) and Bauerle Danzman and Meunier (2024) trace the EU's turn toward geoeconomic instruments, while Kastner (2016) finds that trade dependence on China buys accommodation on economic questions more reliably than on political ones, which bears on the size of China's offer more than on its threats.

3.2 Alliance coercion and dependence

Snyder (1984, 1997) distinguishes the fear of abandonment from the fear of entrapment, and a protégé's need for its patron from the patron's own strategic interest in defending it; when that interest is well known, the stronger state "cannot credibly threaten defection or realignment" (Snyder, 1984, 472–473). Morrow (1991) treats asymmetric alliances as trades of autonomy for security, and Lake (2009) treats the patron's authority as relational, a social contract in which the patron provides order and protection and the subordinate complies. On those accounts compliance needs no fresh threat, a rival reading we take up in Section 8.1. Drezner (1999) finds that allies rarely coerce each other but can win significant concessions when they do.

The coercion literature disagrees about which instrument worked in the canonical case. Gerzhoy (2015) credits conditional threats of military abandonment, paired with assurance, for stopping West Germany's nuclear ambitions. Lanoszka (2018) finds alliance coercion hard and argues that an ally's economic and technological reliance on Washington is the stronger instrument. Pauly (2024) explains why the pairing is fragile: the steps that make a threat credible undermine the coercer's assurance that compliance will not be punished. His case is an adversary, Iran, and his assurance is owed to the target; our cost of conditionality falls mainly on other allies (Section 5.3). Blankenship (2021) finds that allies in locations Washington values bear less of the common burden, consistent with less credible abandonment threats, and Blankenship (2023, 111–119) shows Iceland extracting compensation and fishing concessions for Keflavik. Wolford and Kim (2017) show that threats

against a politically valuable ally can lack credibility, which is our channel of self-harming sanctions.

Base politics studies a host that holds an asset its patron needs. Cooley (2008) traces base stability to political change in hosts and Calder (2007) to compensation, while Cooley and Spruyt (2009) treat sovereignty transfers as incomplete contracts open to renegotiation. In 2009 Kyrgyzstan announced the closure of the US base at Manas alongside a Russian aid package, collected the first Russian payment, and then signed a new deal with Washington at a higher rent (Cooley, 2012, ch. 7).

Nonproliferation work studies the reverse problem, a patron restraining an ally's sensitive exports, and it splits on motive. Kroenig (2009) finds that sensitive nuclear transfers follow suppliers' strategic position and no support for commercial motives. Gray (2012) shows Bonn putting commercial rivalry first in its 1975 sale to Brazil, keeping the deal and promising only future caution.

3.3 Rival bidders and geoeconomics

That a contested client earns what the losing bidder would pay is old. It is the one-voter case of sequential vote buying (Groseclose and Snyder, 1996), extended to finite legislatures by Banks (2000); Dekel et al. (2008, Prop. 2) show that the winner pays the loser's value net of the client's preference, a price pinned down only in undominated strategies. Menu auctions (Bernheim and Whinston, 1986; Grossman and Helpman, 1994) give the simultaneous-offer analogue, and Bueno de Mesquita and Smith (2016) show that a dominant aid donor captures the surplus of an aid-for-policy deal while a rival donor's entry raises what it must pay. The outside-option principle supplies the threshold: a threat to walk away matters only when the alternative beats continued bargaining (Shaked and Sutton, 1984; Sutton, 1986; Binmore et al., 1989). In the two-hegemon extension of Clayton et al. (2026a, Supplement B.2), hegemons whose threats overlap bid transfers down to zero. Lemma 1 restates these results; we claim no novelty for it. Lacy and Niou (2004) model sanctions as coercive linkage of an economic relationship to a political demand under two-sided incomplete information: threats likely to succeed work before sanctions are imposed, so imposed sanctions are a selected sample.

Clayton et al. (2026a) show that a hegemon gains by pooling threats across relationships, bounded by the target's participation constraint, and Clayton et al. (2026b) state the principle we build on: "Threats generate power to the extent that they generate slack in the participation constraint." The term slack is theirs. Their pooled threats are enforcement threats that raise the target's inside option and are never spent; threats to cut off a target that refuses appear only in their Supplement B.3.1, where they are partly self-defeating. In Clayton et al. (2026c) a hegemon gains by committing to limit its own coercion, and Broner et al. (2025) and Becko et al. (2025) model alignment among many small countries facing committed instruments. The effect of a rival is contested. Rival offers come when a protégé's relations with its patron sour (Yang, 2026), yet Camboni and Porcellacchia (2025) find that China's growth can lead countries in the US sphere to offer Washington more.

We combine the two strands. Our patron's threats are cut-off threats aimed at an ally for

whom a rival bids, each must be sequentially rational, and an executed threat is spent; as a result only conditional parts are capped by slack, conditional bites share one budget, and a patron that cannot commit to stop demanding obtains less (Proposition 2). Bidders in Bueno de Mesquita and Smith (2016) have carrots only. Ours have sticks and counter-threats, and China executes a counter-threat only to crowd out another demand (Proposition 3). Once the holder can propose and talks can rupture, the patron's dependence returns through agenda power, which the passive client of Groseclose and Snyder (1996) and Dekel et al. (2008) lacks (Proposition 1). Where the hegemon of Clayton et al. (2026c) gains by binding itself, our patron chooses how conditional to make its guarantee, and it is the holder that may bind itself (Proposition 4 and Remark 2).

3.4 Commitment, control and prevention

Anticipated shifts in power create commitment problems that can make bargaining fail under complete information (Fearon, 1995; Powell, 2006), and Krainin (2017) shows that slow but persistent shifts suffice. Fearon (1996), an unpublished manuscript, studies bargaining over objects that shape future bargaining power, in which a discontinuous shift can cause breakdown. Control of ASML is such an object. Work on investment in future power divides. Debs and Monteiro (2014) find that uncertainty about a rising state's investment can itself produce preventive war, while in Bas and Coe (2016) an observer that monitors imperfectly waits until the shift looks imminent, and Spaniel (2019) shows that potential power can instead be bought off with concessions. Benson and Smith (2023) find that the conditions that make prevention attractive also make it unlikely to be carried out. Section 6.2 sets our preventive demand for control against these models.

Control rights come from the theory of the firm: ownership allocates residual rights over an asset when contracts are incomplete (Grossman and Hart, 1986; Hart and Moore, 1990). Lake (1996) orders relations from alliance to empire by these rights, and Frieden (1994) ties colonial control to investments whose rents are easy to seize. In Harstad (2016) leasing beats buying when the seller's consumption value and the buyer's protection cost are large, which bears on how far a transfer binds (Corollary 3). Sold tools are a wasting asset, since a durable-goods seller cannot commit (Coase, 1972; Bulow, 1982), while control of maintenance works like leasing and restores commitment (Morita and Waldman, 2004); servicing is therefore a renewable source of hold-up (Section 7.4). Leaders can tie their hands with audience costs and other public stakes (Schelling, 1960; Fearon, 1994, 1997). In our model every threat and promise is staked this way, under one commitment technology (Section 4.2).

4 A Model of Chokepoint Bargaining

The model has three players: Europe (E), the United States (U) and China (C). Europe decides whether ASML's most advanced tools, and the engineers who keep them running, reach Chinese fabs. We treat Europe as one actor and separate the Dutch licensing authority, the firm and the European Union in Section 7.

4.1 Setup

Alignment. Europe chooses an alignment $a \in \{A, X\}$. Under Atlantic alignment (A) ASML complies with US denial requirements. Exit (X) is a broad realignment of the chokepoint: Europe lets ASML sell and service EUV and advanced immersion DUV tools for Chinese fabs against US objection. Hedging, the continued sale and servicing of legacy immersion tools in China, has been the margin of dispute since 2023 (Section 8.3); it is a secondary choice in Proposition 3(b). Section 9.1 states the scope conditions.

Stakes. China gains $W > 0$ from Europe's exit, of which a share $\theta \in [0, 1]$ can be blocked by US jurisdiction over China-bound flows (the foreign direct product rule, de minimis rules and controls on US-made subsystems). That jurisdiction leaves ASML's supply to US-aligned fabs untouched, so we treat it as credible. The United States loses $V > 0$ if Europe exits. V is the patron's dependence on the chokepoint, what it gives up in the AI race when Chinese fabs gain access to ASML's frontier tools, and we measure it net of what flow jurisdiction still blocks: V is Washington's loss from the share $1 - \theta$ of access that exit would give. So θ scales both China's gain and Washington's loss.

Instruments and the baseline. Washington's instruments j are the core defence guarantee, force posture and enablers, access to frontier AI, and ASML's access to US inputs, persons and customers. After Europe chooses a, the United States sets each at a level $x_j(a)$ that is sequentially rational given its commitments, and Europe values it at $v_j(x_j)$. The conditional part of instrument j is $d_j = v_j(x_j(A)) - v_j(x_j(X)) \geq 0$. The baseline is $x_j(A)$, Washington's ex post optimal provision to a loyal ally. A transfer is a promised increment above this baseline, and a threat is a withdrawal below it. Each instrument is thus counted once: frontier AI enters through d_{ai}, and only access beyond $x_{ai}(A)$ would be a transfer.

4.2 Commitment

One commitment technology governs every promise and every threat. Before Europe moves, each great power i may attach a stake $m \leq \bar{m}_i$ to any public conditional commitment, which it forfeits if the condition is met and it does not act. The stake is a reduced form for the audience costs of backing down, which tie hands rather than sink costs (Fearon, 1994, 1997). The cap $\bar{m}_i$ applies to each commitment separately.

A promise to pay τ after Europe has moved has no ex post value to the promiser, so it is honoured if and only if $\tau \leq \bar{m}_i$, and a threat whose execution costs the threatener e ex post is credible if and only if $e \leq \bar{m}_i$. Creating a stake costs nothing. A threat that deters is never carried out and costs its author nothing. One that is executed costs e, and $e \geq 0$ for every withdrawal from a loyal Europe because the baseline is Washington's ex post optimum.

Security uses the same technology: Washington can stake $m_S \leq \bar{m}_U$ on a public threat to abandon a realigned Europe. Such a rule makes the guarantee less valuable to loyal allies before any crisis arrives, and that assurance cost (Section 5.3) makes conditionality an interior choice.

4.3 Security, Other Instruments and the Exit Premium

The core guarantee. In a crisis, Washington defends Europe if its net stake in doing so exceeds a crisis-specific cost ξ, drawn from a continuous distribution G with density g. Its stake in defending a realigned Europe is $s_X = s_0 + s_B$, where s_0 is its own interest in European security and s_B the value to it of European territory, airspace and bases. Its stake in defending a loyal Europe is $s_A = s_X + s_L$, where $s_L \geq 0$ is the added value of a loyal ally, for example Washington's reputation with other chokepoint holders. After exit, defending Europe also forfeits the stake m_S. The conditionality of the core guarantee is

$$\kappa = G(s_A) - G(s_X - m_S), \qquad s_A = s_X + s_L, \quad s_X = s_0 + s_B, \tag{1}$$

and its largest value, at $m_S = \bar{m}_U$, is $\bar{\kappa} = G(s_X + s_L) - G(s_X - \bar{m}_U)$. With S_k Europe's crisis-weighted stake in core defence, the conditional part of the guarantee is κS_k. Conditionality has two sources, the value of loyalty (s_L) and the staked threat of abandonment (m_S); Europe's dependence enters only through S_k.

Access and the silicon shield. The term s_B carries the access channel. A less committed United States "may find itself relying on hefty cash payments to try to buy access" (Carson et al., 2025), so a Washington that needs such access has a reason to defend Europe whatever ASML does. If Washington would defend the territory on which ASML sits because it needs the firm, s_B rises with V, a silicon shield. The shield is contested: writing about Taiwan, Weil et al. (2025) call the argument "largely a hopeful narrative", and we treat a rising $s_B(V)$ as an assumption. Its effect on conditionality has no fixed sign: by Proposition 4(d), a higher s_X lowers $\bar{\kappa}$ when the density g is nonincreasing on $[s_X - \bar{m}_U,\, s_X + s_L]$, for example when G is unimodal with its mode at or below $s_X - \bar{m}_U$. Where a staked abandonment makes defence of an exiter unlikely, a higher s_X can raise $\bar{\kappa}$. Europe's own power to deny access is a hold-up instrument (Section 7.4).

Posture and enablers. Washington also provides a flow of posture and enablers: stationed forces, rotations, intelligence sharing, munitions and spare parts. The flow is cheap to adjust and can be withdrawn from one ally at a time, so it can be made conditional without touching the treaty commitment. Its conditional part is d_p.

The AI instrument. Washington allocates frontier models, inference and advanced chips continuously, and the AI instrument's conditional part is $d_{ai} = v_{ai}(x_{ai}(A)) - v_{ai}(x_{ai}(X))$. Scarce compute makes it cheap to cut an exiter's allocation, which raises d_{ai}. Scarcity could also lead Washington to ration loyal allies, which removes the conditional part only if they are cut to the exiter's level. Tiering loyal allies is itself costly: the AI Diffusion Rule of January 2025 tiered EU members and was withdrawn when BIS announced its rescission two days before the compliance date, citing in part the downgrading of allies (Section 2). We read the rescission as evidence that d_{ai} stays positive when compute is scarce.

Firm-level penalties. Washington can also penalize ASML itself, by entity listing or by denying US-origin inputs and US-person support for all of the firm's production. This costs Europe $f \geq 0$ and, by disrupting supply to US and allied fabs, harms the United States too. Washington's ex post gain from penalizing an exiting ASML is $\gamma - \varphi V$, where $\gamma \geq 0$ is the added denial benefit and φV, with $\varphi > 0$, the self-inflicted cost. The threat is credible if and only if its execution cost is within the cap, so

$$d_f = \chi_f f, \qquad \chi_f = \mathbf{1}\{\varphi V \leq \gamma + \bar{m}_U\}.$$

As in Wolford and Kim (2017), threats against a politically valuable ally lack credibility; here that applies only to penalties on the firm, and flow jurisdiction remains credible.

China's offer. China can promise Europe a transfer if Europe exits: market access, credit, co-production or relief from its own export controls. A promise of τ_C costs China τ_C, and Europe values it at $\lambda_C \tau_C$ with $\lambda_C \in (0, 1]$, the lower value to Europe of what China offers in kind. China's promises are capped at $\bar{m}_C$. China may also value Washington's loss: with a relative-gains weight $\rho \geq 0$, the most it would pay for exit is $W_C = (1 - \theta)W + \rho V$, and both terms shrink as θ rises. China's credible offer, in units of European utility, is

$$\omega = \lambda_C \min\{W_C, \bar{m}_C\}. \tag{2}$$

If the race is close to zero-sum, ρV is large and the cap binds; ω is then set by how far Europe trusts and values Chinese promises (λ_C and $\bar{m}_C$).

Chinese punishment of alignment. We set China's untargeted punishment of alignment to zero. Such a threat would be carried out whenever Europe aligned anyway, so China would make it only if punishing paid for itself, and it would then enter Q. China's observed coercion targets specific acts: informal sanctions after South Korea's THAAD deployment (Lim and Ferguson, 2022), the 2025 Nexperia export ban after a Dutch order (Section 8.3), and the listing of Japanese entities, none of which makes chipmaking equipment or photoresists, after Prime Minister Takaichi's statement on Taiwan in November 2025 (Section 8.4). We found no case in which China punished a holder for aligning its chokepoint controls with Washington. Chinese coercion therefore enters as counter-threats attached to particular concessions (Proposition 3) and to a transfer of control (Section 6).

The exit premium. Exit costs Europe the conditional parts of Washington's instruments plus an intrinsic cost $\eta \geq 0$, the value Europe places on keeping frontier capabilities from a rival that supports Russia's war:

$$c = \kappa S_k + d_p + d_{ai} + d_f + \eta.$$

Definition 1 (Exit premium)**.** Europe's exit premium is China's credible offer net of the credible

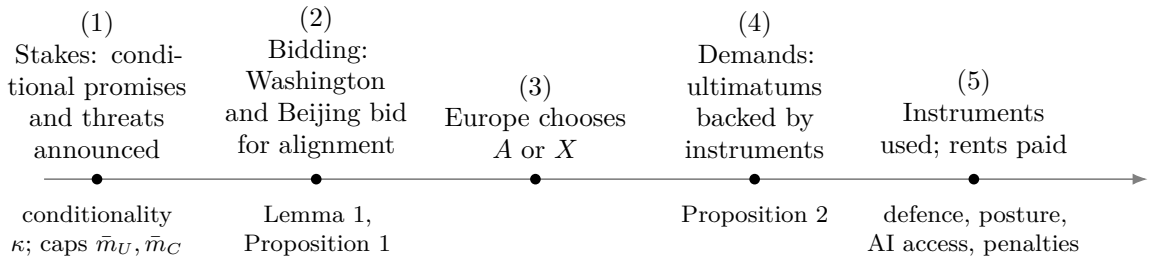


Figure 1: Timing of one period. Each threat in stage (5) is carried out only if doing so is sequentially rational given the stakes announced in stage (1). Section 6 links periods through control rights.

cost of exit,

$$Q = \omega - c = \lambda_C \min\{(1-\theta)W + \rho V,\ \bar{m}_C\} - \kappa S_k - d_p - d_{ai} - \chi_f f - \eta. \tag{3}$$

Europe's slack is $\Lambda = [-Q]^+$.

The premium follows Hirschman (1945), for whom a state's influence over a partner depends on what the partner would lose by leaving. Slack is the amount by which Europe's alternative to alignment falls short, in the sense of Clayton et al. (2026b) (Section 3). Table 1 collects the notation.

Timing. Figure 1 shows one period. The great powers announce conditional promises and threats and attach stakes. Washington and Beijing bid for Europe's alignment, and Europe chooses. Washington may then issue demands backed by its instruments (Section 5). Finally, instruments are used and rents paid, and each threat is carried out only if that is sequentially rational.

4.4 The Bidding Benchmark

In the benchmark, Washington and Beijing bid for Europe's alignment in an ascending auction, so either can raise its bid until neither wants to.

Lemma 1 (Bidding benchmark)**.** *Suppose the United States and China bid for Europe's alignment in an ascending (clock) auction in which bids are staked promises, Europe values China's promise τ_C at $\lambda_C \tau_C$, Europe chooses its alignment after the auction, and instruments are then used. Let $\bar{z}_U = \min\{V, \bar{m}_U\}$ and $Q = \lambda_C \min\{W_C, \bar{m}_C\} - c$.*

(i) After alignment a, the United States provides $x_j(a)$ for every instrument j, so exit lowers Europe's payoff, before transfers, by $c = \sum_j d_j + \eta$.

(ii) A promise of τ by power i is honoured if and only if $\tau \le \bar{m}_i$. Bidding up to one's maximal credible bid (in units of Europe's payoff, $\bar{z}_U$ for the United States and Q for China) is weakly dominant. In weakly undominated strategies, with an indifferent Europe aligning:

(a) if $Q < V$ and $Q \le \bar{m}_U$, Europe aligns and the United States pays $[Q]^+$;

Table 1: Notation for the baseline model

Symbol	Meaning	Symbol	Meaning
E, U, C	Europe, United States, China	$a \in \{A, X\}$	Atlantic alignment or exit
V	US loss from exit, net of flow jurisdiction (dependence)	W, θ	China's gain from exit; share blocked by US flow jurisdiction
ρ	China's relative-gains weight	W_C	most China would pay, $(1-\theta)W + \rho V$
$\bar{m}_U, \bar{m}_C$	largest stake on one commitment	λ_C	Europe's value of a unit of China's promise
ω	China's credible offer, equation (2)	$x_j(a)$, d_j	provision of instrument j; its conditional part
s_0, s_B	US own interest in European security; access value	s_L	added US value of a loyal ally
s_A, s_X	US stake in defending a loyal, realigned Europe	m_S	US stake on abandoning a realigned Europe
G, g	distribution and density of crisis costs ξ	κ, $\bar{\kappa}$	conditionality of the core guarantee, equation (1); its maximum
S_k	Europe's crisis-weighted stake in core defence	d_p, d_{ai}	conditional parts of posture and of the AI instrument
f, γ, φ	firm penalty; US denial benefit; self-harm per unit of V	χ_f	credibility of firm penalties
η	Europe's intrinsic cost of exit	c	credible cost of exit
Q	exit premium, equation (3)	Λ	slack, $[-Q]^+$
β	Europe's probability of proposing	λ, r	rupture hazard; discount rate
ψ	rupture weight, $\lambda/(\lambda + r)$	P^*	equilibrium transfer flow to Europe

(b) if $Q > \min\{V, \bar{m}_U\}$, Europe exits, and China's promise gives Europe $\min\{V, \bar{m}_U\}$ net of c.

In both cases Europe's payoff is $[\min\{Q, V, \bar{m}_U\}]^+$.

(iii) If $\bar{m}_U < Q < V$, every binding transfer $\tau \in [Q, V)$ from the United States to Europe would make both Europe and the United States strictly better off than the equilibrium (inefficient exit from limited US commitment).

The losing bidder sets the price. The lemma restates known results (Section 3), and Clayton et al. (2026a, Supplement B.2) derive a two-hegemon version. The weak-dominance argument is for the clock auction. In an alternating-raise auction (Appendix B), subgame perfection alone does not pin down the price. Europe still aligns in every subgame-perfect equilibrium when Q is more than one grid step below $\bar{z}_U$, and the price is within one grid step of $[Q]^+$ when the losing bidder bids up to its maximal credible bid. What the lemma adds is the content of Q: China's bid enters Europe's participation constraint, and every cost of exit in c is a conditional part that survives sequential rationality.[1]

[1]We value transfers at par. If delivering value P to Europe costs Washington $a_U P$ with $a_U \leq 1$, as for an exemption from an extraterritorial rule, then, when its commitment cap does not bind, Washington keeps Europe aligned if and only if $a_U Q < V$. Cheap in-kind transfers widen the range in which Washington wins without changing the price to Europe (Section 8.2).

4.5 Agenda Power and Rupture Risk

We now give Europe a voice in the price by letting it bargain with Washington, with China's credible offer ω, and hence Q, fixed and open to Europe as an exit option throughout. The two bargain over a perpetual signed transfer flow P: $P > 0$ means Washington pays Europe and $P < 0$ means Europe concedes. Agreements are binding.[2] Every Δ units of time, Europe proposes with probability $\beta \in [0, 1)$ and Washington otherwise, and the responder accepts or rejects. After rejecting a US offer, Europe may exit, which yields flows Q to Europe and $-V$ to Washington forever. If Europe does not exit, the dispute ruptures into exit with probability $\mu = 1 - e^{-\lambda\Delta}$, where $\lambda \geq 0$ is the rupture hazard; otherwise the status quo continues with flows $(0, 0)$ and both discount at rate $r \geq 0$. Rupture is realignment that neither government chooses, for instance after a change of government in The Hague or an unintended escalation: a threat that leaves something to chance (Schelling, 1960). The disagreement point is the aligned status quo with no transfer, so P replaces the bidding stage's payment. Write $\upsilon = e^{-(r+\lambda)\Delta}$ and $\psi_\Delta = \mu/(1 - \upsilon)$, which tends to the rupture weight $\psi = \lambda/(\lambda + r)$ as $\Delta \to 0$.

Proposition 1 (Bargaining with an outside option and rupture risk)**.** *Let $Q < V$, $\beta \in [0, 1)$, $r, \lambda \geq 0$ and $r + \lambda > 0$.*

(i) For every $\Delta > 0$, every subgame-perfect equilibrium gives Europe the same payoff $x^*(\Delta)$ and the United States $-x^*(\Delta)$ (closed forms in Appendix B).

(ii) If $\lambda > 0$, the subgame-perfect equilibrium is unique (up to Europe's exit choice at one point of indifference, which does not affect payoffs), and agreement is immediate. If $\lambda = 0$, there are also stationary equilibria with delay, with the same payoffs.

(iii) As $\Delta \to 0$, $x^*(\Delta)$ and the equilibrium offer and demand converge to

$$P^* = \max\{Q,\ \psi(Q + \beta(V - Q))\}, \qquad \psi = \frac{\lambda}{\lambda + r} \in [0, 1]. \tag{4}$$

If $r > 0$, the second term is the larger one if and only if $Q < \hat{Q} \equiv \lambda\beta V/(r + \lambda\beta)$.

Part (iii) describes bargaining in which offers come much faster than rupture or discounting. Europe's exit right is placed after it rejects a US offer.[3] Figure 2 plots P^* against Q for three parameter sets.

Corollary 1 (Threshold and channels)**.** *Under the conditions of Proposition 1:*

(a) $P^* \geq Q$: Europe never concedes more than its slack, $-P^* \leq \Lambda = [-Q]^+$. (For $\Delta > 0$ the value $x^*(\Delta)$ can be below Q, by at most $\beta(1 - \upsilon)\left[Q - \psi_\Delta V\right]^+/(1 - \beta\upsilon)$.)

[2] Agreements are enforced by the stake technology of Section 4.2. We assume that Washington's cap covers the agreed transfer ($\bar{m}_U \geq V$ suffices) and that a European concession is enforceable, for example because it is a policy Europe would have to reverse publicly.

[3] If Europe could also exit right after Washington rejects Europe's own demand, the stationary equilibrium would have the same limit P^*, but when the outside option binds there would also be nonstationary equilibria in which Europe obtains up to V. Allowing exit only after Europe rejects a US offer pins down the outside-option principle (Binmore et al., 1989). On the timing of outside options see Shaked (1994) and Ponsatí and Sákovics (1998).

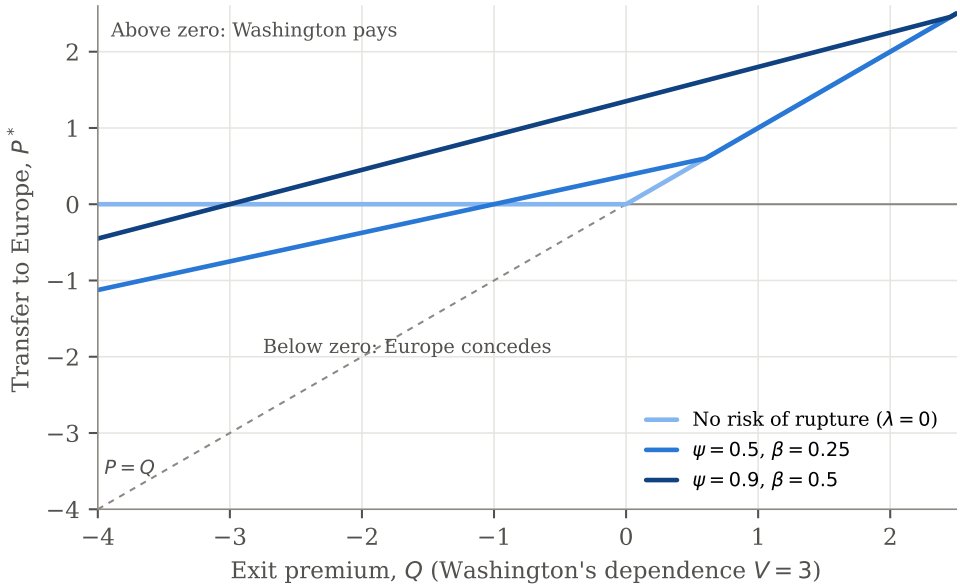


Figure 2: Equilibrium transfer to Europe in the master bargaining game (Proposition 1) as a function of the exit premium, with $V = 3$. Without rupture risk ($\lambda = 0$) the transfer is $[Q]^+$. With rupture weight $\psi = 0.5$ and $\beta = 0.25$, Washington pays Europe whenever $Q > -1$; with $\psi = 0.9$ and $\beta = 0.5$, whenever $Q > -3$. Below those values Europe concedes. Every curve lies on or above the dashed line $P = Q$ (Corollary 1(a)).

(b) If $\lambda = 0$, $P^* = [Q]^+$; if $\beta = 0$, $P^* = \max\{Q, \psi Q\}$. In either case $Q \leq 0$ implies $P^* \leq 0$.

(c) $P^* > 0$ if and only if $Q > 0$, or $\lambda > 0$ and $Q + \beta(V - Q) > 0$.

(d) Holding Q fixed, $\partial P^*/\partial V = \beta\psi$ where the second term binds ($Q < \hat{Q}$) and 0 where $Q > \hat{Q}$. If Q itself depends on V, then $dP^*/dV = \psi[\beta + (1 - \beta)\, dQ/dV]$ in the first case and dQ/dV in the second.

(e) V enters P^* directly only through the term $\beta\psi V$ (agenda power under rupture risk), and indirectly only through Q: through the rival's valuation, ρV in W_C when $W_C < \bar{m}_C$ ($\partial\omega/\partial V = \lambda_C \rho$); through self-harming sanctions, $\chi_f = \mathbf{1}\{\varphi V \leq \gamma + \bar{m}_U\}$ (a rise in V past $(\gamma + \bar{m}_U)/\varphi$ raises Q by f); and through the silicon shield, $\kappa = G(s_X + s_L) - G(s_X - m_S)$ with $s_X = s_0 + s_B(V)$ (which equals $\bar{\kappa}$ when $m_S = \bar{m}_U$), with a sign that depends on g. Proposition 6 adds the fifth channel, hold-up.

Nesting: $\lambda = 0$ gives $[Q]^+$ (the outside-option principle); $r \to 0$ with $\lambda > 0$ gives $Q + \beta(V - Q)$; $\beta = 0$ gives $\max\{Q, \psi Q\}$ (Washington proposes); $\beta \to 1$ and $\lambda \to \infty$ give V.

The threshold in part (b) is the outside-option principle (Shaked and Sutton, 1984; Sutton, 1986; Binmore et al., 1989), and its substance is Wagner's (Section 3). Without rupture risk, a holder whose exit premium is negative gets nothing for its chokepoint, whoever proposes. V then drops out because Europe lacks agenda power under rupture risk. A Europe that proposes in almost every

round still cannot charge for V, since its only threat is exit and exit is credible only when $Q > 0$. When $\lambda > 0$, a rejected demand risks a rupture that costs Washington V, so Europe's demands carry weight in proportion to $\beta\psi$, a binary instance of the identity-dependent externalities through which a seller extracts surplus from parties that do not buy (Jehiel et al., 1996).

As $r \to 0$ the game becomes a random-proposer version of the breakdown model of Binmore et al. (1986). The corner $\beta \to 1$, $\lambda \to \infty$, in which Europe obtains all of V, is the bargaining implied when the summary of Juijn et al. (2026) calls ASML "the only card Europe still holds" (Section 1). It requires take-it-or-leave-it demands whose rejection ends in realignment with near certainty, a commitment Europe has not shown it can make.

Rupture risk cuts both ways. When $Q < 0$, a rupture would leave Europe worse off than the aligned status quo, and Washington can extract part of the value to Europe of avoiding it: $P^* < 0$ if and only if $\lambda > 0$ and $Q + \beta(V - Q) < 0$. By Corollary 1(a), though, Europe never concedes more than its slack. Section 5 shows that in the bidding benchmark slack also caps what Washington obtains by any combination of conditional threats. Under rupture risk the cap is Europe's surplus over exit, $P^* - Q$, which rises with V while agenda power binds. Whatever agenda power wins Europe above $[Q]^+$, Washington's credible conditional threats can claw back, down to the exit premium itself (Remark 1). Washington's dependence thus raises both what Europe obtains in bargaining and the slack Washington can coerce.

5 The Slack Principle

Lemma 1 and Proposition 1 price the chokepoint when Europe's exit premium is positive. On our reading of the record since 2019 (Section 8), it has not been, and Washington has still asked Europe for a great deal. A patron facing an ally whose premium is negative can obtain the ally's slack, $\Lambda = [-Q]^+$, in the sense of Clayton et al. (2026b), plus whatever it can threaten that does not depend on the ally's alignment. We work in the bidding benchmark, without rupture risk, until Remark 1.

5.1 Bite, budgets and assurance

Consider one period in which Europe is aligned, its premium is Q, and it receives $[Q]^+$ from the bidding stage. Washington has finitely many demands. Demand k costs Europe $t_k > 0$ and is worth $w_k > 0$ to the United States. A concession is *reversible* if exit would undo it, so that it lowers Europe's aligned payoff and leaves its exit payoff unchanged. A limit on servicing Chinese fabs is reversible, because a Europe that exited would resume servicing; so are tariff cuts and export-control rules. A concession is *sunk* if it lowers both payoffs: payments, purchases, and transfers of assets or of control over them.

Washington backs its demands with instruments (Section 4.3). Withdrawing a set J of instruments from an aligned Europe for the period lowers Europe's aligned payoff by $\Delta^A(J)$ and its exit

payoff by $\Delta^X(J)$, with $0 \le \Delta^X(J) \le \Delta^A(J)$ and both nondecreasing in J. We call

$$u(J) = \Delta^X(J) \qquad \text{and} \qquad d(J) = \Delta^A(J) - \Delta^X(J)$$

the unconditional and conditional parts of J. The conditional part is what Europe would lose by exiting anyway (for a single instrument, the d_j of equation (3)). The unconditional part is what Washington would still provide to an exited Europe and can threaten to take away. We impose no additivity: the AI instrument and US force posture may be complements or substitutes in Europe's payoff.

Washington's credible set J^c is fixed at the start of the period, and every subset of it is a credible threat, as with one stake per instrument under the commitment technology of Section 4.2. Washington issues *adaptive ultimatums*: any sequence of demands, each naming unconceded demands and a threat set of instruments not yet withdrawn, each chosen after seeing Europe's earlier responses. If Europe refuses, Washington executes the threat at cost $e(J) \ge 0$, and executed instruments stay withdrawn for the period, so a threat once carried out is *spent*. A threat that deters is not spent and can back the next demand. At the end of the period the bidding contest is replayed. If J_w has been withdrawn and Europe has made reversible concessions T_{r}, Washington must pay $[Q + d(J_w) + T_{\mathrm{r}}]^+$ to keep Europe aligned. For a set S of concessions, $T_{\mathrm{r}}(S)$ and $T_{\mathrm{s}}(S)$ are its reversible and sunk totals and $W(S) = \sum_{k \in S} w_k$ its value to Washington.

Proposition 2 (Slack principle)**.** *Let $\Lambda = [-Q]^+$, $u(J) = \Delta^X(J)$, $d(J) = \Delta^A(J) - \Delta^X(J)$ and $B(J) = u(J) + \min\{d(J), \Lambda\} = \min\{\Delta^A(J), \Delta^X(J) + \Lambda\}$. Every subset of J^c is a credible threat, no other instrument can be withdrawn during the period, executing a withdrawal of J costs the United States $e(J) \ge 0$, and the United States retains Europe after every history.*

(a) Bite. *Refusing an ultimatum backed by J, issued when nothing has been conceded or withdrawn, costs Europe $B(J)$, net of the rent the United States must then pay. B is nondecreasing in J. If $Q \ge 0$, then $B(J) = u(J)$: only unconditional parts bite.*

(b) Reversible concessions. *Europe's net loss from reversible concessions totalling T is* $\min\{T, \Lambda\}$. *Faced with a single ultimatum that asks for reversible concessions T_{r} and sunk concessions T_{s} and is backed by J, Europe accepts if and only if* $\min\{T_{\mathrm{r}}, \Lambda\} + T_{\mathrm{s}} \le B(J)$.

(c) One budget.

(i) In every subgame perfect equilibrium of every adaptive sequence of ultimatums, and for every Q, Europe's total net loss is at most $B(J^c)$. Conditional instruments share one budget: $B(J) - u(J) \le \Lambda$ for every J, and reversible concessions cost Europe at most Λ in total, however they are backed.

(ii) The United States' equilibrium payoff is at most $V_1 = \max\{W(S) - [Q + T_{\mathrm{r}}(S)]^+ : \min\{T_{\mathrm{r}}(S), \Lambda\} + T_{\mathrm{s}}(S) \le B(J^c)\}$, the payoff of the best single ultimatum backed by J^c.

(iii) If the United States can commit to a single ultimatum, one bundled ultimatum backed by

J^c yields V_1. Europe's loss then equals $B(J^c)$ if the United States has a divisible sunk demand of positive value and of size at least $B(J^c)$.

(d) Assurance. *Without that commitment the United States still obtains V_1 if all demands are reversible and $B(J^c) \geq \Lambda$, or if some best single ultimatum contains all of its demands. (In every equilibrium if $e(J) > 0$ for every nonempty J, as with additive execution costs that are positive for each instrument; in some equilibrium otherwise.) In general it can obtain strictly less. Threats that deter are not spent, so the United States reuses them after Europe concedes, and Europe anticipates this. With one instrument with $d = 0$ and n identical sunk demands of cost τ, the United States obtains exactly $n \bmod (b+1)$ of them, where $b = \lfloor B(J^c)/\tau \rfloor$.*

Appendix B has the proof and Appendix C the brute-force checks. Withdrawing a conditional part from an aligned Europe takes away something Europe would lose by exiting anyway, so it raises Europe's premium by $d(J)$. Up to Λ the withdrawal hurts Europe. Beyond Λ the premium turns positive and Washington must pay rent to keep Europe aligned, which hands the excess back. A reversible concession works the same way, which is part (b): it lowers only the aligned payoff, so past Λ Europe would exit unless compensated. Sunk concessions and unconditional parts escape the cap because they hurt Europe whatever it does.

Proposition 2 takes Europe's payoff from alignment to be $[Q]^+$, the corner of Proposition 1 in which Washington's dependence does not matter. Where rupture risk raises Europe's payoff above $[Q]^+$, the budget is larger.

Remark 1 (Slack under rupture risk). Suppose Europe's payoff from alignment is the bargaining outcome $P^*(Q)$ of Proposition 1 rather than the bidding payment $[Q]^+$. Withdrawing a threat set J then costs Europe $B_\psi(J) = \Delta^A(J) - [P^*(Q + d(J)) - P^*(Q)]$, and $B_\psi(J) \leq u(J) + \min\{d(J), \Lambda_\psi\}$ with $\Lambda_\psi \equiv P^*(Q) - Q$, which exceeds Λ exactly when $P^*(Q) > [Q]^+$. The bound holds with equality when the outside option binds after the withdrawal ($Q + d(J) \geq \hat{Q}$), and $B_\psi = B$ when $\lambda = 0$. Part (c)(i) of Proposition 2 holds with B_ψ and Λ_ψ in place of B and Λ.

The proof is in Appendix B. The budget Λ_ψ, Europe's surplus over exit, grows with V while agenda power binds. It can also be smaller than Λ: when rupture risk makes Europe concede in the bargaining ($P^*(Q) < 0$), threats can extract only the rest of its surplus over exit. Take $Q = -1$, $V = 4$, $\beta = 0.5$, $\lambda = 1$ and $r = 0.25$. Then $\Lambda = 1$ and $\Lambda_\psi = 2.2$, and withdrawing a conditional part of 3 costs Europe 1.8. Whatever agenda power under rupture risk wins Europe above $[Q]^+$, credible conditional threats can claw back, down to the exit premium. The agenda channel of Corollary 1(e) therefore survives only to the extent that Washington lacks credible conditional threats large enough to push Europe to its outside option.

In panel (a) of Figure 3 the unconditional threat is the smaller one, yet it is the only one that still works as Λ falls toward zero. Panel (b) illustrates part (c)(i). A coercer that prices each conditional threat on its own counts the same slack twice: at $\Lambda = 1$ the separate bites sum to 2 and the joint bite is 1. The bound in (c)(i) needs a fixed credible set (Appendix C), which the stake technology supplies: an instrument is credible when its execution cost is at most $\bar{m}_U$, whatever has

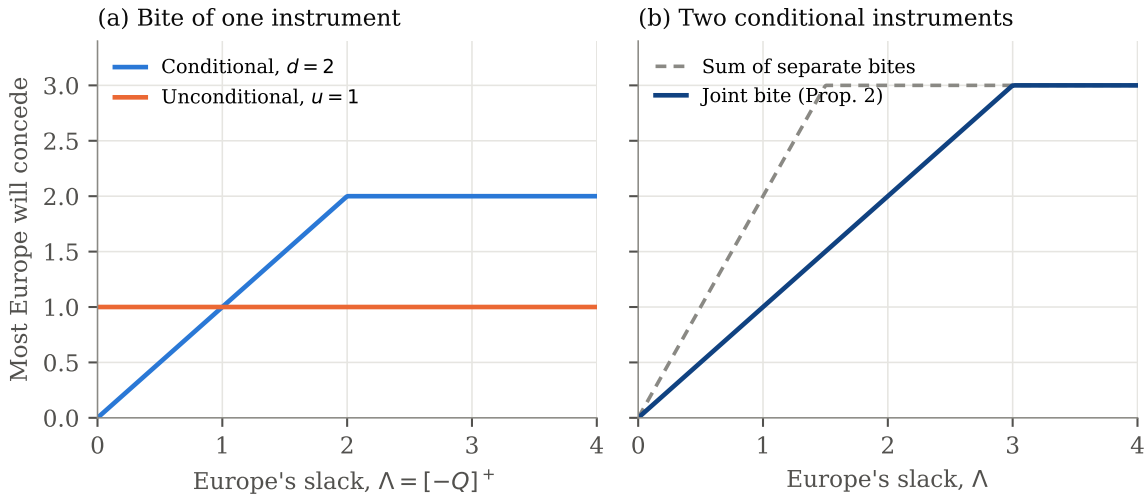


Figure 3: The slack budget. Panel (a): the bite of a conditional instrument with $d = 2$ rises one for one with Europe's slack Λ until $\Lambda = 2$; an unconditional instrument with $u = 1$ bites the same at every Λ. Panel (b): two conditional instruments with $d = 1.5$ each. The sum of their separate bites (dashed) overstates their joint bite $\min\{3, \Lambda\}$ at every $\Lambda < 3$. Bites from Proposition 2(a).

happened before.

Part (d) is the assurance problem of a coercer that cannot commit to stop demanding. Europe concedes only if everything the unspent threat will extract later stays within the bite, so with $b + 1$ identical sunk demands a single ultimatum obtains b of them, and the adaptive sequence obtains none. A threat meant to induce compliance works only if the target believes compliance will end it (Schelling, 1966), and Pauly (2024) shows that the steps that make a threat credible weaken that belief. A staked promise to make no further demand in the period buys the commitment, and stakes cost nothing, so part (d) describes a patron whose cap $\bar{m}_U$ is small relative to the value of its later demands. Such a patron can gain by issuing an ultimatum it expects Europe to refuse, which burns an instrument that would otherwise back later demands (Appendix C).

Corollary 2 (Rationing)**.** *Suppose all demands are reversible and either the United States can commit to a single ultimatum or $B(J^c) \geq \Lambda$. Then its optimal demand set solves*

$$\max_{y_k \in \{0,1\}} \sum_k w_k y_k - [\textstyle\sum_k t_k y_k - \Lambda]^+ \quad s.t. \quad \min\Big\{ \sum_k t_k y_k, \Lambda \Big\} \leq B(J^c),$$

with one y_k for each demand k. This is the knapsack $\max \sum_k w_k y_k$ s.t. $\sum_k t_k y_k \leq \min\{\Lambda, B(J^c)\}$ if $B(J^c) < \Lambda$, or if demands are divisible ($y_k \in [0, 1]$) and $w_k \leq t_k$ for every k. In general the United States may exceed Λ and pay the excess back as rent. The free capacity $\min\{\Lambda, B(J^c)\}$ falls as Q rises toward 0. Demands from every issue area draw on it.

Export-control alignment and limits on servicing in China are reversible concessions (Section 2). So are the trade terms of August 2025, under which the EU said it would eliminate tariffs on US

industrial goods and align its technology-security requirements with US ones (The White House, 2025c). The NATO spending target for 2035 (North Atlantic Treaty Organization, 2025) and the EU's stated intention to buy more US military equipment (The White House, 2025c) are closer to sunk, since they cost Europe whatever it later does with ASML. The reversible demands share the capacity $\min\{\Lambda, B(J^c)\}$, so a concession extracted on tariffs leaves less to extract on servicing, and the sunk ones, which slack does not cap, draw on the whole bite. At The Hague in June 2025 Trump tied Spain's trade terms to its defence spending: "We're negotiating with Spain on a trade deal. We're going to make them pay twice as much" (RTÉ News, 2025).

5.2 Counter-threats and hedges

As in Section 4.3, Chinese coercion enters only as counter-threats attached to particular concessions, announced before Washington makes its demands. We also consider a hedge: legacy-tool sales and servicing worth y_H to Europe and w_H to Washington if stopped. Washington can demand that Europe stop the hedge, backed by J^c (linkage), or threaten to block the hedged flows with its own jurisdiction, at net ex post cost e_F.

Proposition 3 (Counter-threats and hedges)**.** *Maintain the assumptions of Proposition 2.*

(a) Counter-threats. *Before the United States moves, China chooses credible counter-threats $r_k \in [0, \bar{r}_k]$. If concession k is made and $r_k > 0$, China executes at cost $e_k^C > 0$ and Europe bears the sunk loss r_k. China loses $z_k \geq 0$ when concession k is made. The United States makes a single ultimatum backed by J^c, and its optimal bundle is unique at the profiles compared below.*

(i) A counter-threat adds r_k to the sunk cost of k: Europe accepts S if and only if

$$\min\{T_{\mathrm{r}}(S), \Lambda\} + T_{\mathrm{s}}(S) + \textstyle\sum_{k \in S} r_k \leq B(J^c).$$

(ii) China does better than with no threats only if the United States' demand set changes.

(iii) If a counter-threat on k is executed in China's optimal profile, then dropping it changes the United States' demand set to one that contains a demand outside the equilibrium set. China executes on path only to crowd out other demands, and this occurs for an open set of parameters.

(iv) The United States' payoff is weakly lower under any profile of counter-threats.

(v) Europe's net loss is at most $B(J^c)$ with or without counter-threats. It equals $B(J^c)$ in both cases if the United States has a divisible sunk demand of positive value that China cannot threaten. The counter-threat then taxes the patron, not the holder. Otherwise Europe's loss can rise or fall.

(b) Policing a hedge. *Stopping a hedge worth y_H to Europe is a reversible concession. If it is the only demand, linkage stops the hedge if and only if $\min\{y_H, \Lambda\} \leq B(J^c)$ and $w_H \geq [y_H - \Lambda]^+$,*

where w_H is its value to the United States. The United States then pays rent $[y_H - \Lambda]^+$. Linkage is free only if $y_H \leq \min\{\Lambda, B(J^c)\}$, and its free part $\min\{y_H, \Lambda\}$ vanishes as $Q \to 0$. A staked threat to block the hedge by flow jurisdiction, with net ex post cost e_F, is credible if and only if $e_F \leq \bar{m}_U$. It leaves Q unchanged and is never executed. Executing a conditional withdrawal J raises Europe's premium to $Q + d(J)$.

Part (a) changes who pays for Chinese coercion. By (a)(i) a counter-threat makes a concession more expensive for Europe exactly as a sunk cost would. If Washington has other demands that China cannot reach, it fills the budget with them, and Europe's loss stays at $B(J^c)$. Washington drops the demand China has made expensive for others it values less, so the counter-threat taxes the patron. China gains only by changing what Washington asks for (part (a)(ii)), and it carries out a threat only to crowd out another demand (part (a)(iii); Appendix B gives an example on the equilibrium path). Section 8.3 reads the Nexperia episode against that requirement.

Part (b) separates the slack logic from a coercer that prices each demand at a fixed cost of the cheapest credible instrument (Section 8.1). The free part of linkage, $\min\{y_H, \Lambda\}$, shrinks as the premium rises toward zero, and beyond it Washington must pay rent. Flow jurisdiction does not depend on slack: it is credible whenever $e_F \leq \bar{m}_U$ and leaves Q unchanged. So linkage against a hedge worth less to Washington than to Europe ($w_H < y_H$) should appear when the ally has slack and fade as its premium approaches zero, while the use of flow jurisdiction should not track the premium at all. A cost minimizer that counted the rent as a cost of linkage would predict the same switch.

Given our reading that Europe's premium has been negative since 2019 (Section 8), the record is consistent with the first half of this prediction. The 2025 National Security Strategy and the August 2025 joint statement tie favourable treatment and US AI exports to export-control alignment (Section 2), a reversible concession. The second half needs variation in premia that the Dutch record alone does not supply (Section 8.5).

5.3 Conditionality as a choice

Washington chooses its conditionality. With $\kappa = G(s_A) - G(s_X - m_S)$ from equation (1), choosing the stake $m_S \leq \bar{m}_U$ is choosing $\kappa \in [\kappa_0, \bar{\kappa}]$, where $\kappa_0 = G(s_A) - G(s_X)$ is the conditionality with no stake and $\bar{\kappa}$ is the largest Washington can create.

Conditionality is costly to the coercer, and we let it cost $H(\kappa)$, increasing and convex with $H(\kappa_0) = 0$. The cost is one of assurance: other allies discount a guarantee that comes with a broad withdrawal rule, and a guarantee tied to Europe's conduct can be triggered by mistake. Allies that doubt the patron also build capabilities of their own (Lanoszka, 2018), which raises their premia and adds to H. With Q_0 the premium at κ_0, the premium is $Q(\kappa) = Q_0 - (\kappa - \kappa_0)S_k$. Washington converts slack into concessions worth $\nu \in [0, 1]$ per unit of European cost, a linear simplification of the concave value of capacity in Corollary 2, and we assume $B(J^c) \geq \Lambda$ so that all slack can be converted. The bound $\nu \leq 1$, which keeps the objective concave, says that a concession is worth less to Washington than it costs Europe; with $\nu > 1$ Washington would buy concessions with rent.

Washington chooses κ before the bidding stage to maximize

$$\Pi(\kappa) = -\left[Q(\kappa)\right]^+ + \nu\left[-Q(\kappa)\right]^+ - H(\kappa).$$

Let H' be the right derivative of H. For $x \geq 0$ let $\kappa_x = \inf\{\kappa \in [\kappa_0, \bar{\kappa}] : H'(\kappa) \geq x\}$, with $\inf \emptyset = +\infty$, and write $\kappa_1 = \kappa_{S_k}$ and $\kappa_\nu = \kappa_{\nu S_k}$, so that $\kappa_0 \leq \kappa_\nu \leq \kappa_1$. The conditionality at which the premium is exactly zero is $\kappa_z = \kappa_0 + Q_0/S_k$.

Proposition 4 (Conditionality is the price of concessions)**.** *The conditionality*

$$\kappa^* = \min\left\{\bar{\kappa},\ \max\{\kappa_\nu,\ \min\{\kappa_z, \kappa_1\}\}\right\}$$

maximizes Π*, uniquely if* H *is strictly convex, and* $Q(\kappa^*) = (\kappa_z - \kappa^*)S_k$*.*

(a) If $\nu = 0$*, then* $\kappa^* = \kappa_0$ *when* $Q_0 \leq 0$*, and* $\kappa^* = \min\{\kappa_z, \kappa_1, \bar{\kappa}\}$ *otherwise. The United States never makes its guarantee more conditional than needed to hold the premium at zero. It creates no slack:* $\Lambda^* = \left[-Q_0\right]^+$*.*

(b) If $\nu > 0$*,* $\kappa_0 \leq \kappa_z < \bar{\kappa}$ *and* $H'(\kappa_z) < \nu S_k$*, then* $\kappa^* = \min\{\kappa_\nu, \bar{\kappa}\} > \kappa_z$ *and* $\Lambda^* = (\kappa^* - \kappa_z)S_k > 0$*: conditionality is the price of concessions. If* $Q_0 < 0$*, then* $\kappa^* = \min\{\kappa_\nu, \bar{\kappa}\}$ *and* $\Lambda^* = (\kappa^* - \kappa_z)S_k \geq -Q_0$*.*

(c) Let H *be strictly convex and differentiable. Away from regime boundaries:*

(i) if $\kappa^* = \kappa_z$*, then* $d\kappa^*/dQ_0 = 1/S_k$ *and the United States pays no rent: a stronger Chinese bid elicits a more conditional guarantee, not a payment;*

(ii) if $\kappa^* = \kappa_1 < \min\{\kappa_z, \bar{\kappa}\}$ *(assurance is costly at the margin before the premium reaches zero),* κ^* *does not depend on* Q_0 *and the rent* $Q(\kappa_1) > 0$ *rises one for one with* Q_0*;*

(iii) if $\kappa^* = \kappa_\nu \in (\max\{\kappa_z, \kappa_0\}, \bar{\kappa})$*, or if* $\kappa^* = \kappa_0 > \kappa_z$*, then* κ^* *does not depend on* Q_0 *and slack falls one for one with* Q_0*: China's bid is paid for out of the United States' concessions;*

(iv) if $\kappa^* = \bar{\kappa}$*, the United States pays* $\left[Q(\bar{\kappa})\right]^+$*, which rises one for one with* Q_0 *when* $Q(\bar{\kappa}) > 0$*. Otherwise slack falls one for one.*

(d) $d\bar{\kappa}/ds_X = g(s_X + s_L) - g(s_X - \bar{m}_U)$*, with* s_L *and* $\bar{m}_U$ *fixed. Hence* $\bar{\kappa}$ *is nonincreasing in* s_X *wherever* g *is nonincreasing on* $[s_X - \bar{m}_U,\ s_X + s_L]$*, for example when* G *is unimodal with its mode at or below* $s_X - \bar{m}_U$*, and it is then nonincreasing in the access value of European territory and, if* s_B *rises with* V*, in* V*. If instead* $g(s_X + s_L) > g(s_X - \bar{m}_U)$*,* $\bar{\kappa}$ *rises with* s_X*. If* s_A *is held fixed when* s_X *rises,* $\bar{\kappa}$ *is nonincreasing in* s_X *for every* G*.*

A patron with no further demands ($\nu = 0$) makes its guarantee just conditional enough to hold the premium at zero, or less if assurance becomes too costly first. Conditionality beyond that point is worth buying only because slack can be spent, so only a patron that plans to make demands makes its guarantee more conditional than the rivalry with China requires. Europe pays for the

extra conditionality with the concessions it then makes, and other allies pay through a weaker assurance. Because the argument uses only a conditional part chosen before the bidding stage at a convex cost, we think it carries over, with d_j in place of κS_k, to the other instruments.

By part (c)(i), when assurance is cheap at the margin Washington answers a stronger Chinese offer by making its guarantee more conditional, by exactly enough to keep the premium at zero, and pays nothing. The December 2025 National Security Strategy reads this way to us: it offers favourable treatment to countries that align their export controls (Section 2), which raises conditional parts relative to the treatment of a country that does not align and is not a transfer (Section 8.2). Where assurance is already costly at the margin, or Washington is already spending slack, it pays for China's bid in rent or in forgone concessions (parts (c)(ii) and (c)(iii)).

An untargeted drawdown, such as the cut of 25,000 or more troops from Europe that the Pentagon was reported in September 2026 to be weighing (De Luce et al., 2026), lowers what Washington provides to a loyal ally, $x_p(A)$, and with it d_p and Europe's slack. The May 2026 order moving troops out of Germany (Section 2) punished an act with no prior demand, which lies outside our demand-and-threat structure. Read as the execution of an implicit threat, it spends the instrument (Section 5.1) and raises the target's premium by the instrument's conditional part (Proposition 3(b)). Carrying it out at all requires the burning logic described after Proposition 2 or incomplete information, and its effect on other allies' beliefs about κ lies outside the model.

Part (d) gives the sign of the silicon shield, the fourth channel of Corollary 1(e). If holding the chokepoint raises Washington's stake in Europe's security whatever Europe's alignment, then s_X rises. Where G is unimodal with its mode at or below $s_X - \bar{m}_U$, so that the most likely crisis costs lie below Washington's net stake in a Europe that has exited, a higher s_X lowers the most conditionality Washington can create. Where a staked abandonment makes defence of an exiter unlikely, $s_X - \bar{m}_U$ lies in the left tail of G, where the density rises, and holding the chokepoint can make the guarantee more conditional. The shield itself is contested (Weil et al., 2025). Blankenship (2021) finds that allies in locations the United States values contribute less to the common burden, which is what part (d) implies when g is nonincreasing on the relevant interval.

6 Control Rights and Preventive Demands

The static model holds Europe's exit premium fixed. Yet the premium moves: Chinese offers and European rearmament raise it, and Chinese indigenization lowers it. A Washington that expects it to turn positive expects to pay rents that Europe cannot credibly promise to forgo. A transfer of control can remove them.

6.1 Setup

There are two periods, $t \in \{1, 2\}$, and a discount factor $\delta \in (0, 1]$. In each period in which Europe holds control, the bidding contest of Lemma 1 is played at that period's premium Q_t, and Europe receives $[Q_t]^+$. We assume that Washington retains Europe at every contest, including after a cut

($Q_1 + d_1 < V$ and $Q_1 + d_1 \leq \bar{m}_U$, and the same in period 2 at the highest premium Europe can reach), and that payments are made at signing.

The ultimatum. At the start of period 1, before the period-1 contest, Washington may issue an ultimatum at an issuing cost $\iota \geq 0$: Europe transfers control of ASML's customers, output and technology for both periods against a payment $T \geq 0$, or Washington withdraws a bundle J of instruments for period 1. The bundle has unconditional part u_1, conditional part d_1 and direct ex post cost $g_1 \geq 0$ to Washington (Section 4.2). If Europe refuses and the threat is carried out, Europe's premium rises to $Q_1 + d_1$, and by Proposition 2(a) the refusal costs Europe the bite

$$B_1 = u_1 + \min\{d_1, \Lambda_1\}, \qquad \Lambda_1 = [-Q_1]^+ ,$$

net of the rent Washington must then pay. Carrying out the threat costs Washington $e = g_1 + [Q_1 + d_1]^+ - [Q_1]^+$, its direct cost plus the extra rent. The threat is credible if and only if $e \leq \bar{m}_U$; otherwise $B_1 = 0$. We assume that no other period-1 demand draws on the bite; with other demands, control enters Corollary 2 as a sunk demand.

We do not expect the core guarantee to back such an ultimatum. Withdrawing protection from a Europe that stays aligned costs Washington its own stake in defending a loyal ally, so g_1 is large unless s_A is small. Posture, enablers and the AI instrument cost less to withdraw for one period.

Control. A transfer lowers Europe's premium at both dates by $\Delta_c \in (0, \infty]$, the enforceability of control. With $\Delta_c = \infty$ Europe can never again choose exit. With Δ_c finite, a Europe whose premium later rises far enough could take back part of what it gave. What is transferred sets Δ_c. Relocating capacity to US soil raises it, and so does control exercised through ASML's US subsidiaries, which are US-law entities (ASML Holding N.V., 2026b). A controlling vote in a Dutch holding company stays under Dutch law and has low Δ_c; we think it would be exposed to emergency powers like the Goods Availability Act order against Nexperia of 30 September 2025 (Minister van Economische Zaken, 2025).

Values and costs. Washington places an intrinsic value $v_c \geq 0$ on control, apart from the rents it saves. Europe bears a sovereignty cost $\sigma \geq 0$ if it transfers control. It includes the domestic cost of surrendering a national champion and hurdles fixed in advance: ASML's preference-share foundation, screening under the Vifo Act (Section 2), and review under the Foreign Subsidies Regulation, which we infer a US-funded acquisition would trigger (European Parliament and Council of the European Union, 2022). Being public, these hurdles raise σ before Washington decides whether to demand anything. China punishes a completed transfer with $r^c \geq 0$, which Europe bears.

Autonomy investment. After the period-1 outcome, and before the period-2 contest, Europe can make an observable investment at cost K that raises its period-2 premium by $\bar{I}$, for instance

rearmament that lowers S_k. Holding a period-2 premium q, it invests if and only if $\delta(\left[q+\bar{I}\right]^+ - [q]^+) > K$.

Ties. An indifferent Europe accepts; Washington issues only if it strictly gains (the limit $\iota \to 0^+$ when $\iota = 0$); Europe invests only if that is strictly profitable. After a transfer Washington keeps its baseline provision, since withdrawing it would cost something and gain nothing.

6.2 Preventive control

Let $I(q) = \mathbf{1}\{\delta(\left[q+\bar{I}\right]^+ - [q]^+) > K\}$ indicate investment at period-2 premium q. Washington's discounted period-2 rent is $\text{Rent}_U(q) = \delta\left[q + \bar{I}\, I(q)\right]^+$, and Europe's period-2 value net of its investment cost is $\text{Val}_H(q) = \text{Rent}_U(q) - K\, I(q)$.

Proposition 5 (Preventive control)**.** *Let*

$$\Delta R_U = [Q_1]^+ - [Q_1 - \Delta_c]^+ + \text{Rent}_U(Q_2) - \text{Rent}_U(Q_2 - \Delta_c),$$
$$\Delta R_H = [Q_1]^+ - [Q_1 - \Delta_c]^+ + \text{Val}_H(Q_2) - \text{Val}_H(Q_2 - \Delta_c).$$

Let $B_1 = u_1 + \min\{d_1, \Lambda_1\}$ if the threat is credible ($e \le \bar{m}_U$, with $e = g_1 + [Q_1 + d_1]^+ - [Q_1]^+$) and $B_1 = 0$ otherwise. In the subgame-perfect equilibrium, the United States issues the ultimatum, and Europe accepts, if and only if

$$\Gamma \equiv v_c - \iota + \min\{\Delta R_U,\ (\Delta R_U - \Delta R_H) + B_1 - \sigma - r^c\} > 0.$$

The payment is $T^* = [\Delta R_H - B_1 + \sigma + r^c]^+$, the United States gains Γ, and Europe's payoff is $\max\{\Delta R_H - B_1,\ -\sigma - r^c\} + [Q_1 - \Delta_c]^+ + \text{Val}_H(Q_2 - \Delta_c)$. An ultimatum that is issued is accepted. The credibility condition $e \le \bar{m}_U$ does not depend on Δ_c.

The proof is in Appendix B. ΔR_U is the rent that control saves Washington and ΔR_H is what Europe gives up by losing it; they differ only by the investment cost a transfer spares Europe. Washington gains the value of control net of the issuing cost, plus the smaller of the rent saved and what it can extract by force. Europe is paid for the rents it surrenders and the deadweight cost $\sigma + r^c$, less the bite. A refused ultimatum never pays Washington, so the threat is never carried out on the path. Its credibility is judged against a Europe that keeps control, which is why Δ_c does not enter.

In the special case $v_c = \iota = 0$, $\Delta_c = \infty$, $\bar{I} = 0$ and $Q_1 < 0$, the gain is $\Gamma = \min\{\delta\,[Q_2]^+,\ B_1 - \sigma - r^c\}$. Washington demands control if and only if Europe's future premium is positive and the bite exceeds $\sigma + r^c$. Since $B_1 \le u_1 + \Lambda_1$, a purely conditional bundle clears that bar only if $Q_1 < -(\sigma + r^c)$. The demand comes while Europe is weak but expected to grow stronger.

Corollary 3 (Motive, investment, enforceability, China)**.**

(a) Preventive motive. $\Gamma > 0$ *if and only if* $\Delta R_U > \iota - v_c$ *and* $B_1 + (\Delta R_U - \Delta R_H) > \sigma + r^c + \iota - v_c$. *If* $v_c \leq \iota$, *issuance requires* $\Delta R_U > 0$ *and* $B_1 + (\Delta R_U - \Delta R_H) > \sigma + r^c$. *For* $\Delta_c > 0$, $\Delta R_U > 0$ *if and only if* $Q_1 > 0$ *or* $Q_2 + \bar{I}\, I(Q_2) > 0$*: Europe's premium is positive at date 1, or its premium after its own investment is positive at date 2.*

(b) Autonomy investment. $\Delta R_U - \Delta R_H = K\left[I(Q_2) - I(Q_2 - \Delta_c)\right] \geq 0$. *Both arguments of the minimum in* Γ *are weakly larger than when Europe has no investment option (*$\bar{I} = 0$*), so an autonomy investment that Europe cannot commit to forgo weakly raises* Γ. *With investment, the United States can obtain control with no bite at all: if* $B_1 = 0$*, the ultimatum is issued if and only if* $\Delta R_U > \iota - v_c$ *and* $K[I(Q_2) - I(Q_2 - \Delta_c)] > \sigma + r^c + \iota - v_c$*, and the transfer then saves Europe's investment cost.*

(c) Enforceability. *Holding* σ, r^c, v_c *and* ι *fixed,* Γ *and* T^* *are nondecreasing in* Δ_c*;* B_1 *and the credibility condition do not depend on* Δ_c.

(d) Counter-threats against a transfer. *Before the United States decides, China announces a punishment* $r^c \in [0, \bar{r}^c]$ *of a transfer and a stake* $m_C \leq \bar{m}_C$. *Carrying out* r^c *after a transfer has ex post value* $-e^C(r^c)$ *to China, where* e^C *is continuous and increasing with* $e^C(0) = 0$*; the threat is credible if and only if* $e^C(r^c) \leq m_C$. *China gets* $z_C \geq 0$ *if Europe keeps control, and the ex post payoff of its punishment decision (*$-e^C(r^c)$ *if it punishes,* $-m_C$ *if it does not) if control passes. Write* $\Gamma(r^c)$ *for* Γ *at punishment* r^c*, and let* $r^c_{\max} = \max\{x \leq \bar{r}^c : e^C(x) \leq \bar{m}_C\}$. *In every subgame-perfect equilibrium:*

(i) if control is transferred, China does not punish: on the path the punishment is zero, the payment is $[\Delta R_H - B_1 + \sigma]^+$ *and* $\Gamma(0) > 0$*;*

(ii) a credible punishment $r^c > 0$ *is announced only if it deters,* $\Gamma(r^c) \leq 0$*;*

(iii) if $z_C > 0$*, control is transferred if and only if* $\Gamma(r^c_{\max}) > 0$*; otherwise China deters with a credible* $r^c \geq v_c - \iota + \Delta R_U - \Delta R_H + B_1 - \sigma$ *whenever* $\Gamma(0) > 0$*;*

(iv) if $z_C = 0$ *and* $\Gamma(0) > 0 \geq \Gamma(r^c_{\max})$*, both the transfer and its deterrence are equilibrium outcomes.*

If punishing is self-rewarding ($e^C(x) < 0$ *for some* x*), (i) fails: China may let the transfer happen and punish it.*

When $v_c = 0$ and Europe has no autonomy investment that a transfer would forestall, an accepted transfer is inefficient. The rents it avoids are transfers between Washington and Europe, while the sovereignty cost and the issuing cost are destroyed, and China's punishment adds to the loss when, being self-rewarding, it is carried out (part (d)). That is the logic of preventive war (Fearon, 1995; Powell, 2006), with a coerced transfer in place of a war. With an investment option the transfer can be efficient, because it spares Europe an investment cost K sunk only to secure rents (part (b)), the one place where the logic of Grossman and Hart (1986) operates in our model. What our setting adds to the models of Section 3 is a ceiling on coercion: the conditional part of the bite cannot exceed Europe's current slack, so the patron must act while the holder is weak.

Part (b) ties the demand to Europe's own choices. An investment in future bargaining power that Europe cannot commit to forgo invites the demand, as an observable nuclear programme invites prevention or a deal (Debs and Monteiro, 2014; Spaniel, 2019). It raises the rents a transfer saves and creates a surplus the two can share, which permits a purchase with no bite. Take $\delta = 1$, $Q_1 = 1$, $Q_2 = -\frac{1}{2}$, $\bar{I} = 3$, $K = 2$, $\Delta_c = \infty$, $\sigma = 1$ and $r^c = v_c = \iota = 0$, with no threat. Europe would invest, so Washington faces a period-2 rent of 2.5 on top of the period-1 rent of 1. It buys control for $T^* = 2.5$ and gains $\Gamma = 1$; the transfer spares an investment of 2 at a sovereignty cost of 1. Rearmament that Europe cannot commit to forgo can therefore prompt an offer for the chokepoint itself.

By part (c), a transfer that Europe could partly undo removes only $\min\{[Q_t]^+, \Delta_c\}$ of the rent at each date, so the commitment problem returns whenever $Q_2 - \Delta_c > 0$, and Washington gains less and pays less, the buy-or-lease logic of Harstad (2016). Relocating production is the high-Δ_c end. The control rights Washington has taken so far all attach to assets on US soil, where enforcement does not depend on a foreign government: consent rights over the ownership, exit and customers of MP Materials, bought with equity, a price floor and an offtake (MP Materials Corp., 2025), a golden share in US Steel as a condition of its acquisition by Nippon Steel (United States Steel Corporation, 2025), and a passive stake in Intel (Intel Corporation, 2025). Toward Taiwan, Washington has sought relocation (Section 8.4).

By part (d), China's threat to punish a transfer is never carried out on the path: once control has passed, punishing costs China something and deters nothing. The threat protects European ownership only by deterring the demand, which requires $\Gamma(r^c_{\max}) \leq 0$. Sovereignty costs work the same way, which makes the hurdles of Section 6.1 part of Europe's protection against a demand.

6.3 Windows

Proposition 5 allows one ultimatum and a one-period cut. When a cut runs over several dates, its bite is the discounted sum of the per-date bites $u_s + \min\{d_s, \Lambda_s\}$, provided Washington retains Europe throughout (Lemma 2 in Appendix A.1). A cut is spent once executed, and extending a purely conditional cut into a date at which Europe's premium is positive adds nothing to its bite. A threat of permanent exclusion therefore bites no harder than one confined to the dates at which Europe has slack.

Corollary 4 (Windows)**.** *Consider N dates with $v_c = 0$ and $\Delta_c = \infty$. At any date t at which Europe still holds control, the United States may issue, at cost $\iota > 0$, an ultimatum: transfer control for all remaining dates against a payment $T \geq 0$, or face a cut at date t only, with bite $b_t = u_t + \min\{d_t, \Lambda_t\}$ and execution cost $e_t = g_t + [Q_t + d_t]^+ - [Q_t]^+$. Call t* eligible *if $e_t \leq \bar{m}_U$, $b_t > \sigma + r^c$, and $Q_{t'} > 0$ for some $t' \geq t$. There is $\bar{\iota} > 0$ such that for every $\iota \in (0, \bar{\iota})$, the United States issues an ultimatum on the equilibrium path if and only if some date is eligible. Every ultimatum issued is accepted, and the first one comes no later than the last eligible date (possibly at a date that is not eligible). The same holds at $\iota = 0$ when the United States issues only if it strictly gains. If $u_t = 0$ at every date, the condition $t' \geq t$ can be replaced by $t' > t$.*

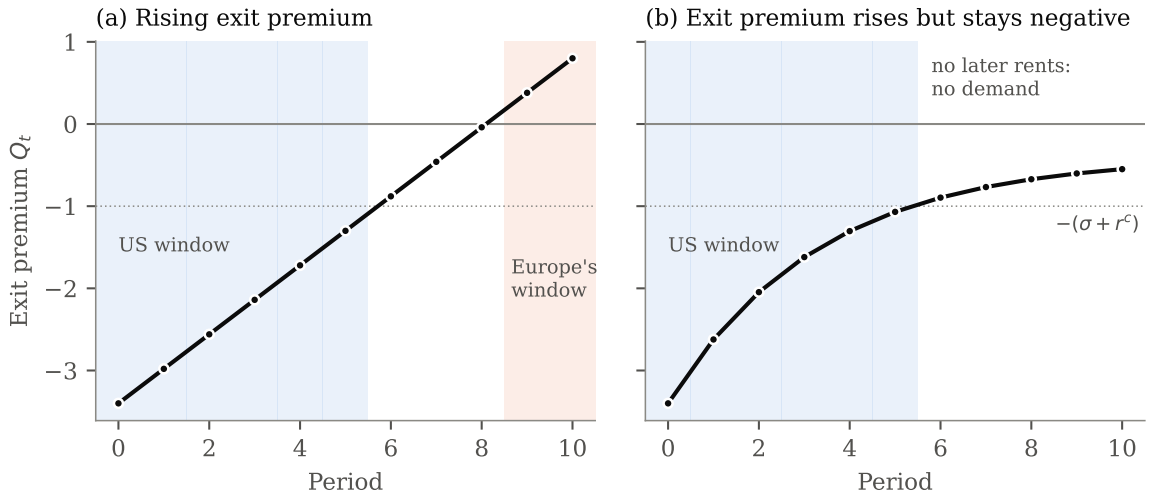


Figure 4: Windows for preventive control (illustrative units). Panel (a): $Q_t = -3.4 + 0.42\,t$. Panel (b): $Q_t = -3.4 + 3(1 - e^{-0.3t})$. In both, $\sigma + r^c = 1$ and Washington holds a conditional instrument with $d_t = 2$ and no unconditional part, so the condition $t' > t$ of Corollary 4 applies; the cut is assumed credible at every date. Blue shading marks dates at which the threat bites by more than $\sigma + r^c$, which here requires $Q_t < -1$. Orange shading marks Europe's window, $Q_t > 0$.

In panel (a) of Figure 4 the threat bites by more than $\sigma + r^c$ through $t = 5$, and the premium turns positive at $t = 9$. Washington acts at or before $t = 5$, long before Europe's window opens. It can act earlier still, because a Europe that refuses remains exposed to a later demand (Appendix C reports how often). In panel (b) the premium rises toward -0.4 but never crosses zero, and with $v_c = 0$ Washington has nothing to prevent; with $v_c > \iota$ it could still demand control (Corollary 3(a)).

In Powell (2006) preventive war requires a shift in power that is large and rapid relative to what the declining state can be offered within the period. Here, under complete information, the speed of the shift affects when Washington acts and what it pays, and not whether it acts, because a transfer with $\Delta_c = \infty$ removes every future rent at once. With finite enforceability a shift far above Δ_c leaves rents that no transfer can reach, and size and speed matter again, as in Powell's account.

The direction of drift is therefore decisive. Rearmament and the rationing of loyal allies invite a demand, while Chinese indigenization, by lowering W, removes the motive. We cannot tell from the record which dominates.

6.4 Promises or threats

Europe could instead remove the commitment problem itself, by binding itself against exit without transferring control. Binding also raises Europe's slack and with it the bite of Washington's threats.

Remark 2 (Self-binding and dual key). Before period 1, Europe can adopt a binding rule (Europeanized or treaty-entrenched controls) that lowers its premium at both dates by $\Delta_b \in (0, \infty]$ at cost σ^b, drawing a Chinese punishment r^b; Europe keeps the key. The game of Proposition 5 is then played with premia $Q_t - \Delta_b$, and the bite of the US threat is computed at the new slack.

(a) With $\Delta_b = \infty$, control removes no rents, and the bite becomes $B_1^b = u_1 + d_1$ if $g_1 \leq \bar{m}_U$ (and 0 otherwise). The United States then issues if and only if $\Gamma^b = v_c - \iota + \min\{0,\, B_1^b - \sigma - r^c\} > 0$, so binding pre-empts the ultimatum if and only if $v_c \leq \iota$ or $v_c - \iota + B_1^b \leq \sigma + r^c$. If it does not, the United States demands control and pays $\left[\sigma + r^c - B_1^b\right]^+$.

(b) Let $Q_1 < 0$, $\Delta_c = \infty$, no investment and $\Delta_b = \infty$. Europe's payoff is $-\sigma^b - r^b - \mathbf{1}\{\Gamma^b > 0\} \min\{\sigma + r^c, B_1^b\}$ if it binds, and $\max\{\delta\,[Q_2]^+ - B_1,\, -\sigma - r^c\}$ if it does not bind and $\Gamma > 0$, or $\delta\,[Q_2]^+$ if it does not bind and $\Gamma \leq 0$. In particular, if the United States would issue the ultimatum without binding and binding pre-empts it, Europe strictly prefers to bind if and only if $-\sigma^b - r^b > \max\{\delta\,[Q_2]^+ - B_1,\, -\sigma - r^c\}$. If the United States would not issue without binding, Europe never strictly prefers to bind.

(c) A partial binding ($\Delta_b < \infty$) lowers the rents control would remove but raises the slack, hence the bite, so it can make Γ positive. Example, with $\Delta_c = \infty$, no investment and $\bar{m}_U \geq 1$ (so that the cut is credible after binding): $\delta = 1$, $Q_1 = -1$, $Q_2 = 10$, $u_1 = 0$, $d_1 = 5$, $g_1 = 0$, $\sigma = 1.5$, $r^c = v_c = \iota = 0$. Without binding $\Gamma = \min\{10,\, 1 - 1.5\} < 0$. With $\Delta_b = 3$, $\Gamma = \min\{7,\, 4 - 1.5\} = 2.5$ and the United States demands control at $T^* = 4.5$.

(d) Dual-key licensing is a partial control transfer with enforceability $\Delta_d < \infty$, sovereignty cost $\sigma_d < \sigma$, value v_d and Chinese punishment r^d. If the United States can demand either arrangement, it demands the one with the larger of Γ_{full} and Γ_{dual} (each given by Proposition 5 with that arrangement's parameters), provided it is positive. Because $\sigma_d < \sigma$, the United States can obtain a dual key when the bite is too small to obtain full control.

Binding pre-empts a demand only if Washington values control little for its own sake, since full binding maximizes the bite and makes control cheaper to coerce for a patron with $v_c > \iota$. By part (b), binding pays Europe only when it pre-empts a demand that would otherwise come, at a lower cost. By part (c), a rule that binds Europe partway removes some of the rents Washington fears but adds to the slack its threats draw on, and can turn a demand that would not pay into one that does.

Washington can also demand binding itself. Proposition 5 prices that demand with $(\Delta_b, \sigma^b, r^b)$ in place of (Δ_c, σ, r^c) and $v_c = 0$, so a patron that values control little for its own sake prefers to demand binding whenever $\sigma^b + r^b < \sigma + r^c$. A preventive demand need not be a demand for ownership. Under binding Europe holds the only key. Under a dual key Washington holds a veto over China-bound licences, a control right over customers that Remark 2(d) prices with its own enforceability and sovereignty cost. The Dutch government's opposition to the extraterritorial reach of the MATCH Act (Section 2) suggests to us that σ_d in The Hague is not far below σ, while parliament's 110-to-40 vote in October 2026 for harmonized EU export control (Bamenga et al., 2026) suggests that σ^b is low (Section 8.3).

We read the European steps of 2024 and 2025 (Section 2) as partial binding. When licensing of the NXT:1970i and NXT:1980i moved from Washington to The Hague in September 2024 (Minister voor Buitenlandse Handel en Ontwikkelingshulp, 2024), Europe's premium fell and the key stayed in Europe, the reverse of a control transfer; Section 8.2 codes the December 2024 Supplement No. 4

status as the matching reward for self-binding. On our reading, Delegated Regulation 2025/2003 (European Commission, 2025a) raised Δ_b, because removing its entry takes an act at Union level that the Netherlands cannot adopt alone. It also had a US-facing cost. Listed tools ship to US fabs under the Union authorisation EU001 (Section 2), so the Netherlands gave up a national licence over their US-bound shipments and, we think, part of Europe's hold-up option (Section 7.4). None of these steps comes close to $\Delta_b = \infty$, and by part (c) they could have strengthened the US motive for a demand as easily as weakened it.

6.5 Private sovereignty costs and breakdown

Under complete information an ultimatum is accepted or never issued. Breakdown requires that Washington be unsure what a transfer would cost Europe. We think that uncertainty is real: since February 2026 the Dutch cabinet has governed with minority support in parliament (NL Times, 2026a; Omroep Brabant, 2026), so the domestic price of a transfer would depend on whichever majority it could find. When σ is Europe's private information, a threat-backed ultimatum is refused with positive probability and the threat is carried out (Proposition 7 in Appendix A.1). Refusal is likelier when China's punishment of a transfer is credible and large, when Europe's future premium is high, and when the threat is cheap for Washington to carry out, the risk-return trade-off of Fearon (1995). A patron that values control for its own sake may switch from a purchase offer to a threat as the bite rises, which can raise the probability of breakdown. These probabilities describe a patron that can commit to one public demand.

7 Who Decides, and the Power to Hurt

So far Europe has been one actor. In fact The Hague licenses ASML's shipments to China, ASML decides whether to make them, and the Union decides whether its tools can be turned against the United States (Section 2). Europe can also impose costs on Washington while the two bargain (Section 7.4).

7.1 The gatekeeper

Exit toward China is a Dutch decision. The licence is granted where the exporter is established, and neither other member states nor the Commission can refuse it (European Parliament and Council of the European Union, 2021). Let $Q_{NL} = \omega_{NL} - c_{NL}$ be the gatekeeper's own exit premium, net of the credible costs of exit that fall on the Netherlands. By Lemma 1, Washington keeps the Netherlands aligned at price $[Q_{NL}]^+$.

Washington can lower that price by divide and rule. A benefit that it provides to a loyal Netherlands in the baseline, and credibly withdraws if The Hague licenses exit, lowers Q_{NL} one for one. It cuts Washington's payment by up to the benefit's value, and once the gatekeeper has slack, every further unit adds to what Washington can spend on other demands (Remark 3, Proposition 2). A side payment promised on top of the baseline saves Washington nothing. On our reading the

results of Sections 5 and 6 carry over to China-bound licensing with Q_{NL} in place of Q and each instrument measured by what the Netherlands bears, but not to US-facing measures, which the Union decides.

European partners move ω_{NL} or c_{NL} only through transfers: compensation, coordinated retaliation or cover against Chinese counter-coercion. Washington can tier member states directly. By our computation from 2026 Council population figures (Council of the European Union, 2025), the ten members in the AI Diffusion Rule's favoured group (Section 2) held 70.46 percent of EU population but too few members for a qualified majority, while the other seventeen held 29.54 percent and enough members to block. BIS withdrew the rule before compliance was due, so tiering is latent, but one rule can reward the gatekeeper and split the Council at once.

7.2 The firm between two jurisdictions

A licence permits a shipment and orders nothing. ASML ships when its Chinese business, plus the penalties China (p_C) and the Union's Blocking Statute (p_B) would impose for complying with US measures instead, at least matches the credible US penalty for shipping. Otherwise the state must compel it at a cost ζ, and Europe does so only if its premium exceeds that cost (Remark 4).

Since April 2026, p_C has a legal basis. Decree 835 lets Beijing blacklist, bar and expose to damages suits and fines those who implement foreign extraterritorial measures (Li et al., 2026). ASML's staff in China who implemented a US servicing ban would be exposed. A higher p_C makes a licensed firm more willing to ship and weakly raises Europe's effective premium, without touching the credibility of Washington's jurisdiction. For a US ban on servicing in China, the same penalty enters Proposition 3(a) as a counter-threat r_k only if Europe internalizes the firm's loss.

The Blocking Statute's annex lists US sanctions on Cuba and Iran and no US export-control law (Council of the European Union, 1996; European Commission, 2018), so $p_B = 0$ today. Even a listing would be bounded: in *Bank Melli* the Court of Justice let national courts weigh the firm's economic loss (Court of Justice of the European Union, 2021), which we read as keeping p_B small for a firm with ASML's US business.

The Hague compelled Nexperia under the Goods Availability Act, China blocked exports from Nexperia's Chinese plants, and the order was suspended after Washington had suspended the rule behind its demand (Minister van Economische Zaken, 2025; Karremans, 2025). If the Chinese ban moved The Hague, ζ is high when China retaliates against compulsion itself, and the record does not tell us which (Section 8.3).

7.3 The Council

Turning ASML against Washington is collective, because its most advanced tools ship to US fabs under the Union general export authorisation EU001 (Section 2). Under the Anti-Coercion Instrument the Council first determines by qualified majority that coercion exists. Response measures are Commission implementing acts; if the examination committee gives no opinion, the act goes to an appeal committee, where the Commission may adopt unless a qualified majority votes

against (European Parliament and Council of the European Union, 2023, 2011). A delegated act removing the United States from EU001 would instead apply at once and stand unless the European Parliament or a reinforced qualified majority of the Council objected (European Parliament and Council of the European Union, 2021; European Union, 2016), but The Hague would still have to deny the individual licences.

With enforceable vote-contingent payments, Washington defeats a countermeasure most cheaply by buying a blocking minority before the Council's determination, since every later stage of either route needs a coalition at least as large (Remark 5). Because Washington can make posture conditional member by member (Section 4.3), the members whose defence depends most on US forward presence join for free. They cannot block alone: the six frontline states hold 15.09 percent of the population, and a block by number needs thirteen eastern and Nordic members (Table 5). We expect opposition to an anti-US measure to start on the frontline and to block only once it spreads to central Europe and the Nordics, or a large state joins. The seven states that *Europe 2031* names as opposing or abstaining on its failed 2029 measure hold 28.06 percent of EU population (Juijn et al., 2026; Council of the European Union, 2025) and cannot block. Any blocking completion needs Germany, France or Spain, or at least three further states (Belgium, Czechia and Portugal would do), or six more of any size to reach thirteen members (Appendix C).

The formal threshold is not what binds. Recital 85 of the instrument commits the Commission, at every stage including appeal, to avoid "going against any predominant position amongst Member States" (European Parliament and Council of the European Union, 2023), and its Statement C/2023/1341 repeats the commitment (European Commission, 2023). A countermeasure that large states oppose should die before any vote: no examination, no proposal, or a package without the ASML element. Member states gave positive committee opinions to rebalancing measures against US goods on 9 April and 24 July 2025 (European Commission, 2025e,b). Neither applied. The April package was suspended after the US tariff pause and the July package after the Turnberry agreement (European Commission, 2025d,c); the suspension was renewed in January 2026 after Washington dropped the tariffs it had threatened over Greenland (Dazio et al., 2026; European Commission, DG Trade, 2026), and again in July 2026 with no end date (European Commission, 2026). The Union arms and then suspends on a US concession. No armed measure has been tested against a Washington that refused to move. Washington has a route cheaper than buying a blocking minority: a concession that triggers suspension.

7.4 Hold-up

Proposition 1 lets the status quo run unchanged while Europe and Washington bargain. In a dispute over control of ASML or over Europe's access to frontier AI, Washington could cut Europe's AI allocation while talks continue, and Europe could interrupt ASML's servicing of US-aligned fabs. Let disagreement carry flow costs $c_E(t)$ to Europe and $c_U(t)$ to the United States, and write $\hat{C}_i(t) = (r+\lambda)\int_0^\infty e^{-(r+\lambda)s} c_i(t+s)\,ds$ for the average discounted cost from date t on.

Proposition 6 (Pain and rupture)**.** *Consider the game of Proposition 1 with $r > 0$ and disagreement flow costs $c_E(t), c_U(t) \geq 0$ that are bounded and piecewise uniformly continuous, with finitely many discontinuities in every bounded interval. Let $M = Q + \beta(V - Q)$, $n(t) = \beta c_U(t) - (1-\beta)c_E(t)$, $N(t) = \beta \hat{C}_U(t) - (1-\beta)\hat{C}_E(t)$ and*

$$x^\circ(t) = \psi M + (1-\psi)N(t).$$

(i) For every $\Delta > 0$, Europe's Markov perfect equilibrium payoff is unique. In every round in which $\lambda > 0$ or $c_E + c_U > 0$, every Markov perfect equilibrium has immediate agreement. If $\lambda = 0$ and $c_E(t) + c_U(t) = 0$ on an interval, there are also payoff-equivalent equilibria with delay during that interval.

(ii) As $\Delta \to 0$, Europe's equilibrium payoff converges to

$$\begin{aligned} P^* &= \sup_{\tau \in [0,\infty]} \Big\{ \int_0^\tau e^{-(r+\lambda)s} \big(\lambda M + r\, n(s)\big)\, ds + e^{-(r+\lambda)\tau} Q \Big\} \\ &= x^\circ(0) + \left[\sup_{t \geq 0} e^{-(r+\lambda)t} \big(Q - x^\circ(t)\big) \right]^+ . \end{aligned}$$

(iii) $P^* = x^\circ(0) = \psi\big[Q + \beta(V-Q)\big] + (1-\psi)\big[\beta \hat{C}_U(0) - (1-\beta)\hat{C}_E(0)\big]$ if and only if

$$Q \leq \inf_{t \geq 0} x^\circ(t) \quad \Longleftrightarrow \quad Q \leq \bar{Q} \equiv \inf_{t \geq 0} \frac{\lambda\beta V + r\, N(t)}{r + \lambda\beta}.$$

Under this condition the limiting equilibrium value at every date t is $x^\circ(t)$, so the exit option is never used. The condition $Q \leq x^\circ(0)$ alone is not sufficient. With constant costs, $P^* = \max\{Q, x^\circ\}$.

When exit does not bind, Europe's transfer averages the rupture value M of Proposition 1, with weight ψ, and the hold-up term N, with weight $1 - \psi$: Washington's average pain weighted by Europe's agenda power, minus Europe's own pain weighted by Washington's. The hold-up term needs no rupture risk. At $\lambda = 0$, Proposition 1 gives $[Q]^+$ whatever V is, while here $P^* = \max\{Q, \beta\hat{C}_U - (1-\beta)\hat{C}_E\}$ with constant costs. Hold-up is the fifth channel of Corollary 1(e).

The proposition prices both the sensitivity and the vulnerability of Keohane and Nye (1977), the cost of a cutoff before anyone adjusts and the cost that remains after adjustment. Let baseline disagreement flows be c_E^0 and c_U^0, which may include Washington's own cost of cutting a loyal Europe's AI access from date zero, and let x_0° be $x^\circ(0)$ under these baseline costs.

Corollary 5 (Interruption)**.** *Let Europe choose, at the start of each round and before the proposer is drawn, whether servicing of US-aligned fabs is interrupted. Interruption costs Europe a flow ℓ, and costs the United States a flow $(1-h)c_S$ once its buffer is exhausted, where h is the share of that servicing Washington can compel. The buffer is a stock of t_U units of interruption time, depleted only in rounds with interruption. Let $D = \beta(1-h)c_S e^{-(r+\lambda)t_U} - (1-\beta)\ell$. In the frequent-offer*

limit:

(i) If exit never binds without interruption (Proposition 6(iii) with the baseline costs), allowing interruption raises P^* by $(1-\psi)[D]^+$. Interruption raises P^* if and only if $\beta(1-h)c_S e^{-(r+\lambda)t_U} > (1-\beta)\ell$, that is, if and only if $t_U < \bar{t} \equiv \ln[\beta(1-h)c_S/((1-\beta)\ell)]/(r+\lambda)$ (with $\bar{t} = \infty$ if $\ell = 0 < \beta(1-h)c_S$).

(ii) With constant baseline costs and the exit option, $P^* = \max\{Q, x_0^\circ, x_0^\circ + (1-\psi)D\}$. Interruption raises P^* if and only if $(1-\psi)D > [Q - x_0^\circ]^+$. If exit binds without interruption, the condition of part (i) is necessary but not sufficient.

(iii) If exit never binds without interruption, as in part (i), the decision is time consistent. At stock b, Europe interrupts if and only if $\beta(1-h)c_S e^{-(r+\lambda)b} > (1-\beta)\ell$. Along an interruption path b falls, so the condition keeps holding. If it fails at $t = 0$, the stock never moves and it fails in every round.

In D, vulnerability is the comparison of $(1-h)c_S$ with ℓ, and sensitivity is the discount $e^{-(r+\lambda)t_U}$ that the buffer buys, which rupture risk and impatience make steep. With constant costs and binding exit, the condition at stock b in part (iii) becomes $(1-\psi)D(b) > [Q - x_0^\circ]^+$, also time consistent. Timing matters (Appendix B proves variants (a) and (c) where exit never binds). If the buffer runs out at a calendar date whatever Europe does, Europe never interrupts before that date (variant (a)). If Europe chooses after an offer is rejected and $\ell > 0$, the unique Markov equilibrium has no interruption, because pausing for one round saves Europe ℓ and changes nothing that determines the outcome (variant (b)). If it chooses after the proposer is drawn but before offers, it interrupts when it proposes, at no cost on the equilibrium path (variant (c)). Interruption therefore helps only under a rule fixed before offers and kept in force for the dispute. We characterize Markov perfect equilibria; when one side chooses its disagreement action every round, complete-information bargaining generally has many subgame-perfect equilibria (Fernandez and Glazer, 1991; Haller and Holden, 1990; Busch and Wen, 1995).

The Union's instruments have the wrong timing for such a rule. The Anti-Coercion Instrument allows urgent, immediately applicable acts only to suspend or amend response measures (Article 12(5)) (European Parliament and Council of the European Union, 2023), so escalation is slow and committee-gated while de-escalation is fast and in the Commission's hands. An interruption decided this way could be lifted after any US offer, the timing of variant (b), as the Union suspended each countermeasure it armed against Washington (Section 7.3). A credible interruption needs a rule that cannot be suspended after offers, which the Union's current instruments do not supply.

Jurisdiction limits hold-up too. ASML's US subsidiaries and their US staff are US persons (Bureau of Industry and Security, 2026; ASML Holding N.V., 2026b), with offices next to US customer fabs (ASML, 2026). No source we found names the employer of US field-service staff, but on this structure Washington could compel most servicing of US-sited fabs, so h is near one there. ASML's US-sited revenue in 2025 was EUR 4.09 billion (ASML Holding N.V., 2026a), so Europe's cost ℓ of an interruption confined to US soil would be at most about EUR 0.34 billion a month, and

Europe could withhold only new optics, modules and software. For a broad interruption covering the EUV fleet in Taiwan and Korea, h depends on Taipei and Seoul as well as on Washington, and Europe's own cost is larger, about EUR 1.9 billion a month of ASML revenue on a 2025 basis (ASML Holding N.V., 2026a). The flows that cannot be commandeered abroad include Zeiss optics and Trumpf drive lasers (ASML Holding N.V., 2026a; Varas et al., 2021), so on our reading Germany would have to take part in any credible interruption.

Other parameters are poorly measured. The buffer t_U runs from about one month to two years or more. The short end takes at face value an anonymously sourced report that without ASML's spare parts EUV tools "quickly stop working" (Bloomberg News, 2024); the long end allows for stockpiles and for parts and repair capacity seized locally. TSMC's 2025 filing implies US$3.9 to 4.8 billion a month of wafer revenue from EUV nodes and US-headquartered customers (Taiwan Semiconductor Manufacturing Company Limited, 2026). With a downstream multiplier of 5 to 12 (Varas et al., 2021) and a post-buffer output loss we assume lies between 10 and 70 percent, c_S for a broad interruption runs from about US$2 billion to US$40 billion a month, with a central value near US$13 billion. These are revenues at risk, not welfare losses. With $\beta = 1/2$ and central costs, interruption pays at $t_U = 0$ only if h is below about 0.85. With an illustrative $r + \lambda$ of 0.55 a year, $\bar{t}$ is about 26 months at $h = 0.5$ and about 15 months at $h = 0.7$. The answer turns more on h than on t_U, and h is the parameter with the least evidence behind it. Since $\bar{t}$ falls as h rises, ASML's plan to have 50 to 70 percent of service parts repaired locally (ASML Holding N.V., 2026a) erodes Europe's hold-up option.

Servicing is not Europe's only hold-up instrument. Denying base access, as Spain did for the Iran war (Section 2), raises $c_U(t)$, and Washington cannot compel access to another state's territory, so on our reading its h is near zero. One member state decides it, outside the Council, at a cost that includes the conditional part of US posture (Section 5.3). By variant (b), a denial improvised after a rejected offer would never be carried out.

Washington's dependence on ASML bites hardest here. In Proposition 1, V is the loss from China gaining the tools, and it reaches Europe's transfer only through agenda power under rupture risk and through Q. In Proposition 6 the relevant loss is c_S, what Washington's own fabs lose without ASML, and it raises Europe's transfer without any rupture risk. On US soil h near one leaves little of that loss for Europe to impose; for a broad interruption the answer turns on an h the record does not pin down (Appendix D applies this to the scenario's claim that Washington could outlast Europe).

8 The Record as a Plausibility Probe

The parameters of the model are not measured, and a negative exit premium can rationalize almost any act of compliance after the fact. Compliance also cannot separate coercion from shared interest, because successful coercion ends at the threat stage (Drezner, 2003). We derive implications that separate the model from its rivals, fix coding rules, and read the Dutch record and seven comparison

Table 2: Observable implications of the model and of rival accounts

Implication (result)	This model	Shared threat (large η)	Hierarchy	Autonomy–security trade-off	Cost minimization
Threshold (Corollary 1)	Holder is paid only if $Q > 0$, or if talks risk rupture and it has agenda power	No payment; holder aligns for its own reasons	No payment; compliance is owed	Protection is the payment	No prediction about payment
Linkage only with slack (Proposition 3(b))	Linkage against a hedge worth less to the patron than to the holder appears with slack and fades as $Q \to 0$; flow jurisdiction does not track Q	Little coercion of any kind	Commands need no threats	Protection is linked to autonomy throughout	Cheapest instrument at fixed cost; linkage shifts with Q only if its cost is defined to include rent
One budget across issue areas (Proposition 2(c), Corollary 2)	Demands on trade, defence and controls are substitutes within a period; the budget shrinks as the rival's offer rises	Issue areas independent	Authority bounded issue by issue by the social contract	One autonomy price for security, set within the alliance	Instruments priced separately at fixed cost
Counter-threats (Proposition 3(a))	Change the patron's demand mix and tax the patron; executed only to crowd out other demands	Holder absorbs them and keeps aligning	Patron shields its subordinate from third parties	No prediction	No prediction
Rival bids (Proposition 4(c)(i))	A stronger bid elicits a more conditional guarantee and no payment, while assurance is cheap	No response needed	Patron must provide more order to keep its authority	Better terms for the protégé	No prediction
Preventive control (Proposition 5, Corollary 4)	Control, or binding, is demanded while the holder is weak but expected to strengthen	No demand	More control where dependence is deeper, with no window	No prediction	Control demanded whenever cheapest, with no window

Note. Entries for the rival accounts are our derivations from Lake (2009) and Morrow (1991) and from the definition of a cost minimizer in the text, not claims their authors make about these cases. Q is the exit premium (Definition 1).

cases with them. None of this is a test (Section 8.5).

8.1 Observable implications

Table 2 sets six implications beside four rival accounts. Under shared threat perception the Netherlands aligns because it shares Washington's security concerns, as the cabinet says it does "in general terms" (Sjoerdsma, 2026, our translation); in the model that is a large intrinsic cost of exit η. In Lake (2009) compliance is owed under a social contract in which the patron provides order and protection. In Morrow (1991) the protégé trades autonomy for security. The fourth rival prices each demand, one at a time, at a fixed cost of the cheapest credible instrument. Drezner (1999) finds that allies rarely coerce one another but can win significant concessions when they do; his account says nothing about a rival's bid.

The threshold separates the model from hierarchy and the autonomy–security trade-off only through holders with a positive premium, which should be paid, as Kyrgyzstan was for Manas (Section 3). The third row discriminates most. Under cost minimization the instruments used on one demand do not change the price of another, while in our model a concession on tariffs leaves less to extract on servicing. The second row separates the two only if instrument costs are fixed, since a coercer that counts the rent of Proposition 3(b) as a cost of linkage predicts the same switch. The fifth row also holds against any coercer that prices demands one at a time, as does the result that an added demand can lower what a patron with small stakes obtains (Proposition 2(d)).

8.2 Coding transfers and threats

Reading the record through Proposition 1 requires a line between a payment and a threat never carried out. We draw it at Washington's ex post optimal provision to an aligned ally, the baseline $x_j(A)$ of each instrument (Section 4.3). A *transfer* exceeds that baseline and enters P^*. A *threat left unexercised* is a baseline benefit that Washington would withdraw after exit, or a sanction it could impose and does not. It enters the exit cost c as a conditional part d_j, however valuable it is to the holder.

Because the baseline cannot be observed, we code a benefit as baseline provision when it is extended to allies that took no comparable step, justified by the recipient's own capacity or alliance status, and revocable by rule. We code it as a transfer when it is created in response to the holder's step, withheld from comparable allies and framed as compensation. Supplement No. 4 meets all three baseline criteria: it covers 33 countries, rests on their governments' authority to control the items, and can be amended by rule (Section 2). We code it as a threat left unexercised, equivalently a reward for self-binding (Remark 2), and code the favourable treatment that the National Security Strategy offers aligning countries the same way (Section 5.3).

The instrument behind a US demand is *flow jurisdiction* if it reaches China-bound flows directly (de minimis and foreign direct product rules, entity listings, licences for US-origin items), *conditional linkage* if it threatens to withdraw something in another domain that an exited holder would lose anyway (posture, AI access, trade terms), and *unconditional* if it hurts the holder whatever it does. Chinese responses are counter-threats if announced before the holder concedes and executed if carried out.

Leverage in the model is the transfer, and on that measure the Netherlands has little to show almost by construction. The model also gives meaning to delay, to the scope of the holder's controls, to retained jurisdiction, which is self-binding with the key kept in Europe (Remark 2), and to tolerated hedges. A tolerated hedge is rationing (Corollary 2) if only linkage could stop it. It is a flow threat not yet credible if Washington could not stake the cost of blocking the flow itself (Proposition 3(b)).

8.3 The Dutch record and Nexperia

Alignment without a price. Since 2019 The Hague has aligned in steps, from 2023 each following a US move (Section 2), and under our rules it received no transfer. By Corollary 1(b), the record is consistent with a premium that has been negative throughout if talks carry no risk of rupture or Washington makes the offers. The absence of payment identifies less than it seems to. Because V is large for ASML, rupture risk combined with Dutch agenda power would imply a positive transfer unless $Q < -\beta V/(1-\beta)$ (Corollary 1(c)). And since P^* approaches $[Q]^+$ continuously as $\lambda \to 0$, a small payment cannot be told apart from none. Reading the record as $Q \leq 0$ therefore assumes that The Hague could not make demands whose rejection risked realignment, or that Washington's credible conditional threats took back what agenda power would have won (Remark 1).

The negative premium needs no security term. US flow jurisdiction over China-bound tools makes θ large, which shrinks both what China gains from exit and the relative gain it values, so ω is small unless China's cap binds (equation (2)). ASML's US entities and staff are US persons (Section 2), so the firm would not ship against a credible US penalty even if The Hague licensed exit (Remark 4). With $\kappa = 0$ these terms alone can make Q negative. We have found no Chinese offer conditioned on Dutch exit, and ASML's revenue in China is not one. The record is therefore consistent with the jurisdictional channel and leaves the security channel untested. We found no documented case that links US security commitments to ASML, although public documents cannot exclude private linkage: the January 2023 talks behind the Dutch and Japanese controls were led by the US national security adviser and also covered Ukraine (Durbin and Madhani, 2023).

Orders of magnitude. The terms of equation (3) can be bounded, not estimated, and we give orders of magnitude in euros a year. The conditional part of the AI instrument, d_{ai}, is Europe's loss from moving from aligned to exited access to US frontier AI. It runs from about EUR 4 billion if European gains from AI are limited by adoption, through 50 billion, to 130 billion if they are limited by frontier capability (Acemoglu, 2025; Filippucci et al., 2025; Epoch AI, 2026). The conditional part of the core guarantee, κS_k, is about 12.5 billion at $\kappa = 0.10$ and a stake of 125 billion, a value between costings of replacing US protection that leave out nuclear deterrence (International Institute for Strategic Studies, 2025; Burilkov and Wolff, 2025); for κ between 0.02 and 0.30 it runs from 2.5 to 37.5 billion. China's input lever over Europe runs from about 3 billion to 77 billion, the high end being the ECB's estimated loss from a sudden halt of critical-dependency imports from the China-led bloc, applied to EU output (Attinasi et al., 2025). The midpoint of 15 billion is geometric, not an estimate.

If Washington's AI threat is credible, then, the AI term is likelier than the core guarantee to decide the sign of Q. China's lever is not a term in Q, since China has punished specific acts and not alignment (Section 4.3). It bears on which concessions Washington can extract (Proposition 3) and on China's punishment r^c of a transfer of control (Corollary 3). China's offer ω has no good measure. And d_{ai} and κ are not independent in practice, since US protection has responded to allies' conduct on unrelated issues (Section 2).

Slack spent and unspent. The sales controls came in steps from 2023, while servicing in China continued at least into mid-2026 (Section 2). If the sales controls were backed by flow jurisdiction, as Table 3 codes them, they drew on no slack (Proposition 3(b)), and the late arrival of demands on servicing reflects the cost of the flow threat. Rationing would require that some steps were backed by conditional linkage, which we have not found. Until 2026 Washington had no credible jurisdictional block on servicing done by ASML staff in China, who are not, we think, US persons, and since Decrees 834 and 835 stopping it carries a Chinese counter-threat (Section 2). The MATCH Act would supply the missing stake. Timing does not separate coercion from shared interest either. A holder that shares Washington's view but would lose sales to Japanese rivals by moving alone would also wait for US rules that bind them, a coordination problem that Farrell and Newman (2019) identify when hubs lie in several jurisdictions.

Nexperia. The Nexperia dispute of 2025 is the one episode in which China executed a counter-threat against a Dutch step taken under US pressure. The Dutch minutes of a meeting with the State Department on 12 June 2025 concluded that it was "almost certain the CEO will have to be replaced to qualify for the exemption from the entity list" (Gerechtshof Amsterdam, Ondernemingskamer, 2025, paras. 2.19–2.20; our translation), a Dutch record of US views, not a formal condition. On 30 September, the day after the Affiliates Rule took effect (U.S. Department of Commerce, Bureau of Industry and Security, 2025b), the economy minister took control of Nexperia under the Goods Availability Act, and the Enterprise Chamber suspended its chief executive in early October (Minister van Economische Zaken, 2025; Gerechtshof Amsterdam, Ondernemingskamer, 2025). On 4 October China announced an export control aimed at Nexperia's Chinese plants (Minister van Economische Zaken, 2025). At Busan on 30 October Washington agreed to suspend the Affiliates Rule for a year from 10 November, and China to "take appropriate measures to ensure the resumption of trade from Nexperia's facilities in China" (Ministry of Foreign Affairs of the People's Republic of China, 2025; The White House, 2025a; U.S. Department of Commerce, Bureau of Industry and Security, 2025c). On 19 November the minister suspended his order, citing Chinese export licences for Nexperia chips. The Enterprise Chamber's separate measures stayed in force (Karremans, 2025).

We read a US-backed demand on the Netherlands met by an executed Chinese counter-threat. Under complete information, Proposition 3(a)(iii) allows execution on the path only when it crowds out another demand. That reading requires a Chinese threat known in The Hague and Washington before 30 September, a US demand that Washington shelved because the Nexperia concession had become expensive, and a Dutch government that conceded expecting to bear the cost. The alternative, outside the model, is that The Hague, and perhaps Washington, misjudged China's resolve. The reversal does not discriminate, since by then the rule behind the US demand had been suspended, and the public record establishes neither reading.

Europe bore the cost and was not compensated. By Proposition 3(a)(v) the counter-threat taxes the patron and not the holder only if Washington has a divisible sunk demand that China cannot reach. The relief came through a US concession to China at Busan, which we do not read as a

transfer to Europe. If China's ban, and not Washington's suspension, moved The Hague, compelling a firm is costly when China retaliates against the compulsion itself (Remark 4). The sequence does not tell us which.

Servicing, MATCH and Dutch politics. A statute is a stake. The MATCH Act would make a flow threat against servicing credible if $e_F \leq \bar{m}_U$, and such a threat leaves Q unchanged and need not be carried out (Proposition 3(b)). The Dutch answer was voice (Section 2), which is consistent both with jurisdiction alone keeping the premium negative and with perceived conditionality having risen after the targeted US moves of 2026. With $Q < 0$, voice and the hold-up of Section 7.4 are all a holder has.

Dutch politics shows which parts of that position would survive a change of government. On 13 October 2026 parliament adopted, 110 to 40, a motion co-signed by members of all three coalition parties that asks the cabinet to push for harmonised EU export control and for joint European action when third countries press a member state (Bamenga et al., 2026). That is support for self-binding through Europeanised controls at a low sovereignty cost σ^b, and against a dual key (Remark 2). On 22 January it had adopted, 103 to 46 and against the caretaker cabinet's advice, a motion to keep countermeasures, including the Anti-Coercion Instrument, in reserve over Greenland (Tweede Kamer der Staten-Generaal, 2026). By our count, within any right-leaning majority in the current house the first motion's supporters outnumber its opponents by 52 seats to 26, while the second's divide 40 to 38. Europeanised controls would survive a change of government, and countermeasures against Washington might not. Only 41 percent of Dutch respondents called the United States a reliable partner in spring 2026, against 88 percent in 2022 (Pew Research Center, 2026). No motion rejects MATCH, and Congress has not posted the stake: the bill has not moved since committee approval in April 2026 (Baumgartner, 2026b), and neither chamber's FY2027 defence authorization contains it.

8.4 Comparison cases

Tables 3 and 4 code four contemporary and three historical cases with the rules of Section 8.2. Taiwan is outside the scope of Section 9.1, and we include it for what it shows about control.

What the comparison shows. No holder was paid in cash for aligning on a chokepoint. The currency is status that BIS grants and withdraws by rule (Supplement No. 4, validated end-user status, annual licences), which we code as exit costs. Every holder in scope has, we think, a nonpositive premium, so the tables cannot test the threshold of Corollary 1, and Manas is the positive-premium contrast (Section 3). They speak to the instruments Washington uses and to where holders resist. Where money moved, it moved to the patron (the offsets) or to a third party, as when Washington offered Brazil, not Bonn, a fuel guarantee in 1977 (Tudda, ed, 2015).

Korea shows flow jurisdiction rebuilt as a renewable status. BIS moved the Samsung and SK hynix fabs in China from validated end-user status to annual licences that allow operation but not

Table 3: US demands on contemporary holders, coded through the model

Holder, period	US demand	Instrument threatened	Compensation	Rival bid or counter-threat	Outcome	Reading in the model
Netherlands (ASML), 2019–2026	End EUV, then advanced and older immersion DUV and metrology sales to China; from 2026, servicing	Flow jurisdiction: zero de minimis (2023), SME FDP rule (2024), MATCH Act (proposed)	None; Supplement No. 4 status, a threat left unexercised	No Chinese offer for exit found; China is ASML's largest market (revenue, not an offer); Nexperia ban executed (2025)	Aligned on sales with lags; kept the licensing pen; servicing in China continued into 2026	$Q \leq 0$ through jurisdiction (Remark 4); hedge tolerated while no flow threat was credible (Proposition 3(b)); security channel untested
Japan (Tokyo Electron), 2023–2026	Control advanced equipment; in 2024, restrict sales and servicing in China	Flow jurisdiction: FDP rule not used but not ruled out	None found, including for the 2025–26 coercion; Supplement No. 4 status	2024: "severe economic retaliation" threatened; 2025–26: 80 entities listed after a Taiwan statement, no toolmaker among them	23 equipment types licensed (2023), "not aligning"; servicing continued into 2026	Counter-threat raised the cost of the servicing concession (Proposition 3(a)); later punishment aimed at a security act
South Korea (Samsung, SK hynix fabs in China), 2025–2026	No expansion or upgrade of Chinese fabs	Flow jurisdiction: VEU status revoked (end-2025)	None; annual licences to operate (2026); November 2025 package silent on controls	None documented	Accepted; fabs operate under annual licence	Hedge policed by flow jurisdiction, rebuilt as a renewable status (Proposition 3(b))
Germany (Zeiss, Trumpf), 2023–2024	Join equipment controls; "primary target is Zeiss"	None documented	None	None documented	Reportedly declined; slower licensing of laser exports to China	Demand of low marginal value while Dutch licences govern the tools that carry Zeiss optics (Corollary 2)
Taiwan (TSMC) and US-sited assets, 2020–2026	Relocate leading-edge capacity; "50-50" floated (2025)	Tariffs; Section 232 preference for firms investing in the US	CHIPS awards; tariff preference tied to US investment	China a threat, not a bidder	TSMC's US commitment from about $12bn to $165bn; US control rights only over US-sited assets	Control sought where Δ_c is high (Corollary 3(c)); outside scope

Sources by row. Netherlands: Section 2 and Bureau of Industry and Security (2024); ASML Holding N.V. (2026a); Zakaria (2026); Baumgartner (2026a); Minister van Economische Zaken (2025); Karremans (2025). Japan: Allen (2023); Scanlan-Dyas et al. (2023); Bloomberg (2024); House of Representatives of Japan (2025); Ministry of Commerce of the People's Republic of China (2026a,b); Teer (2026); Zakaria (2026). South Korea: U.S. Department of Commerce, Bureau of Industry and Security (2025d); Ha et al. (2025); SK hynix Inc. (2026); Reuters (2025); U.S. Census Bureau (2026); The White House (2025b). Germany: Barkin (2024); Burger and Hübner (2023); Kahle (2023). Taiwan and US-sited assets: Younger (2025); Focus Taiwan (CNA); Taiwan Semiconductor Manufacturing Company Limited (2020, 2025); The President of the United States (2026); Taiwan Semiconductor Manufacturing Company Limited (2026); MP Materials Corp. (2025); United States Steel Corporation (2025); Intel Corporation (2025). "None documented" and "None found" mean that the sources we consulted record none; they cannot exclude private linkage or compensation.

Table 4: Historical cases, coded through the model

Holder, period	US demand	Instrument threatened	Compensation	Rival bid or counter-threat	Outcome	Reading in the model
West Germany, 1961–1967 (offsets)	Offset the foreign-exchange cost of US troops	Conditional linkage: troop levels, explicit in 1966	Holder paid: arms purchases, $500m in US securities, gold pledge	German option to convert dollars into gold, given up	Redeployment of up to 35,000 US personnel agreed (1967); Erhard fell (1966)	Linkage extracting sunk payments; repeated renegotiation (Proposition 2(d))
United Kingdom, 1957 (CoCom China differential)	Keep stricter controls on China than on the Soviet bloc	None; Washington chose no sanctions	None	China trade; 12 of 16 members for abolition	Differential abolished; US kept its own embargo	Secondary demand left without linkage (Corollary 2)
West Germany, 1975–1977 (Brazil nuclear deal)	Defer or stop enrichment and reprocessing transfers	High-level pressure; no explicit linkage, though Bonn feared for the US commitment	Offered to Brazil (fuel guarantee), not to Bonn	Brazil as buyer; Bonn cited its contract and the NPT	Contract kept; informal restraint on future transfers by 1978	Concession on the forward margin; sunk deal kept

Sources by row. Offsets: Gerakas et al., eds (1995); Sampson, ed (1995); Miller, ed (1999); Deutsche Bundesbank (n.d.). United Kingdom: McMahon et al., eds (1989); House of Commons (1957). Brazil deal: Berndt et al., eds (2015); Rasmussen, ed (2014); Tudda, ed (2015).

expansion or upgrades (Ha et al., 2025; SK hynix Inc., 2026; Reuters, 2025), and the trade and security package of November 2025 did not mention export controls (The White House, 2025b). Japan shows Chinese punishment aimed at a security act. After Prime Minister Takaichi told the Diet on 7 November 2025 that a Taiwan contingency could become a survival-threatening situation for Japan (House of Representatives of Japan, 2025), China listed eighty Japanese entities, none of them a maker of chipmaking equipment or photoresists (Ministry of Commerce of the People's Republic of China, 2026a,b). Beijing spared the holders whose inputs it needs. We found no US compensation (Teer, 2026), which Corollary 1 leads us to expect while Japan's premium stays negative. Coordination among holders runs through Washington: Japan and South Korea have signed Pax Silica, the US-led initiative on AI chip supply chains and export controls, and in June 2026 the EU was reported to be set to join (Stanciuc, 2026).

Taiwan shows what a patron does when it can move production. TSMC's US commitment rose from about USD 12 billion in 2020 to USD 165 billion in 2025 (Taiwan Semiconductor Manufacturing Company Limited, 2020, 2025), and the control rights Washington has taken all attach to assets on US soil (Section 6.2). We found no US demand for equity in a foreign chipmaking-equipment firm. A demand for a controlling vote in ASML presupposes that relocating EUV production and servicing is infeasible.

Resistance clusters on the installed base and on signed deals: Tokyo and The Hague kept servicing Chinese fabs, and Bonn kept its contract with Brazil. In the model these concessions cost the holder much relative to the remaining bite, but a cost minimizer facing expensive enforcement would also tolerate them.

Explicit links across issues appear in Johnson's tie between troops and offsets in 1966, in the

posture cut in Germany and the threat to cut off trade with Spain after it denied base access in 2026 (Martinez, 2026; Roque, 2026; O'Mullony, 2026), and in the BIS rule of July 2026 that cited the UAE's role in Operation Epic Fury when it granted licence-free chip access (U.S. Department of Commerce, Bureau of Industry and Security, 2026). None concerns a European holder's chip controls.

Where the comparison strains the model. The offset payments were sunk concessions, which escape the slack cap (Proposition 2(b)), and renegotiation recurred from 1961 to 1967 (Gerakas et al., eds, 1995; Miller, ed, 1999; Sampson, ed, 1995), as Proposition 2(d) leads us to expect of a patron whose stakes are small relative to its later demands. But a complete-information patron does not both collect and cut, and in 1967 Washington agreed to redeploy up to 35,000 personnel. If the troops were above Washington's ex post optimum, as balance-of-payments pressure and the Mansfield resolution suggest, keeping them was a transfer and the offsets were its price (Gavin, 2004; Zimmermann, 2002). If not, the cut is unexplained, and FRUS does not let us choose. In 1957 Washington declined to back its China differential with linkage, as Corollary 2 implies for a demand worth little, but twelve of sixteen CoCom members backed abolition (McMahon et al., eds, 1989; Mastanduno, 1992), a collective exit our single-holder model does not represent. In the Brazil case Bonn feared for a US commitment that Washington did not tie to its pressure (Rasmussen, ed, 2014), so conditionality existed only in the holder's beliefs.

8.5 Designs that could test the model

An event dataset. The unit would be a US demand on a holder of a chipmaking-equipment chokepoint (the Netherlands, Japan, South Korea, Germany) from 2019 to 2026, coded from gazettes, parliamentary records and filings for the instrument threatened (Section 8.2), the response, any compensation and any Chinese response. It should be built around the cross-issue budget. A concession in one issue area should lower what Washington extracts in another within the same period (Proposition 2(c), Corollary 2), which no coercer that prices demands one at a time implies. Linkage against hedges should fade as the premium rises toward zero, while flow jurisdiction does not track it (Proposition 3(b)).

An elite survey experiment. Conditionality is a property of US behaviour, so the respondents should be US foreign-policy elites. Vignettes would vary whether an ally holding a chokepoint complied with US controls, refused, or complied on sales while servicing Chinese fabs. The gap in support for defending the ally and keeping forces there estimates κ and d_p. Further arms would vary a public presidential stake and whether Washington uses bases there (Proposition 4).

Process tracing. Requests under the Dutch Open Government Act (Woo) for records of export-control talks with US counterparts since 2019 would show whether security commitments, posture or the F-35 programme were raised. They would also date Dutch drafting against US demands,

which separates coercion from shared interest, and show what The Hague expected from China before 30 September 2025.

The Busan expiry. If the suspensions agreed at Busan lapse (Section 2), the rule behind the Nexperia demand and China's extraterritorial rare-earth controls return together (U.S. Department of Commerce, Bureau of Industry and Security, 2025c; Ministry of Commerce of the People's Republic of China and General Administration of Customs of the People's Republic of China, 2025b). Proposition 3(a) then leads us to expect Washington to shift its demands on The Hague toward demands China cannot reach or flow jurisdiction it exercises itself. A second executed counter-threat against a Dutch step would be hard to square with misjudgment, which learning should correct.

9 Implications and Scope

9.1 Scope

Four conditions define the class of cases: (i) *security dependence*, protection the holder cannot replace on the relevant horizon ($S_k > 0$); (ii) *a partly conditional guarantee*, some of which the patron can stake to withhold after exit, at an assurance cost ($\kappa S_k + d_p > 0$; Proposition 4); (iii) *a rival that bids* ($\omega > 0$); and (iv) *a nested chokepoint*, partly inside the patron's jurisdiction while the holder's licence still decides exit ($0 < \theta < 1$). The Netherlands, Japan and South Korea meet the first, second and fourth. The third holds only weakly for all three: China spared Japan's equipment makers while punishing a security statement (Section 8.4), but we found no Chinese offer for any holder's exit, one reason we think ω is small. Taiwan (threatened by China, not courted) and the Gulf partners (no treaty guarantee) fall outside. The model also assumes complete information about conditionality, unitary great powers and one rival (Section 9.3).

9.2 Implications for Europe

Each implication holds in a region of parameter space. We place Europe at a nonpositive premium with little rupture risk: The Hague has received revocable statuses, which we do not code as transfers (Section 8.2), and has made no demand whose rejection risked realignment (Section 8.3). That placement rests on jurisdiction. If Washington's AI threat is credible, the AI instrument is likelier than the guarantee to decide the sign of Q, and the guarantee's conditionality is untested (Section 8.3).

Reduce the conditional part of dependence. Against the NATO target of 5 percent of GDP by 2035 (North Atlantic Treaty Organization, 2025), the model counts a euro only insofar as it replaces something Washington could withhold after exit: substitutes for posture and enablers more than duplicates of the statutory core (Section 4.3), and AI capacity Washington cannot withhold, which lowers d_{ai}. While $Q < 0$ a gain in premium removes slack one for one and brings no transfer (Corollary 2). Where assurance is cheap at the margin, Washington answers it with a more

conditional guarantee and pays nothing (Proposition 4(c)(i)). Europe gains only where assurance is already costly, slack already spent or conditionality at its ceiling. US statements tying protection to European conduct that multiply as European capability grows would place Europe in the first regime. For Washington, conditionality pays only if slack is spent, and untargeted drawdowns and rationing of loyal allies shrink it (Section 5.3).

Reduce the nesting. Moving US-origin content and US-person servicing out of US jurisdiction lowers θ, which raises ω only while China's cap does not bind (equation (2)). It also raises V, which reaches the transfer directly only through agenda power under rupture risk (Corollary 1(d)), a channel we think closed to Europe. The surer gain is a lower share h of servicing that Washington can compel (Section 7.4).

Interrupt servicing only under a rule that cannot be suspended. Interruption raises Europe's transfer only if the US buffer is short, only for servicing Washington cannot compel, and only under a rule fixed before offers and kept in force for the dispute (Corollary 5). On US soil we think h is near one (Section 7.4). A credible interruption needs a rule that cannot be suspended after offers, and the Union's current instruments do not supply one: escalation under the Anti-Coercion Instrument is slow and de-escalation fast (Section 7.4), and each countermeasure the Union armed against Washington was suspended once Washington moved (Section 7.3). Under present institutions hold-up adds nothing to Europe's transfer, and Washington need not price a threat to interrupt, however long or short its buffer. A credible option would need a legal basis, adopted before any dispute, that no EU body could suspend on a US offer, and German participation, since Zeiss optics and Trumpf lasers cannot be commandeered abroad (Section 7.4).

Expect a preventive demand. Washington demands control when the premium is negative but expected to turn positive and its credible bite exceeds $\sigma + r^c$, or when it values control for its own sake (Corollary 3(a), Corollary 4). Autonomy investments Europe cannot commit to forgo strengthen the motive and permit a purchase with no bite (Corollary 3(b)). Rearmament raises the premium's drift and Chinese indigenization lowers it, and we cannot tell which dominates (Section 6.3), so we cannot say whether Europe is in a window. Demands for relocation, the most enforceable form (Corollary 3(c)), board seats or a licensing veto that follow visible European investment would show that it is.

Bind, or keep the key. Binding pays only when it pre-empts a demand that would otherwise come, and it pre-empts only if Washington values control little for its own sake; partial binding can invite a demand by raising slack (Remark 2). Delegated Regulation 2025/2003 (European Commission, 2025a) bound Europe partway (Section 6.4). Whether that weakened the US motive turns on v_c. US demands shifting from alignment toward ownership or a licensing veto would indicate a high v_c, and binding would then have raised the bite and pre-empted nothing.

Chinese counter-threats. De-risking, the aim of the Critical Raw Materials Act (European Parliament and Council of the European Union, 2024), removes counter-threats. If Washington has a divisible sunk demand China cannot reach, Europe's loss stays at $B(J^c)$ and Washington weakly gains (Proposition 3(a)). We read the spending target and purchases of US equipment as such demands (Section 5.1), so for Europe de-risking pays mainly where China retaliates against Europe's own acts, as in Nexperia (Section 8.3), and it weakens China's protection of European ownership (Corollary 3(d)).

9.3 Limitations

Complete information is the strongest assumption. If κ were private, Washington would gain from overstating it, European beliefs would set Q, and the 2025 National Security Strategy would be partly cheap talk. Refusals and executed counter-threats could then come from misjudgment, the reading of Nexperia outside the model (Section 8.3), and conditionality could exist only in the holder's beliefs, as Bonn's fears in 1975–77 suggest (Section 8.4). Because the autonomy investment is observable, the information problem of Debs and Monteiro (2014) and Bas and Coe (2016) is absent. The assurance cost H is a reduced form; if it is Washington's reputation with Tokyo and Seoul, conditionality toward Europe depends on what they observe. The great powers are unitary, although Congress can set stakes the executive would not, as the MATCH Act's 150-day deadline would (Section 2). With one rival and holders bargaining alone, the model is silent on whether a coalition of the Netherlands, Japan and South Korea would help.

10 Conclusion

A chokepoint holder's exit premium is a floor on what it obtains from its patron. Unless talks risk rupture while the holder has agenda power, or the holder can impose costs on the patron while the two bargain, its leverage is the positive part of that premium, and a holder with a negative premium gets nothing for its chokepoint however much its patron needs it. The patron's demands are bounded by the holder's slack. Under rupture risk Washington's dependence on ASML raises both what Europe obtains in bargaining and the slack that Washington's conditional threats can claw back (Remark 1).

The result adds a condition to the theory of weaponized interdependence. Farrell and Newman (2019) require that a state have jurisdiction over a hub and the domestic institutions to use it. When the hub's holder bargains with a state that depends on the hub, a third condition applies: the holder's exit premium must be positive, unless talks carry a risk of rupture while the holder can make proposals (Corollary 1(c)) or the holder can hurt the dependent state while they bargain (Proposition 6). The Netherlands licenses ASML's exports, so it has the institutions. It has jurisdiction only in part, because Washington reaches the firm through US-origin inputs and US persons (Section 2). Whether it meets the third condition turns first on that shared jurisdiction, which on our reading can keep the premium negative without any security term, then

on China's credible promises and, if Washington's threat to withhold frontier AI is credible, on the AI instrument, which on our orders of magnitude weighs more than the conditional part of the guarantee (Section 8.3). We think the substitutes for a positive premium are weak. We see no sign in the record that The Hague can make demands whose rejection would risk realignment, and an interruption of servicing would help only under a rule that could not be suspended after a US offer, which on our reading the Union's current instruments do not supply (Corollary 5).

The record so far is consistent with the jurisdictional half of the argument. The Hague's one benefit, a place on Supplement No. 4, is a status that BIS can revoke by amending a rule (Bureau of Industry and Security, 2024). What The Hague opposed in May 2026 was the extraterritorial reach of the MATCH Act, on the ground that "every country is responsible for its own export control legislation" (Sjoerdsma, 2026, our translation).

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A Supplementary Results

This appendix states formally the results that the main text describes informally: the breakdown of a preventive ultimatum under private sovereignty costs and the bite of a multi-period cut (Section 6), and the gatekeeper, firm and Council remarks with the population arithmetic (Section 7). Proofs are in Appendix B.

A.1 Cuts and breakdown

Proposition 5 allows one ultimatum and a cut that lasts one period. A threat of permanent exclusion withdraws instruments at many dates, and Corollary 4 lets Washington issue ultimatums at several dates. The lemma below gives the bite and the cost of such a cut. Its two scope conditions push in opposite directions. If a cut drives Europe out at some date, Europe's loss exceeds the formula. If Europe responds to a cut it learns of in advance, for instance with the autonomy investment of Section 6.1, the loss falls short of it.

Lemma 2 (Spent and self-limiting cuts)**.** *Let dates $s = 1, \dots, N$ have premia Q_s, slack $\Lambda_s = [-Q_s]^+$ and discount factor δ. A cut over a set S of dates withdraws at each $s \in S$, before the date-s contest, a bundle with unconditional part u_s, conditional part d_s and direct ex post cost $g_s \geq 0$ to the United States. Suppose that at every $s \in S$ the United States still retains Europe after the cut ($Q_s + d_s < V_s$ and $Q_s + d_s \leq \bar{m}_U$), and that the cut does not change the premium at any other date. Then:*

(i) the bite of the cut, Europe's loss net of the rents the United States must then pay, is $\sum_{s \in S} \delta^{s-1}(u_s + \min\{d_s, \Lambda_s\})$;

(ii) the cost to the United States of carrying it out is $\sum_{s \in S} \delta^{s-1} e_s$ with $e_s = g_s + [Q_s + d_s]^+ - [Q_s]^+$; with a separate stake at each date, the date-s withdrawal is carried out if and only if $e_s \leq m_s \leq \bar{m}_U$;

(iii) extending a cut into a date with $Q_s \geq 0$ adds $\delta^{s-1} u_s$ to its bite and $\delta^{s-1}(g_s + d_s)$ to its cost to the United States. A purely conditional instrument ($u_s = 0$) extended into such a date adds nothing to the bite.

Part (iii) governs threats of permanent exclusion. Whether such a threat is credible depends on how much Washington can stake, but a longer cut adds no bite once Europe's premium is positive. At those dates a conditional cut hands Europe exit as its alternative, and Washington pays the difference as rent.

Under complete information an ultimatum is accepted or never issued (Proposition 5). Breakdown, a refusal followed by the executed threat, requires that Washington be uncertain about what a transfer would cost Europe. The proposition below lets the sovereignty cost σ be Europe's private information and lets Washington choose between a threat-backed ultimatum and an offer to buy control without staking the threat. We keep the other assumptions of Proposition 5 and, to isolate the screening problem, set $\Delta_c = \infty$, rule out the autonomy investment, and assume that Europe's

period-1 premium is negative and that the threat is credible. We write F' for the density of σ and Π_U for Washington's objective, to keep them apart from the firm penalty f of equation (3) and the objective of Proposition 4.

Proposition 7 (Breakdown)**.** *Assume the conditions of Proposition 5 with $\Delta_c = \infty$, no investment, $Q_1 < 0$ and $e \le \bar{m}_U$, and let σ be Europe's private information, distributed with CDF F on $[0, \bar{\sigma}]$, density $F' > 0$ on $(0, \bar{\sigma})$, and F log-concave. Let $R = \delta [Q_2]^+$, $\tilde{b} = B_1 - r^c$ and $\Pi_U(s; A, x) = F(s)(A - s) - (1 - F(s))x$. For $A + x > 0$ let $s^*(A, x)$ be the unique solution in $(0, \bar{\sigma})$ of $A + x - s = F(s)/F'(s)$ if one exists and $\bar{\sigma}$ otherwise; let $s^*(A, x) = 0$ if $A + x \le 0$.*

(a) Threat-backed ultimatum. *With payment T, Europe accepts if and only if $\sigma \le T + \tilde{b} - R$. The optimal cutoff for the United States is $\hat{s} = \max\{\tilde{b} - R, \min\{\bar{\sigma}, s^*\}\}$ with $s^* = s^*(\tilde{b} + v_c, e)$, its payment is $\hat{s} - \tilde{b} + R \ge 0$, and its gain is $\Pi_B = \Pi_U(\hat{s}; \tilde{b} + v_c, e) - \iota$. Conditional on this ultimatum, Europe refuses and the threat is carried out with probability $P_b = 1 - F(\hat{s})$. For uniform F, $s^* = \min\{\bar{\sigma}, (\tilde{b} + v_c + e)/2\}$.*

(b) Purchase offer. *If the United States instead offers to buy control without staking the threat, Europe accepts if and only if $\sigma \le T - r^c - R$, the optimal cutoff is $s_P = s^*(v_c - r^c, 0)$, and the gain is $\Pi_P = \Pi_U(s_P; v_c - r^c, 0) - \iota$. If $v_c \le r^c$, then $\Pi_P \le -\iota$.*

(c) The United States makes an offer if and only if $\max\{\Pi_B, \Pi_P\} > 0$ and backs it with the threat if and only if $\Pi_B \ge \Pi_P$. If $v_c \le r^c$ (in particular if $v_c = 0$), it issues if and only if $F(\hat{s})(\tilde{b} + v_c - \hat{s}) - (1 - F(\hat{s}))e > \iota$, and every offer is threat-backed.

(d) Across parameter values at which the United States issues a threat-backed ultimatum, P_b is nonincreasing in B_1, e and v_c, nondecreasing in r^c and R, and independent of ι. These comparative statics hold for every continuous F (not necessarily log-concave) when $\hat{s}$ is the largest, or the smallest, maximizer.

(e) If $v_c > r^c$ and the United States chooses between (a) and (b), then across parameter values at which it makes an offer the refusal probability is nonincreasing in B_1 and v_c, and the breakdown probability (refusal followed by the executed threat) is nonincreasing in e and nondecreasing in R. The refusal probability is nondecreasing in r^c on any range where the threat-backed payment is positive. The other directions can fail: the refusal probability can rise with e and fall with R, and the breakdown probability can rise with B_1 and v_c and fall with r^c.

Given a threat-backed ultimatum, breakdown is more likely when China's punishment of a transfer is credible and large, when Europe's future premium is high, and when the threat is cheap for Washington to carry out. The last is the risk-return trade-off of Fearon (1995) and Powell (1999): a patron that loses little from a refusal makes a tougher offer and is refused more often. A larger bite makes refusal costlier for Europe, and a higher value of control makes Washington willing to pay more, so both lower the chance of refusal. When $v_c > r^c$, a patron whose threat is expensive to carry out may prefer to offer money alone, since a refused purchase offer costs it nothing, and a rise in the bite can then move it to the threat and raise the probability of breakdown (part (e)). A

credible Chinese punishment is now carried out on the path with positive probability, because the types that accept are punished.

The result rests on a one-shot ultimatum, which is a commitment assumption. A patron that could not commit to walk away after a refusal, and could renew its offer quickly, would compete with its own later offers. With one-sided private information and offers made by the uninformed side, that side loses most of its power to screen as the time between offers shrinks (Fudenberg et al., 1985; Gul et al., 1986). We expect such a patron to extract less and delay to replace some of the refusals that Proposition 7 counts as breakdown. Each refusal would also execute a cut, which brings the repeated game closer to a war of attrition; we do not solve it. The probabilities in Proposition 7 are those of a patron that can commit to one public demand, a commitment that a staked promise to make no further offer could supply (Section 5.1).

A.2 Who decides

Section 7 separates the three European deciders and states the results informally. This subsection states them formally and gives the Council arithmetic. Proofs are in Appendix B and the enumeration checks in Appendix C.

The gatekeeper. The Hague licenses exit toward China, and $Q_{NL} = \omega_{NL} - c_{NL}$ is its own exit premium: the value to the Netherlands of China's credible offer, net of the credible costs of exit that fall on the Netherlands. By Lemma 1, Washington keeps the Netherlands aligned at price $[Q_{NL}]^+$ when $Q_{NL} < V$ and $Q_{NL} \leq \bar{m}_U$. The remark asks what a benefit aimed at the gatekeeper alone does to that price.

Remark 3 (Gatekeeper). Let $x_h \geq 0$ be the value to the Netherlands of a benefit that the United States provides to a loyal Netherlands in the baseline and credibly withdraws if the Netherlands licenses exit. Then the gatekeeper's premium becomes $Q'_{NL} = Q_{NL} - x_h$. If $Q'_{NL} < V$ and $Q'_{NL} \leq \bar{m}_U$, the United States' payment falls from $[Q_{NL}]^+$ to $[Q_{NL} - x_h]^+$, that is, by $\min\{x_h, [Q_{NL}]^+\}$. The gatekeeper's slack rises from $[-Q_{NL}]^+$ to $[x_h - Q_{NL}]^+$, that is, by $\left[x_h - [Q_{NL}]^+\right]^+$. If $Q_{NL} > \min\{V, \bar{m}_U\}$, the benefit restores alignment if and only if $Q_{NL} - x_h \leq \bar{m}_U$ (and $Q_{NL} - x_h < V$). A benefit paid as a promised increment above the baseline lowers no payment, since the total outlay is then $\max\{x_h, Q_{NL}\} \geq [Q_{NL}]^+$.

The firm. A licence permits a shipment and orders nothing. The remark adds the firm's choice to the exit decision, with the US penalty discounted by its credibility and the two penalties for complying with US measures that China and the Union can impose.

Remark 4 (Firm between jurisdictions). Suppose the Netherlands licenses exit. The firm earns y from its Chinese business and faces the credible US penalty $f^F = \chi_f f$ if it ships. If it complies with US measures instead, it faces a Chinese penalty $p_C \geq 0$ for that compliance and a penalty $p_B \geq 0$ under the EU Blocking Statute, which is positive only if the US measures are listed in the statute's annex. The firm ships if and only if $y + p_C + p_B \geq f^F$ (ties ship). If it would not ship, exit requires

the state to compel it at cost $\zeta \geq 0$, where ζ excludes the firm's own net loss (already counted in ω and c through y and d_f). Europe's effective exit premium is $Q - \zeta\,\mathbf{1}\{y + p_C + p_B < f^F\}$, and Europe compels the firm only if $Q - \zeta > 0$.

The Council. Let q_i be member i's net gain from adopting an anti-US countermeasure. A qualified majority needs at least 15 of 27 members representing at least 65 percent of the population, and abstention counts against. Under the Anti-Coercion Instrument the Council first determines by qualified majority that coercion exists, and response measures then pass the examination and, if needed, the appeal committee (European Parliament and Council of the European Union, 2023, 2011). A delegated act removing the United States from EU001 stands unless the European Parliament or a reinforced qualified majority of the Council (20 members representing 65 percent of the population) objects (European Parliament and Council of the European Union, 2021; European Union, 2016).

Remark 5 (Blocking minorities). Let $\mathcal{B}$ be the family of coalitions that block the Council's determination at the first stage: at least 13 members, or at least four members with more than 35 percent of the population. With enforceable vote-contingent payments and no rival bidder, the United States defeats the determination at cost $\Phi(0) = \min_{J \in \mathcal{B}} \sum_{i \in J} [q_i]^+$. Members with $q_i \leq 0$ join any blocking coalition at zero cost, so $\Phi(0)$ is the cheapest completion of $F = \{i : q_i \leq 0\}$ into a blocking coalition, and $\Phi(0) = 0$ if and only if $F \in \mathcal{B}$. The six frontline states (15.09 percent of the population) do not block. Thirteen eastern and Nordic members (26.77 percent) do, by number. Defeating a response measure in the appeal committee needs a qualified majority against, and defeating a delegated act by Council objection needs a reinforced qualified majority against. Each such coalition is in $\mathcal{B}$, so it costs at least $\Phi(0)$.

The cost function is the one-voter logic of Lemma 1 applied to a weighted body, as in Groseclose and Snyder (1996). We assume no one bids for adoption; with a counter-bidder that moves last, Washington would have to buy more than a minimal blocking coalition (Groseclose and Snyder, 1996; Banks, 2000). Because every later stage of either route requires a coalition at least as large as a first-stage block, a Washington that wants to stop a countermeasure buys its blocking minority before the determination. Table 5 gives the arithmetic for the coalitions that matter.

The frontline core falls short on both counts, and a population block without Germany, France, Italy and Spain needs at least twelve members. If all thirteen eastern and Nordic members in the fourth row had $q_i \leq 0$, Washington would defeat a determination for free. That group is not a fixed bloc: Hungary changed government in May 2026, though Slovenia or Greece can replace it and the group still has thirteen members.

Informal norms and procedural defaults pull the outcome away from this arithmetic, in opposite directions. Council practice is consensual. About four in five legislative acts passed with no recorded negative vote or abstention in 1994–2002 (Heisenberg, 2005); Häge (2013) models that consensus as a by-product of negotiators organizing around blocking minorities before any vote, which is our reason for keeping the formal threshold as the primitive; and recorded votes understate opposition, because

Table 5: Blocking arithmetic in the Council, 2026 population weights

Coalition	Members	Population (%)	Blocks determination	Keeps delegated act in force
Frontline core: PL, LT, LV, EE, FI, RO	6	15.09	no	no
Core plus SE, DK	8	18.75	no	yes (members)
Bucharest Nine plus FI, SE, DK	12	25.91	no	yes
Core plus SE, DK, CZ, SK, HU, BG, HR	13	26.77	yes (members)	yes
the same with SI for HU	13	25.13	yes (members)	yes
the same with SI and GR for HU and HR	13	26.57	yes (members)	yes
Eight smallest members	8	3.18	no	yes (members)
DE, FR, IT	3	46.84	no	yes (population)

Note: A coalition blocks the determination (first stage of the Anti-Coercion Instrument) if it has at least 13 members, or at least four members with more than 35 percent of the population. It keeps a delegated act removing the United States from EU001 in force if it has at least 8 members or more than 35 percent. No coalition in the table can defeat a response measure in the appeal committee, which takes a qualified majority against. Population shares from Council Decision 2025/2427 (Council of the European Union, 2025); blocking status is our computation from the treaty rules, checked by enumeration (Appendix C).

a vote would disclose who lost (Novak, 2013). These data predate the Lisbon voting rules, so we treat them as indicative. Reversed defaults pull the other way. On 4 October 2024 the trade-defence committee gave no opinion on countervailing duties on battery electric vehicles from China, with ten member states in favour, five against and twelve abstaining, and the Commission adopted the duties on 29 October (Agence France-Presse, 2024; European Commission, 2024; European Parliament and Council of the European Union, 2011). The Anti-Coercion Instrument's no-opinion clause reverses that default at the committee stage, so the vote shows only what the Commission does when a procedure puts the burden on opponents, as the instrument's appeal stage and the delegated-act route both do. The Commission's commitment not to go against a predominant position among member states (Section 7.3) bears at every stage.

B Proofs

This appendix proves the results in the order in which the main text discusses them. The results stated formally in Appendix A are proved at the point where the main text describes them. Where a statement leaves a modelling point open, the proof begins by fixing it. Symbols introduced inside a proof are local to that proof. Throughout, $[x]^+ = \max\{x, 0\}$ and $\mathbf{1}\{\cdot\}$ is the indicator function. Appendix C describes the brute-force check of each result.

B.1 Proof of Lemma 1

Assumptions. We fix four points. (1) A winning Chinese promise of τ_C costs China τ_C, Europe values it at $\lambda_C \tau_C$ with $\lambda_C \in (0, 1]$, China values Europe's exit at W_C, and a losing bid gives China zero. This is the reading under which $\omega = \lambda_C \min\{W_C, \bar{m}_C\}$. If λ_C were instead the probability that China honours a promise, and China paid only when it did, China would bid up to $\min\{W_C/\lambda_C, \bar{m}_C\}$ and its credible offer would be $\min\{W_C, \lambda_C \bar{m}_C\}$. (2) Each announced conditional action carries its own stake of at most $\bar{m}_U$, so a promise and a threat are separate commitments (Section 4.2). If $\bar{m}_U$ were a budget shared by the bid and the threats, the two would compete and the lemma would change. (3) Washington's ex post payoff from its instruments does not depend on whether it has paid its promise, and payoffs are additive in transfers and instruments. (4) An indifferent Europe aligns, and a power that is indifferent between paying a staked promise and forfeiting the stake pays.

Proof of Lemma 1. (i) Stakes on threats are attached before Europe moves and are separate from stakes on promises. Once Europe's alignment a is fixed, Washington sets each instrument to maximize its ex post payoff given these stakes, and by assumption (3) this choice does not depend on whether it pays a promise. So it provides $x_j(a)$, and Europe's payoff before transfers is $\sum_j v_j(x_j(a)) - \eta\, \mathbf{1}\{a = X\}$. The difference between $a = A$ and $a = X$ is $\sum_j d_j + \eta = c$.

(ii) *Credibility.* At the payment stage, paying a promise of τ costs τ and has no ex post value, while not paying forfeits the stake $m \le \bar{m}_i$. The promiser pays if and only if $\tau \le m$, so the largest promise that is honoured is $\bar{m}_i$. Europe anticipates this and assigns zero value to larger promises.

The clock auction. Measure the clock $z \ge 0$ in units of Europe's payoff relative to the aligned baseline. While the United States is active at z it stands behind the promise $\tau_U = z$. While China is active at z it stands behind $\tau_C = (z + c)/\lambda_C$, which gives Europe $\lambda_C \tau_C - c = z$ if it exits. A bidder that drops cannot re-enter. When one bidder drops at z (or never enters, in which case $z = 0$), the other wins at z. If both drop at the same level, Europe is indifferent and aligns. A strategy is a drop level z_i.

Washington's drop level. Washington's maximal credible bid is $\bar{z}_U = \min\{V, \bar{m}_U\}$: above $\bar{m}_U$ its promise is worthless, and above V winning is worse than losing. Fix China's drop level z_C. If $z_C \le \bar{z}_U$, dropping at $\bar{z}_U$ wins (ties go to alignment) at the price $\max\{z_C, 0\} \le V$. Every strategy that wins pays at least this price, and losing yields $-V \le -\max\{z_C, 0\}$, so dropping at $\bar{z}_U$ is a best response. If $z_C > \bar{z}_U$, dropping at $\bar{z}_U$ loses and yields $-V$. To win, Washington would have to stand behind a promise of at least z_C. If $\bar{z}_U = \bar{m}_U$, such a promise exceeds $\bar{m}_U$ and is worthless, so Washington cannot win. If $\bar{z}_U = V$, winning costs at least $z_C > V$, which is worse than losing. Hence dropping at $\bar{z}_U$ is a best response to every z_C, that is, weakly dominant.

China's drop level. China wins at level z only if it stays strictly longer than the United States. It then earns $W_C - (z + c)/\lambda_C$, which is nonnegative if and only if $z \le \lambda_C W_C - c$, and its promise is credible if and only if $z \le \lambda_C \bar{m}_C - c$. Its maximal credible bid is therefore $\lambda_C \min\{W_C, \bar{m}_C\} - c = Q$. Fix Washington's drop level z_U. If $z_U < Q$, dropping at Q wins at the price z_U, with a credible promise and a positive payoff. If $z_U \ge Q$, dropping at Q loses and yields zero, while winning would

need a level above Q, at which China's promise is either not credible or unprofitable. So dropping at Q is weakly dominant. If $Q < 0$, China cannot offer Europe even the baseline and stays out.

Outcome. With both bidders at their dominant drop levels, Washington wins if and only if $\bar{z}_U \geq Q$ (ties go to alignment), and it pays the level at which China drops, $\max\{Q, 0\} = [Q]^+$. If $Q < V$ and $Q \leq \bar{m}_U$, then $Q \leq \bar{z}_U$, which gives (a). If $Q > \bar{z}_U$, China wins at the level $\bar{z}_U \geq 0$ at which Washington drops. Europe receives $\lambda_C \tau_C - c = \bar{z}_U$, and China's payment $\tau_C = (\bar{z}_U + c)/\lambda_C$ is credible and profitable because $\bar{z}_U < Q$. This gives (b). In case (a) Europe's payoff is $[Q]^+ = [\min\{Q, \bar{z}_U\}]^+$, and in case (b) it is $\bar{z}_U = [\min\{Q, \bar{z}_U\}]^+$; in both, $\min\{Q, \bar{z}_U\} = \min\{Q, V, \bar{m}_U\}$. The case $Q = V \leq \bar{m}_U$, which the statement excludes, leaves Washington indifferent between winning and losing, and both outcomes occur.

(iii) If $\bar{m}_U < Q < V$, then $\bar{z}_U = \bar{m}_U$ and by (ii)(b) Europe exits with payoff $\bar{m}_U$ while Washington receives $-V$. A binding transfer $\tau \in [Q, V)$ with alignment gives Europe $\tau \geq Q > \bar{m}_U$ and Washington $-\tau > -V$. □

The alternating-raise auction. The clock auction builds in the assumption that the losing bidder stays in until its maximal credible bid. We now drop that assumption. Bidders alternate. At its turn a bidder may raise its standing bid to a point on a grid or pass, and two consecutive passes end the auction, after which Europe chooses. Washington's bids lie on a grid of mesh ε in τ_U, so their value to Europe lies on a grid of mesh ε. China's lie on a grid of mesh ε in τ_C, so their value to Europe, $\lambda_C \tau_C - c$, lies on a grid of mesh $\lambda_C \varepsilon$. Bids worth more to Europe than the bidder's maximal credible bid ($\bar{z}_U$ for Washington, Q for China) are excluded; they are weakly dominated. We claim the following. (a) Every subgame-perfect equilibrium has Europe aligned when $Q < \bar{z}_U - \varepsilon$ and exiting when $Q > \bar{z}_U + \lambda_C \varepsilon$. (b) Subgame perfection does not pin down the price: when $0 < Q < \bar{z}_U - \varepsilon$, every price in $[0, Q]$, up to one grid step, is the outcome of some subgame-perfect equilibrium. (c) If a losing bidder that is indifferent between raising and passing raises, the price is within one grid step of $[Q]^+$. (d) Part (iii) of Lemma 1 holds in every subgame-perfect equilibrium for ε small.

Proof of the claim. The game is finite. (a) Let $Q < \bar{z}_U - \varepsilon$ and suppose that some equilibrium path ends with China leading. China's last raise was followed by a move of Washington, which passed. At that node Washington could instead raise to the smallest grid bid $b \geq Q$, which satisfies $b < Q + \varepsilon < \bar{z}_U$, so $b < V$ and $b \leq \bar{m}_U$. No Chinese bid can beat b, because China's bids are worth at most $Q \leq b$ to Europe and ties go to alignment. The raise therefore guarantees Washington $-b > -V$, while passing gave it $-V$, a contradiction. Now let $Q > \bar{z}_U + \lambda_C \varepsilon$ and suppose that some path ends with Washington leading, at a bid worth at most $\bar{z}_U$ to Europe. At China's last move China could raise to a grid bid whose value to Europe lies in $(\bar{z}_U, Q)$, an interval longer than $\lambda_C \varepsilon$. Washington cannot top it, and China's payoff from it is strictly positive because its value to Europe is below Q, whereas losing gives zero. Again a contradiction.

(b) Let p be a point of China's grid with value to Europe in $[0, Q]$. Consider the following profile. China, as long as it has not yet bid, raises to a bid worth p whenever no bid worth p or more stands,

and it passes at every other node. Washington answers any Chinese lead with its smallest grid bid worth at least as much to Europe, and passes whenever it leads or no bid stands. Since China's bids are worth at most Q and $Q+\varepsilon < \bar{z}_U$, Washington's answer is credible and below V. China wins in no subgame, so it earns zero whatever it does and its strategy is optimal. When China leads, answering is optimal for Washington: China then passes and the auction ends with Washington paying the least it can pay to win, while passing would hand the win to China and give Washington $-V$. When Washington leads, or no bid stands, passing is optimal. If China has already bid, it passes next and the auction ends. If it has not, a US bid worth less than p would be met by China's bid worth p and answered again, at a price no lower than on the path, and a US bid worth p or more costs at least as much. On the path Washington pays its smallest grid bid worth at least p, which is within ε of p. The price 0 arises from the same profile with China never bidding: no bid is made, and Europe aligns, since without a Chinese promise its exit payoff is $-c \leq 0$.

(c) Let $0 < Q < \bar{z}_U - \varepsilon$. By (a) China loses and earns zero whatever it does, so it is indifferent between raising and passing whenever a raise to a bid worth at most Q is available. If it raises in that case, the auction cannot end while Washington's standing bid is worth less than $Q - \lambda_C \varepsilon$, so the price is at least $Q - \lambda_C \varepsilon$. Washington never needs to bid more than the smallest grid bid worth at least Q, which is below $Q + \varepsilon$. If $Q \leq 0$, China never bids and the price is 0.

(d) Let $\bar{m}_U < Q < V$, so $\bar{z}_U = \bar{m}_U$, and let ε be small enough that $Q > \bar{z}_U + \lambda_C \varepsilon$. By (a) Europe exits. China's last raise is optimal only if no cheaper bid would also win for sure. A bid worth to Europe a value in $(\bar{z}_U, \bar{z}_U + \lambda_C \varepsilon]$ does, since Washington's bids are worth at most $\bar{z}_U$, and China's payoff falls with its bid. So Europe receives at most $\bar{m}_U + \lambda_C \varepsilon < Q$, and every binding transfer $\tau \in [Q, V)$ makes both Europe and Washington strictly better off. □

B.2 Proof of Proposition 1

Discrete closed forms. Let $\mu = 1 - e^{-\lambda \Delta}$, $\delta = e^{-r\Delta}$, $\upsilon = \delta(1-\mu) = e^{-(r+\lambda)\Delta}$ and $\psi_\Delta = \mu/(1-\upsilon)$, and define

$$x_A = \psi_\Delta(Q + \beta(V - Q)), \qquad x_B = \frac{(1-\beta)Q + \beta\mu V}{1 - \beta\upsilon}, \qquad x^*(\Delta) = \max\{x_A, x_B\}.$$

Part (i) of Proposition 1 holds with this $x^*(\Delta)$, and there is a stationary equilibrium in which Washington offers $P_U = \max\{Q, \mu Q + \upsilon x^*\}$, Europe demands $P_E = \mu V + \upsilon x^*$, each offer is accepted at once, and Europe would exit after rejecting a US offer if and only if $Q > \delta x^*$. When $\lambda > 0$ this equilibrium is the unique one (part (ii)). Part (iii) refers to P_U and P_E as the equilibrium offer and demand.

Assumptions. After a rejection without voluntary exit, rupture is realized at once with probability μ, and exit flows start immediately. Otherwise the round's status-quo flows $(0, 0)$ accrue and the next round starts after discounting by δ. Offers are real numbers with no sign restriction. Agreements are binding: Washington's cap covers every agreed flow ($\bar{m}_U \geq V$ suffices, since every equilibrium offer

and demand is below V), and a European concession is enforceable. We assume $Q < V$, $\beta \in [0,1)$, $r, \lambda \geq 0$ and $r + \lambda > 0$, so that $\upsilon < 1$.

Proof of Proposition 1. We measure payoffs as flows, so that a constant flow z forever is worth z. Agreement on P gives $(P, -P)$, exit gives $(Q, -V)$, and the status quo gives $(0, 0)$ per period.

Step 1 (feasible payoffs). Every outcome gives a total payoff of 0 (agreement or perpetual status quo) or $Q - V < 0$ (exit), and a mixture of these, so in every subgame the two players' payoffs sum to at most zero. Payoffs are bounded. Washington can reject every demand and make offers Europe may accept or reject, which gives it at least $\min\{0, -V\} = -V$, and Europe can reject every offer without exiting and demand 0, which gives it at least $\min\{Q, 0\}$. The upper bounds follow from the sum. Because $\upsilon < 1$, payoffs are continuous at infinity, and the one-shot deviation principle applies.

Step 2 (the value map). Let $\mathcal{R}(v) = \max\{Q, \mu Q + \upsilon v\}$ be Europe's value of rejecting a US offer when the continuation value is v, and let $\mathcal{T}(v) = \beta(\mu V + \upsilon v) + (1-\beta)\mathcal{R}(v)$. $\mathcal{T}$ is nondecreasing with Lipschitz constant $\upsilon < 1$, so $\mathcal{T}(v) - v$ is strictly decreasing and $\mathcal{T}$ has a unique fixed point. Write $\mathcal{T} = \max\{\mathcal{T}_A, \mathcal{T}_B\}$ with $\mathcal{T}_A(v) = \mu(Q + \beta(V - Q)) + \upsilon v$ and $\mathcal{T}_B(v) = \beta\mu V + (1-\beta)Q + \beta\upsilon v$, whose fixed points are x_A and x_B. If $x_A \geq x_B$, then $\mathcal{T}_B(x_A) - x_A \leq \mathcal{T}_B(x_B) - x_B = 0$ because $\mathcal{T}_B(v) - v$ is strictly decreasing, so $\mathcal{T}(x_A) = x_A$. The case $x_B \geq x_A$ is symmetric. Hence the fixed point is $x^* = \max\{x_A, x_B\}$. Finally, $\mathcal{R}(x^*) = Q$ if and only if $Q \geq \mu Q + \upsilon x^*$, that is, $(1-\mu)Q \geq \delta(1-\mu)x^*$, that is, $Q \geq \delta x^*$.

Step 3 (a stationary equilibrium). Consider the profile in which Europe accepts a US offer P if and only if $P \geq \mathcal{R}(x^*)$ and, after rejecting, exits if and only if $Q > \mu Q + \upsilon x^*$; Washington accepts a demand P if and only if $P \leq P_E = \mu V + \upsilon x^*$; Washington offers $P_U = \mathcal{R}(x^*)$; and Europe demands P_E. Under this profile Europe's value is $\beta P_E + (1-\beta)P_U = \mathcal{T}(x^*) = x^*$ and Washington's is $-x^*$. A responding Europe gets P by accepting and $\mathcal{R}(x^*)$ by rejecting (with its exit choice optimal), and a responding Washington gets $-P$ by accepting and $-\mu V - \upsilon x^* = -P_E$ by rejecting, so the acceptance rules are optimal. A proposing Washington gets $-\mathcal{R}(x^*)$ from P_U. A rejected offer gives it $-V$ if Europe then exits, which happens only when $\mathcal{R}(x^*) = Q < V$, and $-\mu V - \upsilon x^* \leq -\mu Q - \upsilon x^*$ if Europe does not exit, which happens only when $\mathcal{R}(x^*) = \mu Q + \upsilon x^*$. In both cases a rejected offer gives at most $-\mathcal{R}(x^*)$. A proposing Europe gets P_E, while a rejected demand gives it $\mu Q + \upsilon x^* \leq P_E$. Both comparisons are strict when $\mu > 0$. By the one-shot deviation principle the profile is a subgame-perfect equilibrium.

Step 4 (all equilibria have the same payoffs). Every subgame that starts a round is a copy of the game. Let $\bar{x}$ and $\underline{x}$ be the supremum and infimum of Europe's equilibrium payoffs, and $\bar{\pi}$ and $\underline{\pi}$ those of Washington. They are finite by Step 1. (a) By Step 1, $\bar{\pi} \leq -\underline{x}$. (b) When Europe proposes, every demand $P < \mu V - \upsilon\bar{\pi}$ is accepted in every equilibrium, because rejecting it gives Washington at most $-\mu V + \upsilon\bar{\pi} < -P$. So Europe's payoff at such a node is at least $\mu V - \upsilon\bar{\pi} \geq \mu V + \upsilon\underline{x}$. (c) When Washington proposes, Europe can reject and obtain at least $\mathcal{R}(\underline{x})$. Hence $\underline{x} \geq \beta(\mu V + \upsilon\underline{x}) + (1-\beta)\mathcal{R}(\underline{x}) = \mathcal{T}(\underline{x})$, and since $\mathcal{T}(v) - v$ is strictly decreasing with root x^*, $\underline{x} \geq x^*$. (d) When Washington proposes, every offer $P > \mathcal{R}(\bar{x})$ is accepted in every equilibrium, because rejecting it gives Europe at most $\mathcal{R}(\bar{x})$. So Washington's payoff at such a node is at least

$-\mathcal{R}(\bar{x})$. When Europe proposes, Washington can reject and obtain at least $-\mu V + \upsilon\underline{\pi}$. Hence $\underline{\pi} \geq \beta(-\mu V + \upsilon\underline{\pi}) - (1-\beta)\mathcal{R}(\bar{x})$, that is, $-\underline{\pi} \leq [\beta\mu V + (1-\beta)\mathcal{R}(\bar{x})]/(1-\beta\upsilon)$. (e) At every node Europe's payoff is at most minus Washington's (Step 1). So Europe gets at most $\mu V - \upsilon\underline{\pi}$ when it proposes and at most $\mathcal{R}(\bar{x})$ when Washington proposes. Using (d),

$$\bar{x} \leq \beta\mu V + (1-\beta)\mathcal{R}(\bar{x}) + \beta\upsilon\,\frac{\beta\mu V + (1-\beta)\mathcal{R}(\bar{x})}{1-\beta\upsilon} = \frac{\beta\mu V + (1-\beta)\mathcal{R}(\bar{x})}{1-\beta\upsilon} \equiv \mathcal{S}(\bar{x}).$$

$\mathcal{S}$ has Lipschitz constant $(1-\beta)\upsilon/(1-\beta\upsilon) < 1$, and $\mathcal{S}(v) = v$ if and only if $\mathcal{T}(v) = v$. So $\mathcal{S}(v) - v$ is strictly decreasing with root x^*, and $\bar{x} \leq x^*$. Therefore $\underline{x} = \bar{x} = x^*$. Then $\bar{\pi} \leq -x^*$ by (a) and $\underline{\pi} \geq -\mathcal{S}(x^*) = -x^*$ by (d), so Washington's payoff is $-x^*$ in every equilibrium. This proves (i).

Step 5 (uniqueness and immediate agreement when $\lambda > 0$). By Step 4, payoffs sum to zero in every equilibrium and in every subgame that starts a round. If a proposal were rejected with positive probability, the total payoff after the rejection would be $Q - V < 0$ (exit) or $\mu(Q-V) + \upsilon s$ with $s \leq 0$ (no exit), which is negative when $\mu > 0$. So agreement is immediate. Since continuation payoffs from every round start are $(x^*, -x^*)$, Europe accepts every offer above $\mathcal{R}(x^*)$ and rejects every offer below it, and Washington accepts every demand below P_E and rejects every demand above it. Washington must offer exactly $\mathcal{R}(x^*)$ and Europe must accept it: if Europe rejected $\mathcal{R}(x^*)$, Washington would have no best offer, and any lower offer is rejected and gives Washington strictly less (Step 3). The same argument pins down Europe's demand P_E and Washington's acceptance of it. Europe's exit choice after rejecting an offer is determined unless $Q = \delta x^*$, and it is never used on the path. This proves the first sentence of (ii). Now let $\lambda = 0$, so $\mu = 0$, $\upsilon = \delta$, $x_A = 0$ and $x_B = (1-\beta)Q/(1-\beta\delta)$. If $Q > 0$, then $x^* = x_B$. Consider the stationary profile in which Europe always makes a demand that Washington rejects (any demand above δx^*) and Washington offers Q, which Europe accepts. Europe's value x satisfies $x = \beta\delta x + (1-\beta)Q$, so $x = x_B = x^*$. Washington's offer is optimal because $\mathcal{R}(x^*) = \max\{Q, \delta x_B\} = Q$ (as $\delta(1-\beta) \leq 1-\beta\delta$), Europe is indifferent between its rejected demand and the acceptable demand δx^*, and Washington is indifferent between accepting and rejecting a demand of δx^*. So the profile is an equilibrium with delay and the payoffs of (i). If $Q \leq 0$, then $x^* = 0$, and perpetual disagreement without exit is an equilibrium with payoffs $(0,0) = (x^*, -x^*)$.

Step 6 (the limit). As $\Delta \to 0$, $\mu = \lambda\Delta + O(\Delta^2)$ and $1 - \upsilon = (r+\lambda)\Delta + O(\Delta^2)$, so $\psi_\Delta \to \psi$ and $x_A \to \psi(Q + \beta(V-Q))$. In x_B the numerator tends to $(1-\beta)Q$ and the denominator to $1-\beta > 0$, so $x_B \to Q$. The maximum is continuous, so $x^* \to P^*$. Finally, $P_U = \mathcal{R}(x^*) \to \max\{Q, P^*\} = P^*$ and $P_E = \mu V + \upsilon x^* \to P^*$. For the threshold, with $r > 0$, $\psi(Q + \beta(V-Q)) > Q$ if and only if $\lambda(Q + \beta(V-Q)) > (r+\lambda)Q$, that is, $Q(r + \lambda\beta) < \lambda\beta V$. □

Exit after a rejected European demand. The footnote to Proposition 1 concerns a game in which Europe may also exit right after Washington rejects Europe's demand. Let $x'_B = \beta V + (1-\beta)Q$. We claim: (1) for each $\Delta > 0$ the stationary equilibrium value is unique and equals $x' = x_A$ if $\delta x_A \geq Q$ and $x' = \min\{x'_B, Q/\delta\}$ otherwise, and, with $r > 0$, $x' \to P^*$ as $\Delta \to 0$; (2) if $\delta x^* > Q$,

which for small Δ is the case $Q < \hat{Q}$, the equilibrium payoffs are those of the base game; (3) if $\beta > 0$ and $\delta x_A \leq Q$, which for small Δ is the case $Q > \hat{Q}$, there are nonstationary equilibria in which Europe's payoff is

$$v_H = \max\Big\{x'_B,\ \frac{\beta V + (1-\beta)\mu Q}{1-(1-\beta)\upsilon}\Big\},$$

the largest equilibrium payoff for Europe, and $v_H \to V$ as $\Delta \to 0$.

Proof of the claim. (1) In a stationary equilibrium with values $(v, -v)$, Europe's continuation after a rejection without exit is $\mu Q + \upsilon v$, so Europe exits after a rejection if $Q > \delta v$ and not if $Q < \delta v$. Washington offers $\mathcal{R}(v)$. Washington's rejection of Europe's demand gives Washington $-V$ if Europe then exits and $-\mu V - \upsilon v$ otherwise, so Europe demands V if $Q > \delta v$, $\mu V + \upsilon v$ if $Q < \delta v$, and $pV + (1-p)(\mu V + \upsilon v)$ if $Q = \delta v$ and it exits with probability p. If $v > Q/\delta$, the value equation is $v = \mathcal{T}_A(v)$, so $v = x_A$. If $v < Q/\delta$, it is $v = \beta V + (1-\beta)Q$, so $v = x'_B$. If $v = Q/\delta$, it requires $\mathcal{T}_A(Q/\delta) \leq Q/\delta \leq x'_B$, that is, $x_A \leq Q/\delta \leq x'_B$. The last case excludes the other two. The first two cannot both hold: $x_A > Q/\delta > x'_B$ would require $x'_B < x_A = \psi_\Delta x'_B$ with $\psi_\Delta \leq 1$, so $x'_B < 0$ and hence $Q < 0$, but then $Q/\delta \leq Q \leq x'_B$. Since one of the three conditions always holds, exactly one value exists, and it is the stated one. (If $\delta x_A < Q$ then $Q > 0$: for $Q \leq 0$, $x_A = \psi_\Delta x'_B \geq \min\{x'_B, 0\} \geq Q \geq Q/\delta$.) As $\Delta \to 0$: if $Q < \hat{Q}$, then $\delta x_A > Q$ eventually and $x' = x_A \to P^*$; if $Q > \hat{Q}$, then eventually $x' = \min\{x'_B, Q/\delta\} \to \min\{Q + \beta(V-Q), Q\} = Q = P^*$; if $Q = \hat{Q}$, both expressions tend to $Q = P^*$.

(2) Steps 4(a) to 4(c) of the proof of Proposition 1 remain valid, because Washington's rejection of a demand now gives it at most $\max\{-V, -\mu V + \upsilon\bar{\pi}\} = -\mu V + \upsilon\bar{\pi}$. So every equilibrium continuation value of Europe is at least x^*. If $\delta x^* > Q$, then $\mu Q + \upsilon v' > Q$ for every continuation value $v' \geq x^*$, exit after a rejection is never optimal, and the game is strategically the base game.

(3) Let L be the stationary equilibrium of (1), with value $v_L = x' \leq Q/\delta$. In state H, Europe demands V and Washington accepts any demand up to V; after Washington rejects a demand, Europe exits, and if it did not, play would continue in L; Washington offers $\mathcal{R}(v_H)$, and after Europe rejects an offer without exiting, play continues in H. Then $v_H = \beta V + (1-\beta)\mathcal{R}(v_H)$, whose solution is the stated maximum (by the argument of Step 2). One-shot deviations: exit after Washington's rejection is optimal for Europe because $\mu Q + \upsilon v_L \leq Q$. Washington is indifferent between accepting V and rejecting it, since exit then gives it $-V$. A demand above V is rejected and gives Europe $Q < V$. A US offer below $\mathcal{R}(v_H)$ is rejected and gives Washington $-V$ or $-\mu V - \upsilon v_H$, both at most $-\mathcal{R}(v_H)$, as in Step 3. Europe's acceptance rule in H is optimal by construction. For the upper bound, Washington can always secure $-V$, so Europe gets at most V when it proposes, and, as in Step 4(d), at most $\mathcal{R}(\bar{x}')$ when Washington proposes, where $\bar{x}'$ is the supremum of Europe's equilibrium payoffs. So $\bar{x}' \leq \beta V + (1-\beta)\mathcal{R}(\bar{x}')$. The right side minus $\bar{x}'$ is strictly decreasing and vanishes at v_H, so $\bar{x}' \leq v_H$. As $\Delta \to 0$, $(\beta V + (1-\beta)\mu Q)/(1-(1-\beta)\upsilon) \to \beta V/\beta = V$. □

Convergence and the order of limits. Since $|\max\{a,b\} - \max\{a',b'\}| \le \max\{|a-a'|, |b-b'|\}$ and $x_B - Q = \beta(1-\upsilon)(\psi_\Delta V - Q)/(1-\beta\upsilon)$,

$$|x^*(\Delta) - P^*| \le \max\left\{|\psi_\Delta - \psi|\,|Q + \beta(V-Q)|,\ \frac{\beta(1-\upsilon)}{1-\beta\upsilon}\,|\psi_\Delta V - Q|\right\}, \qquad \frac{\beta(1-\upsilon)}{1-\beta\upsilon} \le \frac{\beta(r+\lambda)\Delta}{1-\beta}.$$

The error is $O(\Delta)$ where the second term of P^* binds and $O(\Delta/(1-\beta))$ where Q binds, so it is not uniform in β. For fixed Δ, $x^*(\Delta) \to \psi_\Delta V$ as $\beta \to 1$ (both x_A and x_B tend to $\psi_\Delta V$), whereas $P^* \to \max\{Q, \psi V\}$. The limits $\Delta \to 0$ and $\beta \to 1$ therefore do not commute when $Q > \psi V$. The formula for P^* describes bargaining in which US proposals arrive much faster than rupture and discounting, $(1-\beta)/\Delta$ large relative to $r + \lambda$, which is why the statement keeps $\beta < 1$.

B.3 Proof of Corollary 1

Proof of Corollary 1. (a) P^* is a maximum one of whose terms is Q, so $P^* \ge Q$. If $Q \ge 0$ then $P^* \ge 0$, and if $Q < 0$ then $-P^* \le -Q$; in both cases $-P^* \le [-Q]^+ = \Lambda$. For $\Delta > 0$, $x^* \ge x_B$ and $x_B - Q = \beta(1-\upsilon)(\psi_\Delta V - Q)/(1-\beta\upsilon)$, so $Q - x^* \le \beta(1-\upsilon)\,[Q - \psi_\Delta V]^+/(1-\beta\upsilon)$.

(b) $\lambda = 0$ gives $\psi = 0$ and $P^* = \max\{Q, 0\}$. $\beta = 0$ gives $P^* = \max\{Q, \psi Q\}$. If $Q \le 0$, both terms are nonpositive in each case.

(c) $P^* > 0$ if and only if $Q > 0$ or $\psi(Q + \beta(V-Q)) > 0$, and $\psi > 0$ if and only if $\lambda > 0$.

(d) Where $\psi(Q + \beta(V-Q)) > Q$, $P^* = \psi(1-\beta)Q + \psi\beta V$; differentiate in V, holding Q fixed or letting it vary. Where Q is the larger term, $P^* = Q$. The regime condition is the threshold $\hat{Q}$ of Proposition 1(iii).

(e) In the formula for P^*, V appears only in $\beta\psi V$, since β, λ and r do not depend on V. This is agenda power under rupture risk. The remaining dependence runs through $Q = \omega - c$ (equation (3)). In $\omega = \lambda_C \min\{W_C, \bar{m}_C\}$ with $W_C = (1-\theta)W + \rho V$, V enters through the rival's valuation, and $\partial\omega/\partial V = \lambda_C\rho$ where $W_C < \bar{m}_C$ (and 0 where the cap binds). In $c = \kappa S_k + d_p + d_{ai} + d_f + \eta$, the components that depend on V are $d_f = \chi_f f$, where $\chi_f = \mathbf{1}\{\varphi V \le \gamma + \bar{m}_U\}$ falls from 1 to 0 as V rises past $(\gamma + \bar{m}_U)/\varphi$, which lowers c and raises Q by f (self-harming sanctions); and κS_k, where $\kappa = G(s_X + s_L) - G(s_X - m_S)$ depends on V through $s_X = s_0 + s_B(V)$ (the silicon shield). There $\partial\kappa/\partial s_X = g(s_X + s_L) - g(s_X - m_S)$, whose sign depends on the density g between the two points (Proposition 4(d) gives the case $m_S = \bar{m}_U$, where $\kappa = \bar{\kappa}$).

Nesting. $\lambda = 0$ gives $[Q]^+$ by (b). As $r \to 0$ with $\lambda > 0$, $\psi \to 1$ and $P^* \to \max\{Q, Q + \beta(V - Q)\} = Q + \beta(V-Q)$ because $V > Q$. $\beta = 0$ gives $\max\{Q, \psi Q\}$. Finally, P^* is continuous in $(\beta, \psi) \in [0,1]^2$ and equals $\max\{Q, V\} = V$ at $(\beta, \psi) = (1,1)$, which is the limit $\beta \to 1$, $\lambda \to \infty$. □

B.4 Proof of Proposition 2

Assumptions. The setup is that of Section 5.1. We fix six points. (1) *Payoffs.* Europe's aligned baseline payoff is 0 and its exit payoff is Q. With conceded set S and withdrawn set J_w at the end of the period, Europe's aligned payoff before rent is $\pi^A = -\Delta^A(J_w) - T_{\mathrm{r}}(S) - T_{\mathrm{s}}(S)$ and its exit payoff is $\pi^X = Q - \Delta^X(J_w) - T_{\mathrm{s}}(S)$. The replayed contest gives Europe $\max\{\pi^A, \pi^X\}$ and

costs Washington the rent $\left[\pi^X - \pi^A\right]^+ = [Q + d(J_w) + T_{\mathrm{r}}(S)]^+$. Europe's payoff without demands is $[Q]^+$, and its *net loss* is $[Q]^+ - \max\{\pi^A, \pi^X\}$. (2) *Washington retains Europe.* V and $\bar{m}_U$ exceed every rent that can arise. Without this, Europe exits after some histories, its payoff is then at least π^X, and every upper bound on its loss below survives. (3) *Credible set.* J^c is fixed at the start of the period, every subset of J^c is a credible threat, and no instrument outside J^c can be withdrawn during the period. (4) *Execution costs.* Executing a withdrawal of J costs Washington $e(J) \geq 0$, because the baseline is its ex post optimal provision to a loyal ally. (5) *Protocol.* Each ultimatum names a nonempty set of unconceded demands and a nonempty threat set of unwithdrawn instruments. A threat that deters is not executed and stays available, and withdrawn instruments stay withdrawn. Washington may stop at any time. Every ultimatum either adds a concession or spends an instrument, so the game is finite. (6) *Ties.* Europe accepts when indifferent unless stated otherwise. The bounds in (c) hold for every tie-breaking rule.

Proof of Proposition 2. Step 0 (final-state payoffs). Write T_{r} and T_{s} for the totals of the conceded set S. Europe's payoff is $\max\{\pi^A, \pi^X\} = -u(J_w) - T_{\mathrm{s}} + \max\{Q, -d(J_w) - T_{\mathrm{r}}\}$. If $Q \geq 0$, the maximum is Q, because $d \geq 0$ and $T_{\mathrm{r}} \geq 0$. If $Q < 0$, then $-\max\{Q, -d - T_{\mathrm{r}}\} = \min\{\Lambda, d + T_{\mathrm{r}}\}$. In both cases Europe's net loss is

$$L(S, J_w) = T_{\mathrm{s}} + u(J_w) + \min\{d(J_w) + T_{\mathrm{r}}, \Lambda\} = T_{\mathrm{s}} + \min\{\Delta^A(J_w) + T_{\mathrm{r}},\ \Delta^X(J_w) + \Lambda\}.$$

The second form shows that L is nondecreasing in J_w and in S, because Δ^A and Δ^X are nondecreasing (the conditional part d need not be). Washington's payoff is $W(S) - [Q + d(J_w) + T_{\mathrm{r}}]^+ - e(J_w)$.

(a) A refusal executes J when nothing has been conceded, so the loss is $L(\emptyset, J) = u(J) + \min\{d(J), \Lambda\} = B(J)$. Monotonicity of B follows from Step 0. If $Q \geq 0$, then $\Lambda = 0$ and $B = u$. (After a history with conceded set S_0 and withdrawn set J_0, the bite of J is $L(S_0, J_0 \cup J) - L(S_0, J_0)$, which in general differs from $B(J)$.)

(b) $L(S, \emptyset) = T_{\mathrm{s}} + \min\{T_{\mathrm{r}}, \Lambda\}$, which equals $\min\{T, \Lambda\}$ when all concessions are reversible. Accepting a single ultimatum gives the loss $L(S, \emptyset)$ and refusing it gives $B(J)$, so Europe accepts if and only if $\min\{T_{\mathrm{r}}, \Lambda\} + T_{\mathrm{s}} \leq B(J)$.

(c)(i) Fix a subgame-perfect equilibrium and let Europe deviate to refusing every ultimatum. On the resulting path nothing is conceded and every withdrawn instrument belongs to J^c, so the loss is $L(\emptyset, J_w) = B(J_w) \leq B(J^c)$. Europe's equilibrium loss is at most its loss from any deviation. The argument uses only that withdrawals come from J^c, so it holds for adaptive strategies, for every Q and for every tie-breaking rule. Next, $B(J) - u(J) = \min\{d(J), \Lambda\} \leq \Lambda$. Finally, for a fixed J_w, reversible concessions raise the loss by $\min\{d + T_{\mathrm{r}}, \Lambda\} - \min\{d, \Lambda\} \leq \min\{T_{\mathrm{r}}, \Lambda\} \leq \Lambda$.

(c)(ii) Let (S, J_w) be an equilibrium outcome. By (i) and Step 0, $L(S, \emptyset) \leq L(S, J_w) \leq B(J^c)$, so S is acceptable as a single ultimatum backed by J^c. Since $d \geq 0$ and $e \geq 0$, Washington's equilibrium payoff is $W(S) - [Q + d(J_w) + T_{\mathrm{r}}(S)]^+ - e(J_w) \leq W(S) - [Q + T_{\mathrm{r}}(S)]^+ \leq V_1$.

(c)(iii) By (b), Europe accepts a single ultimatum backed by J^c that asks for a maximizer S^* in the definition of V_1, so the ultimatum yields V_1. Now let Washington also have a divisible sunk

demand: any amount $\phi \in [0, \bar{\phi}]$ of it costs Europe ϕ and is worth $\alpha\phi$ to Washington, with $\alpha > 0$. If $L(S, \emptyset) < B(J^c)$, adding $\phi = B(J^c) - L(S, \emptyset)$ keeps the bundle acceptable, leaves the rent unchanged (the demand is sunk) and raises Washington's payoff by $\alpha\phi > 0$. With $\bar{\phi} \geq B(J^c)$, every optimal bundle therefore has $L = B(J^c)$.

(d) The claim for every equilibrium assumes $e(J) > 0$ for every nonempty J, which holds when execution costs are additive and positive for each instrument. *All demands reversible and* $B(J^c) \geq \Lambda$. Every set S satisfies $L(S, \emptyset) = \min\{T_{\mathrm{r}}(S), \Lambda\} \leq \Lambda \leq B(J^c)$, so $V_1 = \max_S\{W(S) - [Q + T_{\mathrm{r}}(S)]^+\}$. Let S^* be an inclusion-maximal maximizer. Consider the subgame after Europe has conceded S^* with nothing withdrawn. Every terminal outcome (S', J_w) of that subgame has $S' \supseteq S^*$ and gives Washington at most $W(S') - [Q + T_{\mathrm{r}}(S')]^+ - e(J_w)$. This is strictly below V_1 if $S' \neq S^*$, by the maximality of S^*, and if $J_w \neq \emptyset$ when $e(J) > 0$ for every nonempty J. So Washington stops, and Europe's loss is $L(S^*, \emptyset) \leq B(J^c)$. Refusing the root ultimatum (S^*, J^c) executes all of J^c, after which no ultimatum can be made, so the refusal loss is exactly $B(J^c)$, and Europe accepts. Washington thus obtains V_1 in every equilibrium, and by (c)(ii) no more. If $e(J) = 0$ for some nonempty J, stopping is still optimal in the subgame, which gives an equilibrium with payoff V_1. *Some best single ultimatum asks for every demand.* Let S^* be such a bundle. After S^* is conceded no ultimatum can be made. Europe accepts (S^*, J^c) because $L(S^*, \emptyset) \leq B(J^c)$, which is the refusal loss. Washington can therefore secure V_1, and by (c)(ii) it obtains no more. *The ratchet.* Let the only instrument have $\Delta^A = \Delta^X = \bar{u}$, so $u = \bar{u} = B(J^c)$ and $d = 0$. Give Washington n sunk demands of cost τ and value $w > 0$ each, and let the execution cost be positive. The rent is always $[Q]^+$. Every ultimatum is backed by the single instrument, and after a refusal no further ultimatum can be made. Let $b = \lfloor \bar{u}/\tau \rfloor$, and let $\mathcal{N}(m)$ be the number of further concessions from a state with m demands left and the instrument intact. We show by induction on m that $\mathcal{N}(m) = m \bmod (b+1)$. Clearly $\mathcal{N}(0) = 0$. An ultimatum asking for $x \in \{1, \ldots, m\}$ demands leads, if accepted, to $x + \mathcal{N}(m - x)$ concessions in total and an additional loss of $(x + \mathcal{N}(m - x))\tau$, while refusal costs $\bar{u}$. So Europe accepts if and only if $x + \mathcal{N}(m - x) \leq b$. By the induction hypothesis $x + \mathcal{N}(m - x) \equiv m \pmod{b+1}$, and this number is at least $x \geq 1$. Its only possible value in $\{1, \ldots, b\}$ is $\bar{n} = m \bmod (b+1)$, and $x = \bar{n}$ attains it. If $\bar{n} \geq 1$, every accepted ultimatum yields $\bar{n}$ concessions in total, and Washington, which values each concession, obtains $\bar{n}$. If $\bar{n} = 0$, every ultimatum is refused, and Washington stops because a refused ultimatum costs it the execution cost and gains nothing. So $\mathcal{N}(m) = \bar{n}$. With $n = b + 1$ Washington obtains nothing, whereas a single ultimatum would yield $\min\{n, b\} = b$ demands. □

B.5 Proof of Remark 1

Assumptions. Those of Proposition 2, except that at the end of the period Europe and Washington play the bargaining game of Proposition 1 in the frequent-offer limit instead of replaying the bidding contest. They bargain at the premium then prevailing, from the aligned status quo net of withdrawals and concessions. Washington retains Europe in the sense that the premium stays below V after every history, so Proposition 1 applies.

Proof of Remark 1. Write $P^*(q) = \max\{q, \psi(q + \beta(V - q))\}$. It is the maximum of two affine functions of q with slopes 1 and $\psi(1-\beta) \in [0,1]$. So P^* is nondecreasing with slope at most one, $P^*(q) \geq q$, and $P^*(q) - q = [\psi\beta V - (1 - \psi + \psi\beta)q]^+$ is nonincreasing in q.

Bite. Without a withdrawal Europe's payoff is $P^*(Q)$. Withdrawing J lowers Europe's aligned payoff by $\Delta^A(J)$ and its exit payoff by $\Delta^X(J) = u(J)$, so the premium becomes $Q + d(J)$ and the bargaining gives Europe $-\Delta^A(J) + P^*(Q + d(J))$. Hence $B_\psi(J) = \Delta^A(J) - [P^*(Q + d(J)) - P^*(Q)]$. Since P^* is nondecreasing, $B_\psi(J) \leq \Delta^A(J) = u(J) + d(J)$. Since $P^*(Q + d) \geq Q + d$, $B_\psi(J) \leq u + d - (Q + d) + P^*(Q) = u(J) + \Lambda_\psi$. Together, $B_\psi(J) \leq u(J) + \min\{d(J), \Lambda_\psi\}$. If $r > 0$ and $Q + d(J) \geq \hat{Q}$, then $P^*(Q + d) = Q + d$ by Proposition 1(iii), so $B_\psi(J) = u(J) + \Lambda_\psi$, and $\Lambda_\psi = P^*(Q) - Q \leq P^*(Q + d) - Q = d$. The bound then holds with equality.

Comparison with Λ. Since $Q + \Lambda = [Q]^+$, $\Lambda_\psi - \Lambda = P^*(Q) - [Q]^+$. So $\Lambda_\psi \geq \Lambda$ exactly when Europe's bargaining payoff is at least its bidding payment, which, because $P^*(Q) \geq Q$, is the case whenever $P^*(Q) \geq 0$. If $\lambda = 0$, then $\psi = 0$ and $P^*(q) = [q]^+$, so $\Lambda_\psi = [Q]^+ - Q = \Lambda$ and $B_\psi(J) = u + d - [Q + d]^+ + [Q]^+ = u + \min\{d, \Lambda\} = B(J)$, by the case analysis in Step 0 of the proof of Proposition 2. If $\lambda > 0$, $Q < 0$ and $Q + \beta(V - Q) < 0$, then $P^*(Q) < 0$ and $\Lambda_\psi < \Lambda$: Europe already concedes $-P^*(Q)$ in the bargaining, and conditional threats can extract only the rest of its surplus over exit.

One budget. With conceded set S and withdrawn set J_w, reversible concessions lower only the aligned payoff and sunk ones lower both, so Europe's net loss is

$$L_\psi(S, J_w) = T_{\mathrm{s}} + \Delta^A(J_w) + T_{\mathrm{r}} - \big[P^*\big(Q + d(J_w) + T_{\mathrm{r}}\big) - P^*(Q)\big].$$

With $x = \Delta^A(J_w) + T_{\mathrm{r}}$ and $y = \Delta^X(J_w)$, $L_\psi = T_{\mathrm{s}} + x - P^*(Q + x - y) + P^*(Q)$, which is nondecreasing in x (the slope of P^* is at most one) and in y (P^* is nondecreasing). So L_ψ is nondecreasing in J_w and in S, and $B_\psi(J) = L_\psi(\emptyset, J)$ is nondecreasing in J. Europe can refuse every ultimatum, which leaves nothing conceded and a withdrawn set inside J^c, so its equilibrium loss is at most $B_\psi(J^c)$, for every tie-breaking rule, as in the proof of Proposition 2(c)(i). The bound $B_\psi(J) - u(J) \leq \Lambda_\psi$ was shown above. For a fixed J_w, let $q = Q + d(J_w) \geq Q$. Reversible concessions raise the loss by $T_{\mathrm{r}} - [P^*(q + T_{\mathrm{r}}) - P^*(q)]$, which is at most T_{r} and at most $P^*(q) - q \leq P^*(Q) - Q = \Lambda_\psi$. Parts (c)(ii) and (c)(iii) carry over with L_ψ in place of L and the payment $P^*(Q + T_{\mathrm{r}}(S))$ in place of $[Q + T_{\mathrm{r}}(S)]^+$ in V_1, by the same arguments: Washington's payoff $W(S) - P^*(Q + d(J_w) + T_{\mathrm{r}}(S)) - e(J_w)$ is at most $W(S) - P^*(Q + T_{\mathrm{r}}(S))$, and a sunk demand leaves the payment unchanged.

Example. Take $Q = -1$, $V = 4$, $\beta = 0.5$, $\lambda = 1$ and $r = 0.25$, so $\psi = 0.8$ and $\hat{Q} = 8/3$. Then $P^*(-1) = \max\{-1, 0.8 \times 1.5\} = 1.2$, so $\Lambda = 1$ and $\Lambda_\psi = 2.2$. Withdrawing a conditional part $d = 3$ with $u = 0$ moves the premium to $2 < \hat{Q}$, where $P^*(2) = 2.4$. So $B_\psi = 3 - (2.4 - 1.2) = 1.8$, which exceeds Λ and is below $\min\{d, \Lambda_\psi\} = 2.2$. □

B.6 Proof of Corollary 2

Proof of Corollary 2. For every Q and every $T \geq 0$, $[Q+T]^+ = [Q]^+ + [T-\Lambda]^+$: if $Q \geq 0$ then $\Lambda = 0$, and if $Q < 0$ then $Q + T = T - \Lambda$. So Washington's payoff from a conceded reversible set S with nothing withdrawn is $-[Q]^+ + W(S) - [T(S) - \Lambda]^+$. By Proposition 2(b), (c)(iii) and (d), Washington obtains the best single ultimatum backed by J^c, so it chooses S to maximize this payoff subject to $\min\{T(S), \Lambda\} \leq B(J^c)$, which is the stated program. If $B(J^c) < \Lambda$, the constraint reads $T(S) \leq B(J^c) < \Lambda$ and the rent term vanishes, which gives the knapsack with capacity $B(J^c) = \min\{\Lambda, B(J^c)\}$. Let demands be divisible with $w_k \leq t_k$ for every k, and let $B(J^c) \geq \Lambda$, so that the constraint is slack. At any y with $T > \Lambda$, lowering y_k by a small ε for some k with $y_k > 0$ changes the objective by $\varepsilon(t_k - w_k) \geq 0$, so some optimum has $T \leq \Lambda$. It solves the knapsack with capacity $\Lambda = \min\{\Lambda, B(J^c)\}$. Without divisibility the reduction fails. Take $\Lambda = B(J^c) = 1$ and one demand with $t = 1.5$ and $w = 1.4$. The knapsack value is 0, but Washington obtains $1.4 - 0.5 = 0.9$ by making the demand and paying rent 0.5. The free capacity $\min\{\Lambda, B(J^c)\}$ is nonincreasing in Q because $\Lambda = [-Q]^+$ is and $B(J^c) = u(J^c) + \min\{d(J^c), \Lambda\}$ is. □

B.7 Proof of Proposition 3

Assumptions. Those of Proposition 2. In part (a), China's payoff is $-\sum_{k \in S} \left(z_k + e_k^C \mathbf{1}\{r_k > 0\}\right)$ when Washington's demand set S is conceded, Washington makes a single ultimatum backed by J^c, and $S(r)$ denotes its optimal bundle under the profile r, unique at the profiles compared. In part (b), Washington demands the hedge when it gains weakly.

Proof of Proposition 3. (a)(i) A counter-threat on k is a sunk loss to Europe when k is conceded, which lowers π^A and π^X equally. Step 0 and part (b) of the proof of Proposition 2 then apply with $T_{\mathrm{s}}(S) + \sum_{k \in S} r_k$ in place of $T_{\mathrm{s}}(S)$.

(iv) Counter-threats leave Washington's payoff from every bundle unchanged, since the rent does not depend on r, and by (i) they shrink the set of acceptable bundles.

(ii) If $S(r) = S(0)$, China's payoff under r is $-\sum_{k \in S(0)}(z_k + e_k^C \mathbf{1}\{r_k > 0\})$, which is at most its payoff with no threats.

(iii) Let r^* be optimal for China, $S^* = S(r^*)$, $k \in S^*$ with $r_k^* > 0$, and let r' equal r^* except that $r'_k = 0$. If $S(r') = S^*$, China gains $e_k^C > 0$ by switching to r', which contradicts the optimality of r^*. So $S' = S(r') \neq S^*$. S^* is still acceptable under r' and gives Washington the same payoff, so by uniqueness Washington strictly prefers S' to S^*. Then S' was not acceptable under r^*, or Washington would have chosen it. Acceptable sets are closed under subsets, because the cost in (i) is monotone in S. Hence $S' \not\subseteq S^*$: some demand $j \in S' \setminus S^*$ has been crowded out. For an example, let $B(J^c) = 2$. Demand 1 is sunk with $t_1 = 0.3$, $w_1 = 1$ and $z_1 = 5$, and China cannot threaten it. Demand 2 is sunk with $t_2 = 1.5$, $w_2 = 20$ and $z_2 = 0$, and China can threaten it with $\bar{r}_2 = 0.4$ at $e_2^C = 0.1$. Without threats Washington demands both (cost $1.8 \leq 2$), and China's payoff is -5. Any $r_2 \in (0.2, 0.4]$ makes the pair cost more than 2, so Washington demands only demand 2, China executes, and its payoff rises to -0.1; threats with $r_2 \leq 0.2$ change nothing and cost China 0.1. So

China's optimal profiles have $r_2 \in (0.2, 0.4]$, and the threat is executed on the path. All inequalities are strict, so the conclusion persists under small perturbations of the parameters.

(v) Europe accepts only bundles whose cost under (i) is at most $B(J^c)$, so its loss is at most $B(J^c)$ under every profile. If Washington also has the divisible sunk demand of the proof of Proposition 2(c)(iii) and China cannot threaten it, the argument there shows that every optimal bundle has loss exactly $B(J^c)$ under every profile, and the counter-threat changes only the mix of demands. Otherwise the loss can move either way. In the example of (iii), Europe's loss rises from 1.8 to 1.9. For a fall, let $B(J^c) = 2$, let demand 1 be sunk with $t_1 = 1.9$, $w_1 = 10$, $z_1 = 5$, threatenable with $\bar{r}_1 = 0.5$ at $e_1^C = 0.1$, and let demand 2 be sunk with $t_2 = 1$, $w_2 = 5$, $z_2 = 0$ and not threatenable. Without threats Washington demands only demand 1 (the pair costs 2.9), and Europe loses 1.9. With $r_1 = 0.5$ demand 1 costs 2.4, so Washington demands only demand 2, the threat is never executed, China's payoff rises from -5 to 0, and Europe's loss falls to 1.

(b) The hedge is a reversible demand with $t = y_H$. By Proposition 2(b), Europe accepts it if and only if $\min\{y_H, \Lambda\} \le B(J^c)$. By the identity in the proof of Corollary 2, Washington's payoff changes by $w_H - [y_H - \Lambda]^+$ when the hedge is stopped, so it makes the demand if and only if $w_H \ge [y_H - \Lambda]^+$, and it pays the rent $[y_H - \Lambda]^+$. The demand is free if and only if $y_H \le \Lambda$ and it is accepted, that is, $y_H \le \min\{\Lambda, B(J^c)\}$, and its free part $\min\{y_H, \Lambda\}$ vanishes as $Q \to 0$ because Λ does. Blocking the hedge by flow jurisdiction is a threat with net ex post execution cost e_F, so under the commitment technology it is credible if and only if $e_F \le \bar{m}_U$. Under complete information it deters and is not executed, and it involves no instrument whose withdrawal changes Europe's exit payoff relative to its aligned payoff, so Q is unchanged. Finally, executing a withdrawal of J lowers Europe's aligned payoff by $\Delta^A(J)$ and its exit payoff by $\Delta^X(J)$, so its premium rises to $Q + d(J)$ (Step 0 of the proof of Proposition 2). □

B.8 Proof of Proposition 4

Assumptions. H is convex and increasing on $[\kappa_0, \bar{\kappa}]$ with $H(\kappa_0) = 0$. H' is its right derivative, which is nondecreasing and right-continuous. $\nu \in [0, 1]$ and $S_k > 0$. The thresholds κ_x, κ_1, κ_ν and κ_z are as defined in Section 5.3, so that $\kappa_0 \le \kappa_\nu \le \kappa_1$.

Proof of Proposition 4. Write $\Pi(\kappa) = \Pi_0(\kappa) - H(\kappa)$ with $\Pi_0(\kappa) = -[Q(\kappa)]^+ + \nu[-Q(\kappa)]^+$, and note that $Q(\kappa) = Q_0 - (\kappa - \kappa_0)S_k = (\kappa_z - \kappa)S_k$. Π_0 is continuous and piecewise linear, with slope S_k for $\kappa < \kappa_z$ and slope νS_k for $\kappa > \kappa_z$. Since $\nu \le 1$, Π_0 is concave, so Π is concave, and strictly concave if H is strictly convex. Its right derivative,

$$\Pi'(\kappa) = S_k \mathbf{1}\{\kappa < \kappa_z\} + \nu S_k \mathbf{1}\{\kappa \ge \kappa_z\} - H'(\kappa),$$

is nonincreasing and right-continuous. So the set $\mathcal{P} = \{\kappa \in [\kappa_0, \bar{\kappa}] : \Pi'(\kappa) > 0\}$ is an interval $[\kappa_0, p)$ (empty when $p = \kappa_0$), Π is strictly increasing on $[\kappa_0, \min\{p, \bar{\kappa}\}]$ and nonincreasing to the right of p. Hence $\min\{\bar{\kappa}, \sup \mathcal{P}\}$ maximizes Π, with $\sup \emptyset = \kappa_0$, and it is the unique maximizer when Π is

strictly concave. In the formulas below a threshold may be $+\infty$, which the outer minimum with $\bar{\kappa}$ handles. By right-continuity of H', $H'(\kappa) < x$ if and only if $\kappa < \kappa_x$.

Case $\kappa_z \leq \kappa_0$. On the whole interval $\Pi' = \nu S_k - H'$, so $\mathcal{P} = [\kappa_0, \kappa_\nu) \cap [\kappa_0, \bar{\kappa}]$ and the maximizer is $\min\{\bar{\kappa}, \kappa_\nu\}$. Because $\min\{\kappa_z, \kappa_1\} \leq \kappa_0 \leq \kappa_\nu$, this is the formula.

Case $\kappa_z > \kappa_0$. Here $\kappa \in \mathcal{P}$ if and only if $\kappa < \min\{\kappa_z, \kappa_1\}$, or $\kappa_z \leq \kappa < \kappa_\nu$. If $\kappa_1 < \kappa_z$, then $\kappa_\nu \leq \kappa_1 < \kappa_z$, the second set is empty, and $\sup \mathcal{P} = \kappa_1 = \max\{\kappa_\nu, \min\{\kappa_z, \kappa_1\}\}$. If $\kappa_1 \geq \kappa_z$, then $\mathcal{P} = [\kappa_0, \max\{\kappa_z, \kappa_\nu\})$ (intersected with $[\kappa_0, \bar{\kappa}]$), and again $\sup \mathcal{P} = \max\{\kappa_\nu, \min\{\kappa_z, \kappa_1\}\}$. In both cases $Q(\kappa^*) = (\kappa_z - \kappa^*)S_k$.

(a) If $\nu = 0$, then $\kappa_\nu = \kappa_0$ because $H' \geq 0$. The formula becomes $\min\{\bar{\kappa}, \max\{\kappa_0, \min\{\kappa_z, \kappa_1\}\}\}$, which equals κ_0 if $Q_0 \leq 0$ (then $\kappa_z \leq \kappa_0$) and $\min\{\kappa_z, \kappa_1, \bar{\kappa}\}$ otherwise. If $Q_0 \geq 0$, then $\kappa^* \leq \kappa_z$, so $Q(\kappa^*) \geq 0$ and $\Lambda^* = 0$. If $Q_0 < 0$, then $\Lambda^* = -Q(\kappa_0) = -Q_0$. In both cases $\Lambda^* = [-Q_0]^+$.

(b) Since $\kappa_z \in [\kappa_0, \bar{\kappa})$, $H'(\kappa_z) < \nu S_k$ means $\kappa_z < \kappa_\nu$, and $\kappa_\nu \leq \kappa_1$. So $\min\{\kappa_z, \kappa_1\} = \kappa_z$ and $\kappa^* = \min\{\bar{\kappa}, \kappa_\nu\} > \kappa_z$. Then $\Lambda^* = -Q(\kappa^*) = (\kappa^* - \kappa_z)S_k > 0$. If $Q_0 < 0$, the first case applies, $\kappa^* = \min\{\kappa_\nu, \bar{\kappa}\} \geq \kappa_0$, and $\Lambda^* = (\kappa^* - \kappa_z)S_k \geq (\kappa_0 - \kappa_z)S_k = -Q_0$.

(c) κ_1, κ_ν and $\bar{\kappa}$ do not depend on Q_0, and $\partial\kappa_z/\partial Q_0 = 1/S_k$. In the interior of each regime the active branch of the formula is locally constant in Q_0, except in regime (i), where it is κ_z. With $Q(\kappa^*) = (\kappa_z - \kappa^*)S_k$, rent $[Q(\kappa^*)]^+$ and slack $[-Q(\kappa^*)]^+$, the derivatives follow. In (i), $Q(\kappa^*) = 0$ and $d\kappa^*/dQ_0 = 1/S_k$. In (ii), and in (iv) with $Q(\bar{\kappa}) > 0$, $\partial Q(\kappa^*)/\partial Q_0 = 1$ and $Q(\kappa^*) > 0$, so the rent rises one for one. In (iii), and in (iv) with $Q(\bar{\kappa}) < 0$, $\partial Q(\kappa^*)/\partial Q_0 = 1$ and $Q(\kappa^*) < 0$, so slack falls one for one.

(d) $\bar{\kappa} = G(s_X + s_L) - G(s_X - \bar{m}_U)$. Differentiating in s_X with s_L and $\bar{m}_U$ fixed gives $g(s_X + s_L) - g(s_X - \bar{m}_U)$. If g is nonincreasing on $[s_X - \bar{m}_U, s_X + s_L]$, the derivative is nonpositive; this holds, for instance, when G is unimodal with its mode at or below $s_X - \bar{m}_U$. That Washington would almost surely defend a loyal Europe is not sufficient: with G standard normal, $s_A = 3$ and $s_X - \bar{m}_U = -4$, $G(s_A) > 0.99$, yet $g(s_A) > g(s_X - \bar{m}_U)$. If $g(s_X + s_L) > g(s_X - \bar{m}_U)$, the derivative is positive. For example, with G standard normal, $s_L = 1$ and $\bar{m}_U = 2$, $\bar{\kappa} = 0.8186$ at $s_X = 0$ and 0.8664 at $s_X = 0.5$. If s_A is held fixed, $\partial\bar{\kappa}/\partial s_X = -g(s_X - \bar{m}_U) \leq 0$ for every G. The statements about access value and about V follow from the chain rule with $s_X = s_0 + s_B$ and s_B nondecreasing in the access value and in V. □

B.9 Proof of Proposition 5

Assumptions. The setup is that of Section 6.1. (1) *Timing.* The ultimatum is issued at the start of period 1, before the period-1 contest. After a transfer, Europe's period-1 premium is $Q_1 - \Delta_c$ and its period-1 rent is $[Q_1 - \Delta_c]^+$. After a refusal, the contest is played after the threat is carried out. (If the period-1 rent were instead settled before the ultimatum and could not be undone, the terms $[Q_1]^+ - [Q_1 - \Delta_c]^+$ would drop out of both ΔR_U and ΔR_H; the two timings differ only when $Q_1 > 0$.) (2) *Contests.* At every contest Washington retains Europe in the sense of Lemma 1: $Q_1 + d_1 < V$, $Q_1 + d_1 \leq \bar{m}_U$, $Q_2 + \bar{I} < V$ and $Q_2 + \bar{I} \leq \bar{m}_U$. The payment T is a spot payment at signing. (3) *Threat.* The threat withdraws in period 1 a bundle with unconditional

part u_1, conditional part d_1 and direct ex post cost $g_1 \geq 0$ to Washington. Its execution cost is $e = g_1 + [Q_1 + d_1]^+ - [Q_1]^+$, and no other period-1 demand draws on the same bite. (4) *Ties.* An indifferent Europe accepts. Washington issues only if it gains strictly (the limit $\iota \to 0^+$ when $\iota = 0$). Europe invests only if that is strictly profitable. After a transfer Washington keeps its baseline provision unless withdrawal is strictly profitable, which it never is. (5) σ, r^c, v_c and ι do not depend on Δ_c; r^c is credible and is executed after any transfer (Corollary 3(d) endogenizes it); $\delta \in (0, 1]$; and for $\Delta_c = \infty$ we set $[-\infty]^+ = 0$ and $I(-\infty) = 0$.

Proof of Proposition 5. Period 2. After the period-1 outcome Europe holds a premium $q \in \{Q_2, Q_2 - \Delta_c\}$. If it invests, the period-2 contest (Lemma 1) gives it $\delta \left[q + \bar{I}\right]^+$, and otherwise $\delta [q]^+$. It invests if and only if $\delta(\left[q + \bar{I}\right]^+ - [q]^+) > K$, that is, if and only if $I(q) = 1$. So Europe's continuation value, in period-1 units and net of K, is $\mathrm{Val}_H(q)$, and Washington's continuation cost is $\mathrm{Rent}_U(q)$. The map $q \mapsto \left[q + \bar{I}\right]^+ - [q]^+$ is nondecreasing, so I is nondecreasing; hence $q + \bar{I}\, I(q)$ and Rent_U are nondecreasing, and $\mathrm{Val}_H(q) = \max\{\delta [q]^+, \delta \left[q + \bar{I}\right]^+ - K\}$ is nondecreasing.

No ultimatum. Europe gets $[Q_1]^+ + \mathrm{Val}_H(Q_2)$ and Washington gets $-[Q_1]^+ - \mathrm{Rent}_U(Q_2)$.

Acceptance. Control passes before the period-1 contest, so Europe's premia are $Q_1 - \Delta_c$ and $Q_2 - \Delta_c$. No stake is triggered, and withdrawing the bundle would cost Washington $g_1 + [Q_1 - \Delta_c + d_1]^+ - [Q_1 - \Delta_c]^+ \geq 0$, so Washington keeps its baseline provision. Europe gets $T - \sigma - r^c + [Q_1 - \Delta_c]^+ + \mathrm{Val}_H(Q_2 - \Delta_c)$, and Washington gets $v_c - \iota - T - [Q_1 - \Delta_c]^+ - \mathrm{Rent}_U(Q_2 - \Delta_c)$.

Refusal. Europe keeps control. If Washington carries out the threat, Europe's aligned payoff falls by $u_1 + d_1$ and its premium rises to $Q_1 + d_1$. By Lemma 1 Europe gets $-(u_1 + d_1) + [Q_1 + d_1]^+$ in period 1, and Washington gets $-g_1 - [Q_1 + d_1]^+$. If Washington does not carry it out, Europe gets $[Q_1]^+$ and Washington gets $-m - [Q_1]^+$, where $m \leq \bar{m}_U$ is the stake. Period 2 is as above with $q = Q_2$ in both cases. Washington carries out the threat if and only if $e \leq m$. Creating a stake costs nothing, so Washington can set $m = \bar{m}_U$, and the threat is carried out if and only if $e \leq \bar{m}_U$. None of these payoffs involves Δ_c, because the threat is carried out only when Europe keeps control. When the threat is carried out, Europe's period-1 loss relative to no ultimatum is $u_1 + d_1 - [Q_1 + d_1]^+ + [Q_1]^+$. It equals u_1 if $Q_1 \geq 0$, $u_1 - Q_1$ if $-d_1 < Q_1 < 0$, and $u_1 + d_1$ if $Q_1 \leq -d_1$, that is, $u_1 + \min\{d_1, \Lambda_1\} = B_1$ in all cases. So Europe's refusal payoff is $[Q_1]^+ - B_1 + \mathrm{Val}_H(Q_2)$, with $B_1 = 0$ if the threat is not credible.

Europe's decision. Europe accepts if and only if $T - \sigma - r^c + [Q_1 - \Delta_c]^+ + \mathrm{Val}_H(Q_2 - \Delta_c) \geq [Q_1]^+ - B_1 + \mathrm{Val}_H(Q_2)$, that is, if and only if $T \geq \Delta R_H - B_1 + \sigma + r^c$.

Washington's decision. With $T \geq 0$ and an indifferent Europe accepting, the cheapest accepted offer is $T^* = [\Delta R_H - B_1 + \sigma + r^c]^+$. Any lower offer is refused, and no optimal offer would exist if Europe refused at indifference. Relative to no ultimatum, the accepted offer gives Washington

$$v_c - \iota + \Delta R_U - T^* = v_c - \iota + \min\{\Delta R_U,\ \Delta R_U - \Delta R_H + B_1 - \sigma - r^c\} = \Gamma,$$

using $x - [y]^+ = \min\{x, x - y\}$. A refused ultimatum gives Washington $-\iota - e$ relative to no ultimatum when the threat is credible, and $-\iota - m \leq -\iota$ otherwise. Since $e \geq 0$, this is never strictly positive.

Hence Washington issues if and only if $\Gamma > 0$, and every ultimatum issued is accepted. Europe's payoff is $T^* - \sigma - r^c + [Q_1 - \Delta_c]^+ + \mathrm{Val}_H(Q_2 - \Delta_c)$, and $T^* - \sigma - r^c = \max\{\Delta R_H - B_1,\ -\sigma - r^c\}$. $\square$

The special case. If $v_c = \iota = 0$, $\Delta_c = \infty$ and $\bar{I} = 0$, then $\Delta R_U = \Delta R_H = [Q_1]^+ + \delta\,[Q_2]^+$, and substituting into Γ gives $\Gamma = \min\{[Q_1]^+ + \delta\,[Q_2]^+,\ B_1 - \sigma - r^c\}$. If also $Q_1 < 0$, then $\Gamma = \min\{\delta\,[Q_2]^+,\ B_1 - \sigma - r^c\}$. These are the expressions in the text after Proposition 5. Since $B_1 \le u_1 + \Lambda_1$, a bundle with $u_1 = 0$ has $B_1 > \sigma + r^c$ only if $\Lambda_1 > \sigma + r^c$, that is, $Q_1 < -(\sigma + r^c)$.

B.10 Proof of Corollary 3

Assumptions. Those of Proposition 5. In part (d), r^c is China's announced punishment, and it enters Proposition 5 only if it is credible.

Proof of Corollary 3. (a) The first two statements restate $\Gamma > 0$, using $\min\{a, b\} > x$ if and only if $a > x$ and $b > x$. For the third, $[Q_1]^+ - [Q_1 - \Delta_c]^+ = \min\{[Q_1]^+, \Delta_c\}$, which is positive if and only if $Q_1 > 0$. Let $q = Q_2$ and $q' = Q_2 - \Delta_c < q$. Since I is nondecreasing, $q' + \bar{I}\,I(q') < q + \bar{I}\,I(q)$, so $\mathrm{Rent}_U(q) - \mathrm{Rent}_U(q') > 0$ if and only if $q + \bar{I}\,I(q) > 0$ (with $\delta > 0$). Both terms of ΔR_U are nonnegative, which gives the claim.

(b) The period-1 terms of ΔR_U and ΔR_H coincide, and $\mathrm{Rent}_U(q) - \mathrm{Val}_H(q) = K\,I(q)$, which gives the identity. It is nonnegative because I is nondecreasing. Without the investment option ($\bar{I} = 0$), both ΔR_U and ΔR_H equal $\Delta R^0 = [Q_1]^+ - [Q_1 - \Delta_c]^+ + \delta([Q_2]^+ - [Q_2 - \Delta_c]^+)$, so the second argument of the minimum in Γ is $B_1 - \sigma - r^c$; with the option it is larger by $K[I(Q_2) - I(Q_2 - \Delta_c)] \ge 0$. For the first argument, let $q = Q_2$, $q' = q - \Delta_c$ and $\mathcal{G}(x) = [x]^+ - [x - \Delta_c]^+$, which is nondecreasing. If $I(q) = I(q') = 0$, the period-2 terms with and without the option are equal. If $I(q) = I(q') = 1$, $\mathrm{Rent}_U(q) - \mathrm{Rent}_U(q') = \delta\,\mathcal{G}(q + \bar{I}) \ge \delta\,\mathcal{G}(q)$. If $I(q) = 1 > I(q')$, $\mathrm{Rent}_U(q) - \mathrm{Rent}_U(q') = \delta(\left[q + \bar{I}\right]^+ - [q']^+) \ge \delta([q]^+ - [q']^+)$. The case $I(q) = 0 < I(q')$ cannot occur. So $\Delta R_U \ge \Delta R^0$. The statement on purchases without bite is (a) with $B_1 = 0$ and the identity just proved. In the example of the text ($\delta = 1$, $Q_1 = 1$, $Q_2 = -\frac{1}{2}$, $\bar{I} = 3$, $K = 2$, $\Delta_c = \infty$, $\sigma = 1$, $r^c = v_c = \iota = 0$, no threat), $I(Q_2) = 1$ because $[2.5]^+ - 0 > 2$, and $I(-\infty) = 0$. So $\Delta R_U = 1 + 2.5 = 3.5$ and $\Delta R_H = 1 + 0.5 = 1.5$, which give $\Gamma = \min\{3.5,\ 2 + 0 - 1\} = 1$ and $T^* = [1.5 + 1]^+ = 2.5$.

(c) $[Q_1 - \Delta_c]^+$, $\mathrm{Rent}_U(Q_2 - \Delta_c)$ and $\mathrm{Val}_H(Q_2 - \Delta_c)$ are nonincreasing in Δ_c, because $[\cdot]^+$, Rent_U and Val_H are nondecreasing. So ΔR_U and ΔR_H are nondecreasing in Δ_c, and so is $\Delta R_U - \Delta R_H = K[I(Q_2) - I(Q_2 - \Delta_c)]$, because I is nondecreasing. Both arguments of the minimum in Γ are therefore nondecreasing in Δ_c, and T^* is nondecreasing because ΔR_H is. B_1 and e are computed in the refusal branch, where Europe keeps control, so they do not depend on Δ_c.

(d) Given an announcement r^c, Europe and Washington play the game of Proposition 5 with that punishment if the threat is credible and with a punishment of zero otherwise. $\Gamma(r^c)$ is nonincreasing in r^c. After a transfer China punishes if and only if $e^C(r^c) \le m_C$. (i) Suppose control passes on the path. Then $\Gamma(r^c) > 0$ for the effective r^c, so $\Gamma(0) \ge \Gamma(r^c) > 0$. If $r^c > 0$, China punishes and gets $-e^C(r^c) < 0$. If the threat is not credible but $m_C > 0$, China forfeits m_C. In either case announcing $r^c = 0$ with no stake leaves the transfer in place, because $\Gamma(0) > 0$, and gives China 0, a profitable

deviation. So on the path China gets 0, the effective r^c is 0, and the payment is $[\Delta R_H - B_1 + \sigma]^+$. (ii) By (i), a credible $r^c > 0$ on the path is not followed by a transfer, so Washington does not issue, which by Proposition 5 means $\Gamma(r^c) \leq 0$. (iii) If $\Gamma(0) \leq 0$, no announcement produces a transfer. If $\Gamma(r^c_{\max}) > 0$, every announcement produces a transfer and, by (i), China announces no credible punishment. If $\Gamma(r^c_{\max}) \leq 0 < \Gamma(0)$, announcing $r^c_{\max}$ with $m_C = \bar{m}_C$ deters and gives China $z_C > 0$, while any announcement that lets the transfer happen gives it at most 0, so China deters. The deterring credible announcements are those with $\Gamma(r^c) \leq 0$, that is, $r^c \geq v_c - \iota + \Delta R_U - \Delta R_H + B_1 - \sigma$; here we use $v_c - \iota + \Delta R_U > 0$, which follows from $\Gamma(0) > 0$. (iv) With $z_C = 0$, China gets 0 both from deterring and from announcing nothing and letting the transfer happen. Finally, if $e^C(x) < 0$ for some x, a transfer followed by punishment gives China $-e^C(x) > 0$, which can exceed z_C, so the argument of (i) fails. □

The argument in (d)(i) relies on complete information. Under the private information of Proposition 7, types that accept are punished on the path, so a punishment with negative ex post value is carried out with positive probability whenever Washington issues and $F(\hat{s}) > 0$.

B.11 Proof of Lemma 2

Proof of Lemma 2. Fix $s \in S$. Without the cut, Lemma 1 gives Europe a date-s payoff of $[Q_s]^+$ above its aligned baseline and costs Washington $[Q_s]^+$. With the cut, Europe's aligned payoff falls by $u_s + d_s$ and its exit payoff by u_s, so its premium becomes $Q_s + d_s$. Since Washington still retains Europe, Lemma 1 gives Europe $-(u_s + d_s) + [Q_s + d_s]^+$ and costs Washington $g_s + [Q_s + d_s]^+$. Europe's date-s loss is $u_s + d_s - [Q_s + d_s]^+ + [Q_s]^+$, which equals u_s if $Q_s \geq 0$, $u_s - Q_s$ if $-d_s < Q_s < 0$, and $u_s + d_s$ if $Q_s \leq -d_s$, that is, $u_s + \min\{d_s, \Lambda_s\}$. Washington's extra date-s cost is e_s. Because the cut changes no other date's premium, the date-s effects add up with weights δ^{s-1}, which gives (i) and the first part of (ii). At date s, once the condition of the threat is met, Washington's continuation does not depend on whether it withdraws, so it withdraws if and only if $-e_s \geq -m_s$. For (iii), $Q_s \geq 0$ gives $\Lambda_s = 0$ and $[Q_s + d_s]^+ - [Q_s]^+ = d_s$. □

Scope. Both conditions in the lemma are needed, and their failures push in opposite directions. (1) If $Q_s + d_s \geq \min\{V_s, \bar{m}_U\}$, the cut drives Europe out at s. By Lemma 1 Europe's date-s loss is then $u_s + d_s - [\min\{Q_s + d_s, V_s, \bar{m}_U\}]^+ + [\min\{Q_s, V_s, \bar{m}_U\}]^+$, which exceeds the formula. (2) If Europe responds to the cut, for instance with the autonomy investment of Section 6.1 when it learns of the cut before deciding, the bite is smaller. Take $\delta = 1$, $Q_2 = -1$, $d_2 = 1$, $u_2 = 0$, $\bar{I} = 3$ and $K = 2.5$. Without the cut Europe does not invest and gets 0. With it the premium is 0, Europe invests and gets $3 - 2.5 - 1 = -0.5$, so the bite is 0.5, while the formula gives 1.

B.12 Proof of Corollary 4

Assumptions. Those of Proposition 5 at every date, with $v_c = 0$, $\Delta_c = \infty$ and no investment option. A refusal leaves Europe in control at the next date. Washington may issue ultimatums at

several dates.

Proof of Corollary 4. Let $R_s = \sum_{\tau \geq s} \delta^{\tau - s} [Q_\tau]^+$ be the value at date s of the rents from s on.

Step 1. We show by backward induction that if no date in $\{s, \ldots, N\}$ is eligible, no ultimatum is issued in the subgame that starts at s with Europe in control. Suppose this holds from $s+1$. Then at s the continuation after a refusal and after no ultimatum is the same: rents worth δR_{s+1} to Europe and $-\delta R_{s+1}$ to Washington. As in the proof of Proposition 5, with the period-2 terms replaced by δR_{s+1}, $\Delta_c = \infty$ and no investment, the cut is carried out after a refusal if and only if $e_s \leq \bar{m}_U$; Europe's refusal loss is b_s if so and 0 otherwise; and an accepted ultimatum gives Washington $-\iota + \min\{R_s, b_s^{\text{eff}} - \sigma - r^c\}$ relative to no ultimatum, where b_s^{eff} is b_s or 0. A refused ultimatum gives at most $-\iota$. Since s is not eligible, $R_s = 0$ or $b_s^{\text{eff}} \leq \sigma + r^c$, so Washington's gain from issuing is at most $-\iota \leq 0$, and it does not issue.

Step 2. Suppose some date is eligible and let t^* be the last one. By Step 1, no ultimatum is issued after t^*, so in the subgame at t^* with Europe in control Washington's gain from issuing is $-\iota + \min\{R_{t^*}, b_{t^*} - \sigma - r^c\}$. Eligibility gives $b_{t^*} > \sigma + r^c$ and $R_{t^*} > 0$ (some $Q_{t'} > 0$ with $t' \geq t^*$, and $\delta > 0$). Let $\bar{\iota} = \min\{R_{t^*}, b_{t^*} - \sigma - r^c\} > 0$. For $\iota < \bar{\iota}$, Washington issues at t^* if Europe still holds control there. Hence on the path an ultimatum is issued at t^* or earlier. If no date is eligible, Step 1 at $s = 1$ shows that none is issued.

Step 3. At any date, a refused ultimatum leaves Europe in control at the next date and costs Washington ι plus the execution cost $e_t \geq 0$ or the forfeited stake, so it is never strictly better than issuing nothing. Hence every ultimatum issued is accepted. With $\iota = 0$ and issuing only on a strict gain, Steps 1 to 3 apply with $\bar{\iota}$ replaced by any positive number.

Step 4. If $u_t = 0$, then $b_t = \min\{d_t, \Lambda_t\} > \sigma + r^c \geq 0$ requires $\Lambda_t > 0$, that is, $Q_t < 0$. So a date $t' \geq t$ with $Q_{t'} > 0$ satisfies $t' > t$. □

The condition $t' \geq t$ cannot be replaced by $t' > t$ when some $u_t > 0$. With $N = 2$, $\delta = 1$, $Q = (1, -5)$, $u_1 = 3$, $d_1 = g_1 = 0$ and $\sigma + r^c = 1$, no date has a later date with positive premium, yet at date 1 Washington gains $\min\{[Q_1]^+ + \delta [Q_2]^+, u_1 - \sigma - r^c\} = 1 > 0$ and issues, because the transfer precedes the date-1 contest and removes the date-1 rent.

B.13 Proof of Remark 2

Assumptions. Those of Proposition 5. In the example of part (c), $\Delta_c = \infty$, there is no investment option, and $\bar{m}_U \geq 1$, so that the cut is credible after binding.

Proof of Remark 2. (a) With $\Delta_b = \infty$, $[Q_t - \Delta_b]^+ = 0$, and Rent_U and Val_H vanish at both premia, before and after a transfer, so $\Delta R_U = \Delta R_H = 0$. Europe's period-1 slack is unbounded, so the bite of the threat is $u_1 + \min\{d_1, \infty\} = u_1 + d_1$, and its execution cost is $g_1 + [-\infty + d_1]^+ - 0 = g_1$. Proposition 5 then gives Γ^b and the payment $\left[0 - B_1^b + \sigma + r^c\right]^+$. Binding pre-empts the ultimatum if and only if $\Gamma^b \leq 0$, that is, $v_c \leq \iota$ or $v_c - \iota + B_1^b \leq \sigma + r^c$.

(b) After binding, Europe pays $\sigma^b + r^b$ and earns no rents. If $\Gamma^b > 0$ it also accepts the ultimatum and gets $\left[\sigma + r^c - B_1^b\right]^+ - \sigma - r^c = -\min\{\sigma + r^c, B_1^b\}$. Without binding, Proposition 5 with $Q_1 < 0$, $\Delta_c = \infty$ and no investment gives $\Delta R_H = \delta\,[Q_2]^+$, and Europe's payoff is $\max\{\delta\,[Q_2]^+ - B_1,\ -\sigma - r^c\}$ if $\Gamma > 0$ and $[Q_1]^+ + \delta\,[Q_2]^+ = \delta\,[Q_2]^+$ if $\Gamma \le 0$. Comparing gives the statements. In the last case, $-\sigma^b - r^b - \mathbf{1}\{\Gamma^b > 0\}\min\{\sigma + r^c, B_1^b\} \le 0 \le \delta\,[Q_2]^+$.

(c) Without binding, $Q_1 = -1$ gives $B_1 = \min\{5, 1\} = 1$ (if the cut is credible; otherwise $B_1 = 0$) and $\Delta R_U = \Delta R_H = [Q_1]^+ + \delta\,[Q_2]^+ = 10$, so $\Gamma \le \min\{10,\ 1 - 1.5\} < 0$. With $\Delta_b = 3$, the premia are -4 and 7, the cut's execution cost is $0 + [-4 + 5]^+ - 0 = 1 \le \bar{m}_U$, $B_1 = \min\{5, 4\} = 4$ and $\Delta R_U = \Delta R_H = 7$. So $\Gamma = \min\{7,\ 4 - 1.5\} = 2.5$ and $T^* = [7 - 4 + 1.5]^+ = 4.5$.

(d) Washington's gain from demanding an arrangement is its Γ, by Proposition 5 applied with that arrangement's parameters. It demands the arrangement with the larger gain and issues only if that gain is positive. Since $\sigma_d < \sigma$, the second argument of the minimum in Γ_{dual} can be positive when that of Γ_{full} is not. □

By the argument of Corollary 3(d), if China's ex post value of punishing a binding rule is negative, a binding rule that Europe adopts on the path is not punished, and r^b matters only through deterrence. We have not checked this step numerically.

B.14 Proof of Proposition 7

Assumptions. Those of Proposition 5, with the conditions in the statement. Washington may also offer to buy control without staking the threat (a stake of zero), which is the purchase offer of part (b). Ties between accepting and refusing have probability zero.

Proof of Proposition 7. Acceptance. A type that accepts gets $T - \sigma - r^c$; no rents remain because $\Delta_c = \infty$. Refusing a threat-backed ultimatum gives $[Q_1]^+ - B_1 + R = R - B_1$ by the refusal step of the proof of Proposition 5, and refusing a purchase offer gives R. So Europe accepts if and only if $\sigma \le T + \tilde{b} - R$, respectively $\sigma \le T - r^c - R$.

Washington's gain. Relative to no offer, an accepted offer gives Washington $v_c + R - T - \iota$. A refused threat-backed ultimatum gives $-e - \iota$, since $e \le \bar{m}_U$ and Washington carries out the threat; a refused purchase offer gives $-\iota$. A stake in $(0, e)$ is never executed and is forfeited after a refusal, so it is dominated by a stake of zero. Write the cutoff as $s = T + \tilde{b} - R$ for the threat and $s = T - r^c - R$ for the purchase offer. Washington's gain is then $\Pi_U(s; \tilde{b} + v_c, e) - \iota$ subject to $s \ge \tilde{b} - R$, respectively $\Pi_U(s; v_c - r^c, 0) - \iota$ subject to $s \ge -r^c - R$, where the constraints are $T \ge 0$.

Shape of Π_U. For $s \le 0$, $\Pi_U(s; A, x) = -x$. For $s \ge \bar{\sigma}$, $\Pi_U = A - s$, which is decreasing. On $(0, \bar{\sigma})$, $\partial\Pi_U/\partial s = F'(s)(A - s) - F(s) + F'(s)x = F'(s)\,[A + x - s - F(s)/F'(s)]$. Log-concavity of F means that $(\log F)' = F'/F$ is nonincreasing, so F/F' is nondecreasing and the bracket is strictly decreasing in s. As $s \downarrow 0$, $\log F(s) \to -\infty$; a concave function with a bounded right derivative near 0 would stay bounded, so $F'/F \to \infty$ and $F/F' \to 0$. Hence the bracket is positive near 0 if and only if $A + x > 0$, it changes sign at most once, and Π_U is nondecreasing on $(-\infty, s^*(A, x)]$ and nonincreasing on $[s^*(A, x), \infty)$, with s^* as defined in the statement.

(a) A function that rises up to s^* and falls after it attains its maximum over $s \geq L$ at $\max\{L, s^*\}$. With $L = \tilde{b} - R$ and $s^* \leq \bar{\sigma}$, this is $\hat{s}$, and the payment is $T = \hat{s} - \tilde{b} + R \geq 0$. Refusal, and with it the execution of the threat, has probability $1 - F(\hat{s})$. For uniform F, $F(s)/F'(s) = s$, so the first-order condition gives $s = (\tilde{b} + v_c + e)/2$ when this lies in $(0, \bar{\sigma})$, and $s^* = \bar{\sigma}$ when it is at least $\bar{\sigma}$; hence $s^* = \min\{\bar{\sigma}, (\tilde{b} + v_c + e)/2\}$.

(b) Here $L = -r^c - R \leq 0$ and Π_U is constant on $(-\infty, 0]$, so the constraint never binds and the cutoff is s_P. If $v_c \leq r^c$, then $\Pi_U(s; v_c - r^c, 0) = F(s)(v_c - r^c - s) \leq 0$ for $s \geq 0$, and $\Pi_U = 0$ for $s \leq 0$, so $\Pi_P \leq -\iota$.

(c) Washington compares no offer (0), the best threat-backed ultimatum (Π_B) and the best purchase offer (Π_P). If $v_c \leq r^c$, then $\Pi_P \leq -\iota \leq 0$, so Washington makes an offer if and only if $\Pi_B > 0$, and every offer is threat-backed; $\Pi_B > 0$ is the stated inequality.

(d) The bracket is increasing in $A + x$, so $s^*(A, x)$ is nondecreasing in $A + x = B_1 - r^c + v_c + e$. Hence $\hat{s} = \max\{B_1 - r^c - R,\, s^*\}$ is nondecreasing in B_1, v_c and e, nonincreasing in r^c and R, and independent of ι, and $P_b = 1 - F(\hat{s})$ moves the opposite way. Without log-concavity: $\partial\Pi_U/\partial A = F(s)$ and $\partial\Pi_U/\partial x = F(s) - 1$ are nondecreasing in s, so Π_U has increasing differences in (s, A) and in (s, x). The feasible set $[B_1 - r^c - R, \infty)$ rises with B_1 (as does A), falls with r^c (as does A) and falls with R (with A unchanged), and it does not depend on v_c or e. Π_U is continuous and decreasing beyond $\bar{\sigma}$, so the set of maximizers is nonempty and compact. By Topkis's monotonicity theorem it moves in the same direction as the parameter in the strong set order, so its largest and smallest elements are monotone.

(e) Two facts. First, $\hat{s} \geq s_P$, because $\hat{s} \geq s^*(\tilde{b} + v_c, e) \geq s^*(v_c - r^c, 0) = s_P$, as $(\tilde{b} + v_c + e) - (v_c - r^c) = B_1 + e \geq 0$. Second, the direction in which Washington can switch. In terms of the payment, the threat-backed gain is $\pi_B(T) = F(T + \tilde{b} - R)(v_c + R - T + e) - e - \iota$. Only $T \leq v_c + R + e$ can be optimal, since a larger T makes acceptance worse for Washington than refusal. On that range $\pi_B(T)$ is nondecreasing in B_1 and nonincreasing in e (its derivative in e is $F - 1 \leq 0$), so Π_B is nondecreasing in B_1 and nonincreasing in e. In the cutoff form, R does not enter $\Pi_U(s; \tilde{b} + v_c, e)$ and only relaxes the constraint $s \geq \tilde{b} - R$, so Π_B is nondecreasing in R. Π_P depends on none of B_1, e and R. For v_c: in the cutoff forms the feasible sets do not depend on v_c, and each objective is affine in v_c with slope $F(s)$, so by the envelope theorem $\partial\Pi_B/\partial v_c = F(\hat{s}) \geq F(s_P) = \partial\Pi_P/\partial v_c$ wherever the derivatives exist, and $\Pi_B - \Pi_P$ is nondecreasing in v_c. So as B_1, R or v_c rises, Washington can switch only from the purchase offer to the threat, and as e rises only from the threat to the purchase offer. Within each option the cutoff moves as in (d); for the purchase offer, s_P is nondecreasing in v_c and independent of B_1, e and R. Combining: as B_1 or v_c rises, the cutoff never falls (a switch moves it from s_P up to $\hat{s}$), so the refusal probability is nonincreasing. As e rises, the breakdown probability is nonincreasing within the threat and drops to zero at a switch. As R rises, it is nondecreasing within the threat and rises from zero at a switch. For r^c: where the threat-backed payment is positive, the constraint does not bind, and the envelope theorem gives $\partial\Pi_B/\partial r^c = -F(\hat{s}) \leq -F(s_P) = \partial\Pi_P/\partial r^c$; the purchase payment $s_P + r^c + R$ is positive when $v_c > r^c$, because then $s_P > 0$. So switches go from the threat to the purchase offer, the cutoff falls from $\hat{s}$ to s_P, and the refusal probability rises;

within each option it is nondecreasing in r^c by (d) and because s_P is nonincreasing in r^c. For the remaining directions, take uniform F on $[0, 10]$, $R = 0$ and $\iota = 0$. Raising e from 0.2 to 20 (with $B_1 = 1$, $r^c = 0$, $v_c = 8$) moves Washington from the threat (refusal probability 0.54) to the purchase offer (refusal probability 0.6). Raising B_1 from 1 to 2 (with $e = 1$, $v_c = 8$, $r^c = 0$) moves it from the purchase offer (breakdown probability 0) to the threat (0.45). Raising v_c from 8 to 10 (with $B_1 = e = 1$, $r^c = 0$) moves it from the purchase offer to the threat (breakdown probability from 0 to 0.4). Raising r^c from 0 to 2 (with $B_1 = e = 1$, $v_c = 10$) moves it from the threat (breakdown probability 0.4) to the purchase offer (0). □

B.15 Proof of Remark 3

Assumptions. The benefit x_h is part of the baseline: it is provision that Washington finds ex post optimal for a loyal Netherlands, such as a preferential tier of the AI instrument, and its withdrawal on exit is credible (its execution cost is at most $\bar{m}_U$). The conditions of Lemma 1 hold before and after the benefit, except where the statement treats the case $Q_{NL} > \min\{V, \bar{m}_U\}$.

Proof of Remark 3. Credible withdrawal on exit adds x_h to the conditional part of Washington's instruments for the Netherlands, so c_{NL} rises by x_h and Q_{NL} falls by x_h. Lemma 1 gives the payment $[Q]^+$ at premium Q. For $x \geq 0$: if $Q \leq 0$, then $[Q]^+ - [Q - x]^+ = 0 = \min\{x, [Q]^+\}$ and $[x - Q]^+ - [-Q]^+ = x = \left[x - [Q]^+\right]^+$; if $Q > 0$, then $Q - [Q - x]^+ = \min\{x, Q\}$ and $[x - Q]^+ - 0 = \left[x - [Q]^+\right]^+$. With $Q = Q_{NL}$ and $x = x_h$, these are the stated changes in the payment and in the slack. If $Q_{NL} > \min\{V, \bar{m}_U\}$, Lemma 1(ii) applied to $Q'_{NL} = Q_{NL} - x_h$ gives alignment if and only if $Q_{NL} - x_h \leq \bar{m}_U$ and $Q_{NL} - x_h < V$. If instead x_h is a promised increment above the baseline, it is a transfer paid on the path: the Netherlands aligns if and only if $x_h + \tau \geq Q_{NL}$, so Washington's cheapest total outlay is $x_h + [Q_{NL} - x_h]^+ = \max\{x_h, Q_{NL}\} \geq [Q_{NL}]^+$. □

B.16 Proof of Remark 4

Assumptions. Under a licence, the firm compares shipping, worth $y - f^F$, with complying with US measures, worth $-p_C - p_B$, and ships when indifferent. The penalty $f^F = \chi_f f$ is the credible one, so it is zero when the threat to penalize the firm is not credible. p_B is positive only if the US measures are listed in the annex of the Blocking Statute. The compulsion cost ζ excludes the firm's own net loss, which already enters ω and c through y and d_f.

Proof of Remark 4. The firm ships if and only if $y - f^F \geq -p_C - p_B$, that is, $y + p_C + p_B \geq f^F$. If it ships, exit needs nothing further and Europe's premium is Q. If it does not, exit requires the state to compel it at cost ζ, which lowers Europe's exit payoff, and hence its premium, by ζ. So Europe's effective premium is $Q - \zeta\,\mathbf{1}\{y + p_C + p_B < f^F\}$. By Lemma 1, Europe exits only if its effective premium is positive, so it compels the firm only if $Q - \zeta > 0$. □

B.17 Proof of Remark 5

Assumptions. The determination is a Council implementing act adopted by qualified majority on a Commission proposal (European Parliament and Council of the European Union, 2023). Under Article 16(4) TEU and Article 238(3)(a) TFEU, a qualified majority needs at least 55 percent of the members (15 of 27) representing at least 65 percent of the population, and a blocking minority must include at least four members, failing which the qualified majority is deemed attained (European Union, 2016). Abstention and absence count as not voting in favour. Response measures are Commission implementing acts under the examination procedure; in the appeal committee the Commission may adopt unless a qualified majority votes against (European Parliament and Council of the European Union, 2011). A Council objection to a delegated act needs at least 72 percent of the members (20 of 27) representing at least 65 percent of the population (Article 238(2) TFEU). Member i withholds its vote from the countermeasure if it is paid $\tau_i \geq q_i$, payments are enforceable and contingent on the vote, and no one bids for adoption. Population shares are those of Council Decision 2025/2427 (Council of the European Union, 2025).

Proof of Remark 5. The blocking family. Let N be the set of members that do not vote in favour and Y its complement. The determination fails if and only if $|N| \geq 4$ and either $|Y| < 15$ or Y holds less than 65 percent of the population. Since $|Y| < 15$ is $|N| \geq 13$, and Y holds less than 65 percent if and only if N holds more than 35 percent, N blocks if and only if $|N| \geq 13$, or $|N| \geq 4$ and N holds more than 35 percent of the population. This is the family $\mathcal{B}$. It is upward closed: adding members raises both the count and the population.

The cost of blocking. A member with $q_i > 0$ must be paid q_i to withhold its vote, and a member with $q_i \leq 0$ withholds it for nothing, so blocking with a coalition J costs $\sum_{i \in J} [q_i]^+$, and the cheapest block costs $\Phi(0) = \min_{J \in \mathcal{B}} \sum_{i \in J} [q_i]^+$. Adding the members of $F = \{i : q_i \leq 0\}$ to any blocking coalition costs nothing and preserves blocking, because $\mathcal{B}$ is upward closed. So some cheapest blocking coalition contains F, and $\Phi(0)$ is the cost of the cheapest completion of F into a member of $\mathcal{B}$. If $F \in \mathcal{B}$, then $\Phi(0) = 0$. Conversely, if $\Phi(0) = 0$, a cheapest blocking coalition consists of members with $[q_i]^+ = 0$, that is, it is a subset of F, and $F \in \mathcal{B}$ by upward closure.

The coalitions. The six frontline states have fewer than 13 members and 15.09 percent of the population, so they do not block. The thirteen eastern and Nordic members of Table 5 hold 26.77 percent and block by number.

Later stages. Defeating a response measure in the appeal committee requires a qualified majority against, hence at least 15 members, and a Council objection to a delegated act requires at least 20. Either coalition has at least 13 members, so it belongs to $\mathcal{B}$, and paying its members to vote against costs at least $\sum_{i \in J} [q_i]^+ \geq \Phi(0)$. □

B.18 Proof of Proposition 6

Assumptions. Rounds start at $t_k = k\Delta$. After a rejection, if Washington proposed, Europe may exit, which gives Europe Q and Washington $-V$ in normalized flow units. Otherwise rupture

occurs with probability $\mu = 1 - e^{-\lambda\Delta}$, with the same values. Otherwise the status quo runs for the period with normalized flows $-c_E(t_k)$ and $-c_U(t_k)$ (weight $1-\delta$, with $\delta = e^{-r\Delta}$), and play moves to round $k+1$. The costs are measured in the same normalized flow units as Q and V. They are bounded, each interval of unit length contains at most N_d of their discontinuity points, and between discontinuities they are uniformly continuous. "Markov" means that strategies depend only on the round index (calendar time) and the current offer. We write $\upsilon = \delta(1-\mu) = e^{-(r+\lambda)\Delta}$, $c_{i,k} = c_i(t_k)$ and $n_k = n(t_k)$, and $\bar{a}$ denotes a constant that does not depend on Δ and may change from line to line.

Proof of Proposition 6. Recursion. Fix Δ and a Markov perfect equilibrium, and let x_k and y_k be Europe's and Washington's values at the start of round k. In a Markov equilibrium the continuation after a rejection does not depend on the rejected offer. If Washington rejects Europe's demand in round k, its value is $\varrho_k = -\mu V + (1-\mu)[-(1-\delta)c_{U,k} + \delta y_{k+1}]$. If Europe rejects Washington's offer and does not exit, its value is $d_k = \mu Q + (1-\mu)[-(1-\delta)c_{E,k} + \delta x_{k+1}]$, so its value of rejecting is $\max\{Q, d_k\}$. A responder accepts an offer exactly equal to its rejection value, since otherwise the proposer would have no best offer. So the proposer's best acceptable offer gives Europe $-\varrho_k$ when Europe proposes and Washington $-\max\{Q, d_k\}$ when Washington proposes.

Every outcome is an agreement (whose payoffs sum to zero), exit (summing to $Q - V < 0$) or disagreement flows (summing to at most zero), so $x_k + y_k \le 0$ for every k. The proposer's gain from its best acceptable offer over an unacceptable one is, when Europe proposes,

$$\begin{aligned} -\varrho_k - d_k &= \mu(V-Q) + (1-\mu)(1-\delta)(c_{E,k} + c_{U,k}) - (1-\mu)\delta(x_{k+1} + y_{k+1}) \\ &\ge \mu(V-Q) + (1-\mu)(1-\delta)(c_{E,k} + c_{U,k}), \end{aligned}$$

and when Washington proposes it is the same expression if Europe would not exit after rejecting, and $V - Q > 0$ if it would. The gain is strictly positive if $\lambda > 0$ (because $Q < V$) or $c_{E,k} + c_{U,k} > 0$, and then agreement in round k is immediate. If the gain is zero, then $\mu = 0$, $c_{E,k} + c_{U,k} = 0$ and $x_{k+1} + y_{k+1} = 0$, and an unacceptable offer gives both players the same payoffs as an acceptable one, so delay is payoff equivalent. In either case the payoffs in round k sum to zero, so $y_k = -x_k$ for every k. Substituting $y_{k+1} = -x_{k+1}$,

$$\begin{aligned} x_k = \mathcal{T}_k(x_{k+1}) &\equiv \beta\big[\mu V + (1-\mu)\big((1-\delta)c_{U,k} + \delta x_{k+1}\big)\big] + (1-\beta)\max\{Q, d_k\} \\ &= \max\{\mathcal{T}_k^A(x_{k+1}), \mathcal{T}_k^B(x_{k+1})\}, \end{aligned}$$

$$\mathcal{T}_k^A(x) = \mu M + (1-\mu)(1-\delta)n_k + \upsilon x, \qquad \mathcal{T}_k^B(x) = (1-\beta)Q + \beta\big[\mu V + (1-\mu)(1-\delta)c_{U,k} + \upsilon x\big],$$

where $M = Q + \beta(V-Q) = \beta V + (1-\beta)Q$ and $n_k = \beta c_{U,k} - (1-\beta)c_{E,k}$.

Part (i). Each $\mathcal{T}_k$ is nondecreasing with modulus $\upsilon < 1$, so the map $(x_k)_k \mapsto (\mathcal{T}_k(x_{k+1}))_k$ is a contraction on bounded sequences in the supremum norm. Equilibrium values are bounded, so every Markov perfect equilibrium has the same values, the unique fixed point. The profile built from the fixed point (each proposer makes its best acceptable offer, each responder accepts exactly

the offers at or above its rejection value, and Europe exits after rejecting if and only if $Q > d_k$) is an equilibrium by the one-shot deviation principle. The statements on immediate agreement and payoff-equivalent delay were shown above.

Part (ii), policies. For a sequence $\sigma = (\sigma_k)$ with $\sigma_k \in \{\mathcal{T}_k^A, \mathcal{T}_k^B\}$, let v^σ be the bounded solution of $v_k^\sigma = \sigma_k(v_{k+1}^\sigma)$. Since $\sigma_k \le \mathcal{T}_k$ and $\mathcal{T}_k$ is a monotone contraction, $v^\sigma \le x$, and the sequence that picks the larger map at each round attains x. So $x_0 = \max_\sigma v_0^\sigma$.

The limit problem. Let $\mathcal{Z}(s) = \lambda M + r\, n(s)$, let $\mathcal{J}_t(\tau) = \int_t^\tau e^{-(r+\lambda)(s-t)}\mathcal{Z}(s)\,ds + e^{-(r+\lambda)(\tau-t)}Q$ for $\tau \in [t,\infty]$, and let $\mathcal{V}(t) = \sup_{\tau \ge t}\mathcal{J}_t(\tau)$. $\mathcal{V}$ is bounded, $\mathcal{V} \ge Q$ (take $\tau = t$), and, by taking $\tau \ge t+\Delta$, $\mathcal{V}(t) \ge \int_t^{t+\Delta} e^{-(r+\lambda)(s-t)}\mathcal{Z}(s)\,ds + \upsilon\,\mathcal{V}(t+\Delta)$. The last inequality and boundedness give $\mathcal{V}(t+\Delta) \le \mathcal{V}(t) + \bar{a}\Delta$.

One-step errors. Since $\mu = \lambda\Delta + O(\Delta^2)$, $(1-\mu)(1-\delta) = r\Delta + O(\Delta^2)$ and $\int_0^\Delta e^{-(r+\lambda)s}ds = \Delta + O(\Delta^2)$,

$$\mu M + (1-\mu)(1-\delta)n_k = \int_{t_k}^{t_{k+1}} e^{-(r+\lambda)(s-t_k)}\mathcal{Z}(s)\,ds + \varepsilon_k, \qquad |\varepsilon_k| \le \bar{a}\Delta^2 + r\Delta\,\mathrm{osc}_{[t_k,t_{k+1}]}n.$$

Let ϖ be the modulus of continuity of n between discontinuities. In a round that contains no discontinuity, $\mathrm{osc}\, n \le \varpi(\Delta)$. A unit interval of time $[m, m+1)$, with m a nonnegative integer, contains at most N_d discontinuities, so at most $2N_d$ of its rounds contain one; for those rounds $\mathrm{osc}\, n \le 2\sup|n|$, and their index is at least $\lfloor m/\Delta \rfloor$. Hence, for weights $w_k \le \bar{\upsilon}^k$ with $\bar{\upsilon} = \max\{\upsilon, \beta\} < 1$,

$$\sum_k w_k|\varepsilon_k| \le \frac{\bar{a}\Delta^2 + r\Delta\varpi(\Delta)}{1-\bar{\upsilon}} + 4N_d\, r \sup|n|\,\Delta \sum_{m=0}^{\infty} \bar{\upsilon}^{\lfloor m/\Delta \rfloor} \;\longrightarrow\; 0 \quad (\Delta \to 0).$$

For the first term, $1 - \bar{\upsilon} \ge 1 - \upsilon$ and $\Delta/(1-\upsilon) \to 1/(r+\lambda)$, so it is $O(\Delta) + \varpi(\Delta)\, r/(r+\lambda)$. For the second, $\bar{\upsilon}^{\lfloor m/\Delta \rfloor} \le \bar{\upsilon}^{-1}(\bar{\upsilon}^{1/\Delta})^m$ and, for $\Delta \le 1$, $\bar{\upsilon}^{1/\Delta} \le \max\{e^{-(r+\lambda)}, \beta\} < 1$, so the sum over m is bounded and the term is $O(N_d\Delta)$.

Lower bound. Fix $\tau < \infty$ and let $K = \lceil \tau/\Delta \rceil$. Consider the policy that uses $\mathcal{T}^A$ in rounds $k < K$ and $\mathcal{T}^B$ afterwards. Using $\mathcal{T}^B$ in every round from K on gives values with $v_k - Q = \beta[\mu V + (1-\mu)(1-\delta)c_{U,k} - (1-\upsilon)Q] + \beta\upsilon(v_{k+1} - Q)$ for $k \ge K$, where the bracket is $O(\Delta)$, so $|v_K - Q| \le \bar{a}\Delta/(1-\beta)$. The policy's value is therefore $\sum_{k<K}\upsilon^k[\mu M + (1-\mu)(1-\delta)n_k] + \upsilon^K v_K = \mathcal{J}_0(K\Delta) + \sum_{k<K}\upsilon^k\varepsilon_k + \upsilon^K(v_K - Q)$, which tends to $\mathcal{J}_0(\tau)$ as $\Delta \to 0$. Hence $\liminf x_0 \ge \mathcal{J}_0(\tau)$ for every finite τ, and the case $\tau = \infty$ is a limit of finite τ. So $\liminf x_0 \ge \mathcal{V}(0)$.

Upper bound. Fix any policy σ and let $e_k = [v_k^\sigma - \mathcal{V}(t_k)]^+$. By the one-step error bound and the inequality for $\mathcal{V}$, $\mathcal{V}(t_k) \ge \mathcal{T}_k^A(\mathcal{V}(t_{k+1})) - \varepsilon_k$, so after a $\mathcal{T}^A$ step $e_k \le \upsilon e_{k+1} + |\varepsilon_k|$. Since $Q \le \mathcal{V}(t_k)$ and $\mathcal{V}(t_{k+1}) \le \mathcal{V}(t_k) + \bar{a}\Delta$, $\mathcal{T}_k^B(\mathcal{V}(t_{k+1})) \le (1-\beta)\mathcal{V}(t_k) + \beta\mathcal{V}(t_k) + \bar{a}\Delta$, so after a $\mathcal{T}^B$ step $e_k \le \beta e_{k+1} + \bar{a}\Delta$. Iterating, e_0 is at most the sum of the step errors, each weighted by the product of the earlier factors, plus a remainder that vanishes because every factor is at most $\bar{\upsilon} < 1$ and e is bounded. The $\mathcal{T}^A$ errors carry weights at most $\bar{\upsilon}^k$ and contribute a term that tends to zero, as shown above. The m-th $\mathcal{T}^B$ step carries a weight at most β^{m-1}, so all of them together contribute at most $\bar{a}\Delta/(1-\beta)$. Hence $e_0 \to 0$ uniformly in σ, $\limsup x_0 \le \mathcal{V}(0)$, and $P^* = \mathcal{V}(0)$,

which is the first expression in (ii).

Part (iii) and the second expression. By definition of $\hat{C}_i$ and N, $\int_\tau^\infty e^{-(r+\lambda)s}\mathcal{Z}(s)\,ds = e^{-(r+\lambda)\tau}\big[\psi M + (1-\psi)N(\tau)\big] = e^{-(r+\lambda)\tau}x^\circ(\tau)$, because $\lambda/(r+\lambda) = \psi$ and $r/(r+\lambda) = 1-\psi$. So $\mathcal{J}_0(\tau) = x^\circ(0) - e^{-(r+\lambda)\tau}\big(x^\circ(\tau) - Q\big)$ and $\mathcal{J}_0(\infty) = x^\circ(0)$, which gives the second expression for P^*, and shows that $P^* = x^\circ(0)$ if and only if $x^\circ(\tau) \geq Q$ for every τ. Next, $(r+\lambda)(x^\circ(t) - Q) = \lambda\beta(V-Q) + r(N(t)-Q)$, so $x^\circ(t) \geq Q$ if and only if $Q(r+\lambda\beta) \leq \lambda\beta V + rN(t)$, which is the condition $Q \leq \bar{Q}$ once the infimum over t is taken. Applying the same argument from any date t shows that the limit value at t is $\mathcal{V}(t) = x^\circ(t)$ for every t exactly under this condition, so the exit option is never used. With constant costs x° is constant and $P^* = x^\circ + [Q - x^\circ]^+ = \max\{Q, x^\circ\}$. The condition at $t=0$ alone is not sufficient. Take $\beta = \lambda = 0$, $r = \frac{1}{2}$, $c_U \equiv 0$, $c_E = 0$ on $[0,2)$ and $c_E = 2$ afterwards, and $Q = -1$. Then $x^\circ(t) = -\hat{C}_E(t)$, so $x^\circ(0) = -2e^{-1} \approx -0.736 \geq Q$, but $x^\circ(t) = -2 < Q$ for $t \geq 2$. The supremum in the second expression is attained at $t = 2$, so $P^* = x^\circ(0) + e^{-1}(Q+2) = -e^{-1} \approx -0.368$, which exceeds $x^\circ(0)$. □

B.19 Proof of Corollary 5

Assumptions. Baseline disagreement flows are c_E^0 and c_U^0, as in Proposition 6; they may include Washington's cost of cutting a loyal Europe's AI access from date zero. In a round with interruption, Europe's disagreement flow rises by ℓ, and Washington's rises by $(1-h)c_S$ only if the buffer stock is exhausted at the start of the round. The stock falls by Δ in each round with interruption and is unchanged otherwise, starting at t_U. We characterize Markov perfect equilibria, in which strategies depend on calendar time, the stock and the current offer. Let $D_0 = \beta(1-h)c_S - (1-\beta)\ell$, the value of D at $t_U = 0$, and let $x_0^\circ(t)$ be $x^\circ(t)$ of Proposition 6 under the baseline costs, with $x_0^\circ = x_0^\circ(0)$.

Timing variants. The text after Corollary 5 describes three other timings. We prove the following claims, (a) and (c) in the setting of part (i), where exit never binds without interruption. (a) If the buffer runs out at calendar date t_U whatever Europe does, Europe never interrupts before t_U when $\ell > 0$, and interruption raises P^* by $(1-\psi)e^{-(r+\lambda)t_U}\,[D_0]^+$, so it helps if and only if $\beta(1-h)c_S > (1-\beta)\ell$, for any finite t_U. (b) If Europe chooses interruption after an offer is rejected and $\ell > 0$, the unique Markov perfect equilibrium has no interruption, and P^* is unchanged. (c) If Europe chooses after the proposer is drawn but before offers, it interrupts in the rounds in which it proposes, at no cost on the equilibrium path, and interruption raises P^* by at least $(1-\psi)\beta(1-h)c_S e^{-(r+\lambda)t_U/\beta}$ whenever $\beta(1-h)c_S > 0$ and $t_U < \infty$, whatever the value of ℓ.

Proof of Corollary 5. Limit problem. The state is the round and the stock b. Interruption is chosen before the proposer is drawn, and both proposers' offers depend on the choice only through the round's disagreement flows and the next state, so Europe chooses it to maximize its own value, and its value satisfies a recursion of the form in the proof of Proposition 6 with an additional maximization over $i_k \in \{0,1\}$. The convergence argument of that proof, applied state by state,

gives as the limit value the supremum over measurable $i : [0,\infty) \to \{0,1\}$ and stopping times τ of

$$\mathcal{J}(i,\tau) = \int_0^\tau e^{-(r+\lambda)s}\big(\mathcal{Z}_0(s) + r\,\tilde{n}(s)\big)\,ds + e^{-(r+\lambda)\tau}Q,$$
$$\tilde{n}(s) = i(s)\big[-(1-\beta)\ell + \beta(1-h)c_S\mathbf{1}\{b(s)=0\}\big],$$

where $\mathcal{Z}_0 = \lambda M + r(\beta c_U^0 - (1-\beta)c_E^0)$ and $b(s) = [t_U - \int_0^s i]^+$. Let T_b be the first date with $b = 0$; $T_b \geq t_U$, and $T_b = \infty$ if the stock is never exhausted.

Key bound. For every i and τ, $r\int_0^\tau e^{-(r+\lambda)s}\tilde{n}(s)\,ds \leq (1-\psi)\,[D]^+$, and if $T_b \leq \tau$ and $D_0 \geq 0$, the left side is at most $(1-\psi)e^{-(r+\lambda)(T_b-t_U)}D$. If $T_b > \tau$, the integrand is $-(1-\beta)\ell\,i \leq 0$. Otherwise, before T_b the set of interruption dates has measure t_U and the integrand is $-(1-\beta)\ell\,i$. Among sets of measure t_U in $[0,T_b]$, the discounted measure $\int_0^{T_b} e^{-(r+\lambda)s}i(s)\,ds$ is smallest for $[T_b - t_U, T_b]$, where it equals $e^{-(r+\lambda)T_b}(e^{(r+\lambda)t_U}-1)/(r+\lambda)$. After T_b the integrand is at most $[D_0]^+$. So the left side is at most

$$\frac{r}{r+\lambda}\,e^{-(r+\lambda)T_b}\Big[-(1-\beta)\ell\big(e^{(r+\lambda)t_U}-1\big) + [D_0]^+\Big].$$

If $D_0 \geq 0$, the bracket equals $\beta(1-h)c_S - (1-\beta)\ell e^{(r+\lambda)t_U} = e^{(r+\lambda)t_U}D$, so the bound is $(1-\psi)e^{-(r+\lambda)(T_b-t_U)}D \leq (1-\psi)\,[D]^+$, since $T_b \geq t_U$. If $D_0 < 0$, the bracket is nonpositive, and $D \leq D_0 < 0$.

Part (i). Since exit never binds without interruption, $\sup_\tau \mathcal{J}(0,\tau) = \mathcal{J}(0,\infty) = x_0^\circ$. By the key bound, $\mathcal{J}(i,\tau) \leq \mathcal{J}(0,\tau) + (1-\psi)\,[D]^+ \leq x_0^\circ + (1-\psi)\,[D]^+$. If $D > 0$, the policy $i \equiv 1$, $\tau = \infty$ attains this: it exhausts the stock at $T_b = t_U$ and adds $\frac{r}{r+\lambda}\big[-(1-\beta)\ell(1-e^{-(r+\lambda)t_U}) + D_0 e^{-(r+\lambda)t_U}\big] = (1-\psi)D$. If $D \leq 0$, $i \equiv 0$ is optimal and interruption does not raise P^*. Since $r > 0$, $1-\psi > 0$, and D is strictly decreasing in t_U when $\beta(1-h)c_S > 0$, so $D > 0$ if and only if $t_U < \bar{t}$.

Part (ii). With constant baseline costs, $\alpha = \mathcal{Z}_0 - (r+\lambda)Q$ is constant and $x_0^\circ - Q = \alpha/(r+\lambda)$. Then $\mathcal{J}(i,\tau) - Q = \int_0^\tau e^{-(r+\lambda)s}(\alpha + r\tilde{n})\,ds$. If $T_b > \tau$, this is at most $\int_0^\tau e^{-(r+\lambda)s}\alpha\,ds \leq [\alpha]^+/(r+\lambda)$. If $T_b \leq \tau$, apply the bathtub step on $[0,T_b]$ and bound the integrand on $[T_b,\tau]$ by $[\alpha + rD_0]^+$, which gives

$$\mathcal{J} - Q \leq \frac{1}{r+\lambda}\Big[\alpha\big(1-e^{-(r+\lambda)T_b}\big) + e^{-(r+\lambda)T_b}\Big([\alpha+rD_0]^+ - r(1-\beta)\ell\big(e^{(r+\lambda)t_U}-1\big)\Big)\Big].$$

If $\alpha + rD_0 \geq 0$, then, since $D_0 - (1-\beta)\ell(e^{(r+\lambda)t_U}-1) = e^{(r+\lambda)t_U}D$, the bound equals $\alpha/(r+\lambda) + (1-\psi)e^{-(r+\lambda)(T_b-t_U)}D$. This is affine in $e^{-(r+\lambda)(T_b-t_U)} \in (0,1]$, so it is at most $\max\{\alpha/(r+\lambda),\ \alpha/(r+\lambda) + (1-\psi)D\}$. If $\alpha + rD_0 < 0$, the bound is at most $[\alpha]^+/(r+\lambda)$. Hence $\mathcal{J} \leq \max\{Q,\ x_0^\circ,\ x_0^\circ + (1-\psi)D\}$, and the three terms are attained by $\tau = 0$, by $(i \equiv 0, \tau = \infty)$ and by $(i \equiv 1, \tau = \infty)$. Without interruption the value is $\max\{Q, x_0^\circ\}$ by Proposition 6, so interruption raises P^* if and only if $(1-\psi)D > [Q - x_0^\circ]^+$. This requires $D > 0$, the condition of part (i), and when exit binds $(Q > x_0^\circ)$ it requires more.

Part (iii). In the setting of part (i), exit never binds from any date, and the problem from state (t,b) is the problem of part (i) with t_U replaced by b. Attaining the bound $x_0^\circ(t) + (1-\psi)\,[D(b)]^+$, where $D(b) = \beta(1-h)c_S e^{-(r+\lambda)b} - (1-\beta)\ell$, requires interrupting continuously from t when $D(b) > 0$,

because any delay multiplies the gain by $e^{-(r+\lambda)(T_b-t-b)} < 1$; and when $D(b) < 0$ every interruption lowers the value. So at stock b Europe interrupts if and only if $\beta(1-h)c_S e^{-(r+\lambda)b} > (1-\beta)\ell$. Along an interruption path the stock falls one for one with time, and the left side rises as b falls, so the condition keeps holding. Without interruption b stays at t_U, so a condition that fails at $t = 0$ fails in every round. (With constant costs and exit binding, the condition at stock b is $(1-\psi)D(b) > [Q - x_0^\circ]^+$ by part (ii), which is also monotone in b, so the decision is time consistent there as well.)

(a) Calendar buffer. Now $\tilde{n}(s) = i(s)\big[-(1-\beta)\ell + \beta(1-h)c_S \mathbf{1}\{s \geq t_U\}\big]$, which does not depend on past choices and can be maximized date by date: $i(s) = 1$ if and only if $s \geq t_U$ and $D_0 > 0$. With $\tau = \infty$ optimal, this adds $r\int_{t_U}^{\infty} e^{-(r+\lambda)s}\,[D_0]^+\,ds = (1-\psi)e^{-(r+\lambda)t_U}\,[D_0]^+$, and interrupting before t_U strictly lowers the value when $\ell > 0$.

(b) Choice after a rejection. Europe then chooses i ex post to maximize its own disagreement value $\mu Q + (1-\mu)[-(1-\delta)(c_E^0 + \ell i) + \delta x(b_i')]$, where $b_1' = [b-\Delta]^+$ and $b_0' = b$; the expression is the same after Europe rejects an offer and after Washington rejects a demand. At $b = 0$ interruption leaves the state unchanged and costs Europe $(1-\mu)(1-\delta)\ell > 0$, so $i = 0$, and $x(0) = y_0$, the value without interruption. Suppose $x(b-\Delta) = y_0$. If Europe interrupted at b, its value there would be x_1, the value of the recursion of Proposition 6 with Europe's disagreement flow raised by ℓ for one round and continuation y_0; since the two recursions differ only in Europe's disagreement value, which enters through the exit-floor term, $y_0 - x_1 \in [0, (1-\beta)(1-\mu)(1-\delta)\ell]$. Interrupting would then change Europe's disagreement value by $(1-\mu)[-(1-\delta)\ell + \delta(y_0 - x_1)] \leq (1-\mu)(1-\delta)\ell\,[-1+\delta(1-\beta)(1-\mu)] < 0$, so interruption is not a best response. Not interrupting at b gives the value y_0 and is a best response. So $i = 0$ is the unique consistent choice and $x(b) = y_0$, and induction over b completes the argument.

(c) Choice after the proposer is drawn. When Europe proposes, its accepted demand equals Washington's disagreement value. Interruption raises Washington's disagreement cost by $(1-\mu)(1-\delta)(1-h)c_S\mathbf{1}\{b=0\}$ plus $(1-\mu)\delta[x(b-\Delta) - x(b)] \geq 0$, where the second term is nonnegative because a lower stock only brings forward the date from which interruption costs Washington something and Europe nothing. Europe's own cost ℓ does not enter, because the demand is accepted. So Europe interrupts in all of its proposer rounds. Under the policy that interrupts in those rounds only, the stock falls by Δ in each of them; by the law of large numbers, in the frequent-offer limit it falls at rate β per unit of time and is exhausted at t_U/β, after which the flow gain is $r\beta(1-h)c_S$ with no cost. This policy therefore adds $r\int_{t_U/\beta}^{\infty} e^{-(r+\lambda)s}\beta(1-h)c_S\,ds = (1-\psi)\beta(1-h)c_S e^{-(r+\lambda)t_U/\beta}$, which bounds the gain from below and is positive whenever $\beta(1-h)c_S > 0$ and $t_U < \infty$. □

C Numerical Verification

We checked every result in Appendix B against solutions computed by brute force from the primitives of the model. The scripts are in `code/checks`, run with `python3 -I` and finish in under thirty seconds each. No solver uses the formula it checks: equilibria come from enumeration, backward induction or direct optimization, and closed forms are computed separately. Where a statement

depends on a tie-breaking rule, the script solves the game under that rule or computes every equilibrium outcome. `code/checks/README.md` maps the names used in the script comments to the labels used here.

Bidding. `v2_m1_lemma1.py` works in exact rational arithmetic. In the clock auction (300 lattice draws) dropping at one's maximal credible bid is weakly dominant in every draw, and Europe's payoff equals $[\min\{Q, V, \bar{m}_U\}]^+$ exactly. In the alternating-raise auction, 800 games are solved for the full set of pure subgame-perfect outcomes with no tie-breaking. The winner matches Lemma 1 in all 580 games with Q more than one grid step below $\bar{z}_U$ and in all 44 with Q more than one step above it. The largest payoff to Europe is within one grid step of the formula in all 800, and in 108 games the smallest is lower by more than a step, the price indeterminacy of the claim in Appendix B. When the losing bidder raises whenever indifferent, the price is within one grid step of $[Q]^+$.

Bargaining. `v2_m1_prop1.py` runs finite-horizon backward induction in which each proposer compares every offer on a 601-point grid with the payoff of an unacceptable offer. Over 500 runs the largest gap between Europe's value and $x^*(\Delta)$ is 1.7×10^{-11}, and with $\lambda > 0$ no proposer weakly prefers delay. Richardson extrapolation from $\Delta = 0.004, 0.002, 0.001$ on 120 draws gives a largest error against P^* of 7.5×10^{-6}. Iterating the supremum and infimum maps of Step 4 from $[-10V, 10V]^2$ converges to $(x^*, x^*, -x^*, -x^*)$ within 4.4×10^{-12} (300 draws). Parts (a) to (c) of Corollary 1 hold in every draw, finite differences in V match $\beta\psi$ to 6.6×10^{-10}, and $\hat{Q}$ classifies every draw. For the timing of the footnote to Proposition 1, the stationary value matches its formula to 1.6×10^{-14} (300 draws), and the nonstationary profile that gives Europe close to V passes a one-shot deviation check in all 67 draws in which it was built.

Slack principle, rationing and hedges. `v2_m3_prop2.py` draws one to four instruments and one to four demands, with additive or general monotone set functions, premia of every sign, reversible and sunk demands, zero or positive execution costs, and integer draws that create ties. The full set of subgame-perfect outcomes of the adaptive protocol comes from set-valued backward induction over states, and a separate solver finds the best single ultimatum. $B(J)$ and $\min\{T, \Lambda\}$ match to 8.9×10^{-16} over 3,000 instances, and the acceptance rule of Proposition 2(b) has no mismatch in 123,300 pairs. The bound of part (c) is never violated in 4,000 adaptive instances, and Washington's payoff never exceeds V_1 (it falls strictly below in 325 of 2,000). Where a sufficient condition of part (d) holds (1,593 instances), the adaptive and single-ultimatum values coincide, and the ratchet count $n \bmod (b+1)$ matches in all 300 draws. In the burning example (two instruments with $u = 1.2$ and 1.0 and $d = 0$, $Q = -5$, sunk demands with $t = 1$, $w = 5$ and $t = 1.5$, $w = 1$, execution cost 0.1 each), Washington first issues an ultimatum that Europe refuses in every equilibrium, which burns one instrument, and then obtains the first demand, for a payoff of 4.9 against 5 under a single ultimatum. When a refusal can make a new instrument credible, Europe's loss exceeds the bite of the initial credible set by up to 4.91 but never that of the enlarged set, which is why the proposition

fixes J^c. A plain knapsack with capacity $\min\{\Lambda, B(J^c)\}$ misses Washington's optimum in 521 of 1,046 reversible-only instances, and the program of Corollary 2 in none. The linkage condition of Proposition 3(b) has no mismatch in 3,000 draws.

Rupture risk. `v2_rupture_bite.py` computes the bite of a withdrawal directly from P^* of Proposition 1 in 13,079 random draws with $Q + d < V$ ($\lambda = 0$ in about half). The bound of Remark 1 has no violation, at $\lambda = 0$ the bite agrees with Proposition 2(a) to 8.9×10^{-16}, and the example of the text gives $\Lambda = 1$, $\Lambda_\psi = 2.2$ and a bite of 1.8.

Counter-threats. `v2_m3_prop2e.py` lets China choose counter-threats from a grid before Washington's single ultimatum. Over 4,000 instances the acceptance rule of Proposition 3(a)(i) has no mismatch in 13,286 checks. China gains in 615 instances, each time by changing Washington's demand set; a counter-threat is executed on the path in 140 instances, and each executed threat crowds out another demand. Washington never gains. Europe's loss rises in 89 instances and falls in 405, and with a divisible sunk demand that China cannot threaten it equals $B(J^c)$ to 8.9×10^{-16}.

Conditionality. `v2_m3_prop3.py` draws 1,500 problems with H from six convex families and maximizes each on a grid of 200,001 points with golden-section polishing. The formula for κ^* is never beaten (largest shortfall 8.9×10^{-16}), all five regimes occur, and finite differences in Q_0 give the derivatives of Proposition 4(c). For part (d), 3,000 draws per distribution show $\bar{\kappa}$ rising with s_X in 1,218 normal, 1,198 logistic, 974 exponential and 973 uniform cases, and the derivative matches finite differences to 3×10^{-11}.

Preventive control. `v2_m4_prop4.py` solves the two-period game by backward induction from primitives on nine edge cases, 20,000 integer draws with dyadic δ (so ties are exact) and 10,000 continuous draws. The issue decision never differs from Proposition 5; the errors in Γ, T^* and Europe's payoff are zero on integer draws and below the grid mesh otherwise; and none of the 7,715 draws with an ultimatum has a refusal on the path. The identity of Corollary 3(b) holds to 2.7×10^{-15}, the purchase without bite gives $\Gamma = 1$ and $T^* = 2.5$, Γ never falls as Δ_c rises (3,000 draws), and the special case after Proposition 5 matches in 1,504 draws. `v2_m4_prop4_china.py` adds China's announcement of r^c (step $\frac{1}{2}$) and a stake, with $e^C(r^c) = \gamma r^c$. In 5,000 integer draws every China-optimal announcement satisfies Corollary 3(d)(i) to (iii), $z_C = 0$ leaves the outcome open in 183 draws, as (iv) allows, and a self-rewarding punishment ($\gamma = -\frac{1}{2}$) is carried out on the path in 968.

Cuts and windows. `v2_m4_cuts_windows.py`, part A, draws 20,000 games with one to five dates. In the 13,927 draws inside the scope of Lemma 2 the largest errors are 3.6×10^{-15} for the bite and 1.8×10^{-15} for the cost; in all 6,073 outside it the true loss exceeds the formula. Part B solves 60,000 multi-date games by backward induction in exact rational arithmetic. The condition with $t' > t$ misclassifies 1,030 draws, all with some $u_t > 0$, and the condition of Corollary 4 matches

every draw. Of the 20,199 games with an ultimatum, 13,923 issue it at the last eligible date and 1,202 at a date that is not itself eligible. No refused ultimatum is ever optimal.

Self-binding. `v2_m4_selfbind.py` solves 6,000 integer draws with $Q_1 < 0$, $\Delta_c = \infty$ and no investment. Parts (a), (b) and (d) of Remark 2 have no mismatch, partial binding invites an ultimatum that would not otherwise be issued in 210 draws, and the example of part (c) is reproduced.

Breakdown. `v2_m4_prop5.py` maximizes Washington's gain directly over the payment, without the first-order condition, for both kinds of offer, with σ drawn from six families (1,500 draws each): uniform, truncated normal, Beta(2,2), triangular, truncated exponential, and a bimodal mixture that is not log-concave. For the five log-concave families the issuance condition of Proposition 7(c) matches every draw away from the threshold, and the largest errors are 3.4×10^{-7} in the cutoff and 2.2×10^{-7} in P_b. For the comparative statics (500 draws per family) part (d) has no violation in any family, the bimodal one included; with the purchase option, violations occur only in the directions part (e) says can fail, and the four examples in the proof are reproduced.

Who decides. `v2_m2_remarks_GFE.py` checks Remark 3 by brute force on a grid of mesh 10^{-3} (20,000 draws; largest errors 4.4×10^{-16} in the payment and 8.9×10^{-16} in the slack) and Remark 4 on 200,000 integer draws, with no mismatch. For Remark 5 it reads the population shares of Council Decision 2025/2427 (Council of the European Union, 2025) from `pop_check.py`. The first-stage blocking rule matches the treaty rule on all 403,168 sets of members tested and the delegated-act rule on 241,708, and the coalitions of Table 5 and $\Phi(0)$ are computed exactly. `council_revealed_blocking.py` applies the same rule in integer arithmetic to 2026 populations and the EUROPOP2023 projections for 2029 to 2031. The seven states that *Europe 2031* names hold 28.06 percent of the population in 2026 and 27.76 percent in 2031. In every year Germany, France and Spain are the only states that complete a block alone, and without them and Romania the smallest completions have three states (for example Belgium, Czechia and Portugal). Section 7.3 and Appendix D use the population shares and completions. Exact Shapley–Shubik and normalized Banzhaf indices, checked against 200,000 random orderings, give Germany 17.54 and 12.03 percent against 3.66 and 3.80 percent for the Netherlands, ratios of 4.8 and 3.2.

Hold-up. `v2_m2_prop6_holdup.py` solves the discrete game by backward induction on the grid of offer dates, with random piecewise-constant cost paths, for 6,000 parameter sets. Where $Q \leq \bar{Q}$, the largest error against $x^\circ(0)$ is 3.1×10^{-7}, and the test $Q \leq \bar{Q}$ classifies every draw. Where exit binds, the value exceeds $x^\circ(0)$ in every draw and converges to the stopping value at first order, and the counterexample in the proof is reproduced (-0.3679 against -0.7358 for $x^\circ(0)$). `v2_m2_interruption.py` solves the stock-buffer game of Corollary 5 and the three timing variants by backward induction over the stock (3,000 parameter sets, exit binding without interruption in 977). The largest error against part (ii) is 8.1×10^{-4}, the decision agrees with $(1-\psi)D > [Q - x_0^\circ]^+$

in all 2,834 draws not near a tie, and every draw that interrupts at the full stock interrupts at every lower stock. No draw of variant (a) interrupts before t_U; in variant (b) the gain is exactly zero in all 2,852 draws with $\ell > 0$; and in variant (c) the gain is positive in 551 draws in which the condition of part (i) fails.

D Europe 2031 Read Through the Model

Europe 2031 stages four rounds of US–European bargaining over ASML (Juijn et al., 2026). Its authors say the storyline "is not meant to be a prediction", and we do not treat it as evidence. With as many free parameters as our model has, the statement that an episode is an equilibrium carries little information on its own. Table 6 therefore states, for each move, the result that applies and what the move assumes about the parameters.

2028. Washington threatens the foreign direct product rule to end ASML's remaining immersion sales and servicing in China; "The Dutch cave" and get nothing in return (Juijn et al., 2026). The demand concerns the hedge, and the threat uses flow jurisdiction, which by Proposition 3(b) is credible whenever its net ex post cost is within Washington's stake, whatever Europe's slack. The text concedes that "such penalties would also harm the US". In our model that self-harm attaches to penalties on the whole firm (χ_f), and jurisdiction over China-bound flows stays credible for every V. The episode needs a nonpositive premium and no effective rupture risk. The crisis comes during a compute crunch, when V is high, so if the dispute carried a hazard of rupture and the Dutch could make proposals, Corollary 1(c) would give Europe a positive transfer whenever $Q + \beta(V - Q) > 0$, unless Washington's credible conditional threats clawed it back (Remark 1). The scenario attributes the outcome to a European position that "fragments before it has properly formed". Licensing is a Dutch decision (Section 7.1), and partners reach the gatekeeper's premium only through transfers, so a united position would not by itself have changed Q.

2029. The Frontier Inference Services Rule caps Tier 2 countries at 25 percent of any provider's frontier inference, which the text says means "roughly halving European allocations", while the Netherlands joins the Five Eyes, Japan, South Korea and Taiwan in Tier 1 (Juijn et al., 2026). Cutting a loyal Europe lowers the conditional part of the AI instrument, because conditional parts are measured from what a loyal ally receives (Section 4.3). Washington spends slack to keep compute, and Europe's premium drifts upward. We read the withdrawal of the real AI Diffusion Rule before compliance was due (Section 2) as a sign that such tiering is costly for Washington. Tier 1 for the Netherlands is the targeted benefit of Remark 3.

The failed vote on the Arizona measure rests on the thinnest assumptions. The states the scenario names as opposing or abstaining cannot block the Council's determination (Section 7.3 gives the arithmetic and the completions), and what fails is a response measure, which after appeal only a qualified majority against can stop (European Parliament and Council of the European Union, 2023, 2011). The story needs more members with $q_i \leq 0$ than it names, or a Commission that will not go

against a predominant position among member states (European Commission, 2023). We think the second is likelier, though it points to a measure that is never tabled, and the Union's record against Washington is to arm countermeasures and suspend them on a US concession (Section 7.3). Even if adopted, the measure, a hold-up instrument aimed at the servicing Washington can most easily compel (Section 7.4), could be lifted after a US offer, the timing under which Corollary 5 gives no interruption.

The stockpile. Washington's claim that through Taiwanese partners it has "amassed enough EUV machines and maintenance know-how" to outlast Europe (Juijn et al., 2026) is a claim about t_U and h in Corollary 5. It lowers c_S, what Washington's own fabs would lose in a dispute, and leaves V, its loss if China gained the tools, where it was. The claim and the ultimatum are therefore compatible: a Washington that can do without ASML's servicing still has a motive to keep China from gaining the chokepoint. Under complete information the claim would be either true or irrelevant; as a bluff it belongs to the incomplete-information extension that Section 9.3 sets aside.

2031. The vice president's offer is a preventive demand of the kind Proposition 5 describes, made "now, while the option still exists" (Juijn et al., 2026). Before it, Chinese state-backed funds extend credit to a Portuguese infrastructure bank, refinance Greek sovereign debt and underwrite a Spanish regional government, and rationing has lowered d_{ai}. Rationing raises Europe's premium directly; the Chinese credit raises it only insofar as it shows what China could credibly promise for exit. A Washington that expects the premium to turn positive acts while its threat still bites (Corollary 4). A controlling vote in a Dutch-American holding stays under Dutch law and has low enforceability Δ_c, which by Corollary 3(c) lowers what control is worth to Washington. Capital for production on US soil raises Δ_c. Cash transfers indexed to US AI windfalls are promises paid after control has passed, credible only up to $\bar{m}_U$ (Section 4.2). If the Tier 3 threat is a lasting cut, Lemma 2(iii) gives it only its unconditional part as added bite at dates when Europe's premium is positive.

Since under complete information an issued ultimatum is accepted, Europe's refusal requires sovereignty costs that Washington cannot observe (Proposition 7). Beijing's warning on rare-earth terms is conditioned on signing, so by Corollary 3(d) it is never carried out if control passes, and it protects European ownership only by deterring the demand. The scenario's further claim that signing with neither power means absorbing "the full displeasure of both great powers at once" needs a Chinese punishment of non-alignment that the text does not state.

The epilogue's counterfactual coalition of eight states, among them Japan and South Korea (Juijn et al., 2026), lies outside the model, which has one rival and lets holders bargain one at a time (Section 9.3).

Table 6: Europe 2031 moves and the conditions under which they are equilibria

Scenario move	Results	Equilibrium when
May 2028: Washington demands an end to ASML's immersion sales and servicing in China and threatens the foreign direct product rule; the Dutch comply and get nothing in return	Proposition 3(b), Corollary 1(c), Remark 1	The jurisdictional threat is credible ($e_F \leq \bar{m}_U$), so it deters without being carried out and leaves Q unchanged. A zero transfer needs $Q \leq 0$ and either no rupture risk in the dispute ($\lambda = 0$), or $Q + \beta(V - Q) \leq 0$, or credible conditional threats that claw back what agenda power would win.
April 2029: the Frontier Inference Services Rule puts the Netherlands in Tier 1 and most of Europe in Tier 2	Section 4.3, Remark 3, Corollary 2	Rationing a loyal Europe lowers $x_{ai}(A)$, hence d_{ai} and Washington's slack. We think Washington rations only if the compute it keeps is worth more to it than the slack it loses. Tier 1 lowers the gatekeeper's price only if it is baseline provision credibly withdrawn on exit.
May 2029: an Anti-Coercion Instrument measure restricting ASML's exports and servicing to US fabs in Arizona falls short of a qualified majority	Remark 5, Corollary 5	The named opponents and abstainer cannot block the determination, and after appeal only a qualified majority against stops a response measure (Section 7.3). Failure needs further members with $q_i \leq 0$ or a Commission that will not go against a predominant position. Even adopted, the measure could be suspended after a US offer, and it adds little where Washington can compel servicing (h near one).
March 2031: Washington says that through Taiwanese partners it can forgo ASML longer than Europe can forgo frontier AI	Proposition 6, Corollary 5	The claim is that the US buffer t_U exceeds $\bar{t}$. It holds for a measure confined to US soil if h is near one; for a broad interruption it depends on h in Taiwan and Korea. It concerns c_S and leaves V unchanged.
March 2031: a joint Dutch-American holding with a US controlling vote, capital for US production and windfall-indexed transfers, made public with a Tier 3 threat	Proposition 5, Corollary 3, Corollary 4, Lemma 2	Issued iff $\Gamma > 0$: a premium expected to turn positive and a bite above $\sigma + r^c$ while slack lasts, or $v_c > \iota$. A vote in a Dutch company has low Δ_c, which lowers Γ. Transfers due after the transfer of control are credible only up to $\bar{m}_U$. A lasting cut adds only its unconditional part at dates with $Q \geq 0$.
March 2031: Europe refuses; Beijing says rare-earth terms will be reviewed if the Dutch sign	Proposition 7, Corollary 3(d)	Under complete information an issued ultimatum is accepted, so refusal needs private sovereignty costs. China's threat is never carried out if control passes; if credible and large enough ($\Gamma(r^c_{\max}) \leq 0$) it deters the demand.